

Goldeneye Final Design Report

Competing Project 41
Turner Fenton Rocketry Association (TFRA)



LAUNCH CANADA CHALLENGE 2026

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ABSTRACT

Goldeneye is a high-power, single-stage rocket developed by the Turner Fenton Rocketry Association (TFRA) for the 2026 Launch Canada Challenge. Designed as a successor to previous team vehicles, Goldeneye serves as a technical demonstrator for supersonic flight and integrated student-researched-and-developed (SRAD) systems. The mission's primary goal is to characterize aerodynamic performance in transonic and supersonic regimes. The project officially began on October 26, 2025, and is scheduled for completion in time for the competition on August 14, 2026, spanning a total duration of approximately 9.6 months.

The vehicle is constructed using a 4-inch diameter commercial-off-the-shelf (COTS) carbon fiber airframe, chosen for its superior strength-to-weight ratio and modularity with existing TFRA hardware. A COTS Von Kármán G10 fiberglass nose cone is utilized to minimize drag. Propulsion is provided by a Cesaroni Technologies Pro75 4864L2375-P (L-class) motor, which generates a total impulse of 4864 Ns over a 1.9-second burn duration. Flight dynamics analysis conducted via OpenRocket indicates that this configuration will propel Goldeneye to a maximum velocity of Mach 1.25 (429 m/s) and an apogee of 12,590 ft AGL.

To ensure the safe retrieval of the vehicle, Goldeneye employs a dual-separation, dual-deployment recovery scheme. A COTS 24-inch drogue parachute is deployed via a 1.7 g black powder charge at apogee to stabilize the descent, followed by an 72-inch main parachute at 1,500 ft AGL to slow the vehicle to a touchdown velocity of approximately 5 m/s (17 ft/sec). The recovery system is governed by a redundant avionics suite consisting of a primary Blue Raven and a secondary RRC3+ flight computer, each utilizing independent power sources and pyrotechnic channels to mitigate the risk of a single-point failure.

The core scientific focus of the mission involves a distinct payload: a student-developed pitot-static system (pitot tube). The pitot tube is designed to acquire high-fidelity airspeed data in subsonic, transonic, and supersonic regimes.

The configuration presented above represents the final design of Goldeneye. Since the submission of the project's initial design presentation (IDP), the vehicle has undergone several significant design iterations. The most substantial change involved the propulsion system. The original design was intended to fly on an M-class motor, however due to competition regulations for high school teams this design was prohibited. After discussion with LC officials, the team elected to compete in the Restricted category while limiting the vehicle to an L-class motor. This decision required a comprehensive redesign of the vehicle to ensure that the airframe, stability, recovery system, and payload remained optimized for the revised flight profile.

The payload was also simplified to better align with the team's available time and resources. While the initial concept incorporated multiple experimental subsystems including an airbraking system, and "Sharkskin" technology to reduce drag, the final payload focuses exclusively on the pitot-static system.

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ABBREVIATIONS

AIAA	American Institute of Aeronautics and Astronautics
AGL	Above Ground Level
CONOPS	Concept of Operations
COTS	Commercial Off The Shelf
DOT	Department of Transportations
EGSE	Electrical Ground Support Equipment
ESA	European Space Agency
GSE	Ground Support Equipment
GPS	Global Positioning System
MGSE	Mechanical Ground Support Equipment
NASA	National Aeronautics and Space Administration
NDT	Non-Destructive Testing

P&ID	Piping and Instrumentation Diagram
SRAD	Student Researched And Developed
STEM	Science, Technology, Engineering and Mathematics
TC	Transport Canada
WBS	Work Breakdown Structure

NOMENCLATURE

m/s^2	Acceleration
CO_2	Carbon Dioxide
I_{sp}	Specific Impulse
ft	Feet
g	Gravitational Constant of 9.81 m/s^2
kg	Kilograms
m	Meters
O/F	Mixture Ratio
N	Newtons
s	Seconds
C_v	Valve Flow Coefficient
m/s	Velocity (also denoted by ft/s)

1 INTRODUCTION

Goldeneye (Project 41) represents the Turner Fenton Rocketry Association's (TFRA) entry into the 2026 Launch Canada Challenge. The mission is a significant technical leap for the team, moving beyond the inaugural flight of Spectre (2025) to focus on the characterization of aerodynamic performance in supersonic flight regimes.

The design philosophy of Goldeneye centered on balancing high-performance flight with reliability. To achieve this, the team practiced design optimization, analysis, and rigorous precision in manufacturing processes.

Goldeneye introduced an advanced experimental payload: a high-precision pitot-static velocity measurement system (pitot tube). Developing this payload pushed the team to conduct specialised research, deepening our knowledge in aerodynamics, and flight systems.

Another major area of focus for this mission is the integration of student-researched-and-developed (SRAD) electronics within the avionics bay. Goldeneye features a custom Data Acquisition (DAQ) system. This system captures high-resolution data from sensors including high-G accelerometers and barometric sensors for post-flight analysis.

Structurally, the vehicle incorporates an SRAD aluminum boat tail at the aft end for drag reduction and motor retention, which serves as the primary thrust transfer mechanism. Through this, the team learned about bending analysis, thrust transfer, and metal manufacturing.

Overall, the complex integration of mechanical and electronic SRAD components on Goldeneye demonstrates the team's maturation and focus on gaining technical expertise.

The development of Goldeneye is managed by a structured team of approximately 13 core authors and sub-team leads, with 52 members in its entirety. To ensure technical continuity, TFRA has implemented a rigorous succession planning strategy, fully transitioning the team structure to recruit younger students, who comprised 75% of the year's applicants. By placing experienced members in leadership roles and involving junior members in onboarding projects, the team ensures that the technical knowledge required for supersonic flight is transferred beyond the founding years. This organizational focus, combined with the technical rigor of the Goldeneye design and the expansion of the NOCSA outreach initiative, positions TFRA for long-term sustainable success in the aerospace field.

1.1 Updates & Lessons Learned

The overall design, scope, and objectives of the Goldeneye flight vehicle have evolved drastically from when the project was conceived in August of 2025, til the present day. During the Initial Design Proposal, Goldeneye was expected to be utilizing an M3400 in order to complete its objectives, and compete at the Launch Canada 2026 competition. The payloads at this time were a locator beacon, a comprehensive DAQ system, and a proprietary drag-reducing “Sharkskin” technology. However, TFRA was placed in the “Junior Launch Category”, and thereby was limited in impulse. After much cooperation and leniency from Launch Canada officials, TFRA was permitted to compete within the “Restricted” category, given that Goldeneye would be powered by an L-class rocket motor. As the design was reinvented and reworked, the payload also shifted into a singular focus on a functional supersonic pitot tube. This allowed the team to focus its resources on one payload, and therefore increased chances of success. Furthermore, the team’s overall design philosophy shifted drastically over the course of the development cycle. It became clear that the competition officials, and the industry at large, prefers well-engineered, substantiated, and reliable engineering. Therefore, TFRA began conducting far more research, and making far more substantiated design decisions, as evident within the designs for the boat tail, avbay, pitot tube, and many other subsystems.

In addition, there were continuous organizational changes and logistical obstacles that made this year difficult but incredibly insightful. In particular, the majority of senior management in the TFRA team were in Grade 12. As such, every single team lead and manager was unable to dedicate time to TFRA as consistently as previous years. Amid university applications, exams, and other grade 12 expectations, several leads stepped down, and left a gap in the team’s organizational structure and knowledge base. With that being said, this gave the opportunity for new members to step up, take responsibility, and lead. This had the added benefit of empowering newer members to embrace the challenges of rocketry, and inadvertently improved the future generations of TFRA.

There were consistent issues with funding as well. Last year, the Peel District School Board (PDSB) was freely funding TFRA and other such initiatives from schools across the district. With that being said, with the provincial government taking supervision over the PDSB, some programs and initiatives faced severe funding cuts. TFRA was one such program.

As such, the administrative department had to pivot strategies and focus on other sources of funding, and the expenditure of the engineering team was scrutinized to a much greater degree in order to prevent waste. Overall, more so than the various technical challenges experienced during this competition year, the financial and administrative challenges were a consistent obstacle to TFRA’s success. Nevertheless, the team’s unrelenting resolve to push forward has kept us on the path to Launch Canada 2026.

1.2 Team Organization

Faculty Advisor: Timothy Shing

Team Captain: Dhyan Soni

Chief Engineer: Nishargo Syeed

Chief Safety Officer: Christopher Hobbs

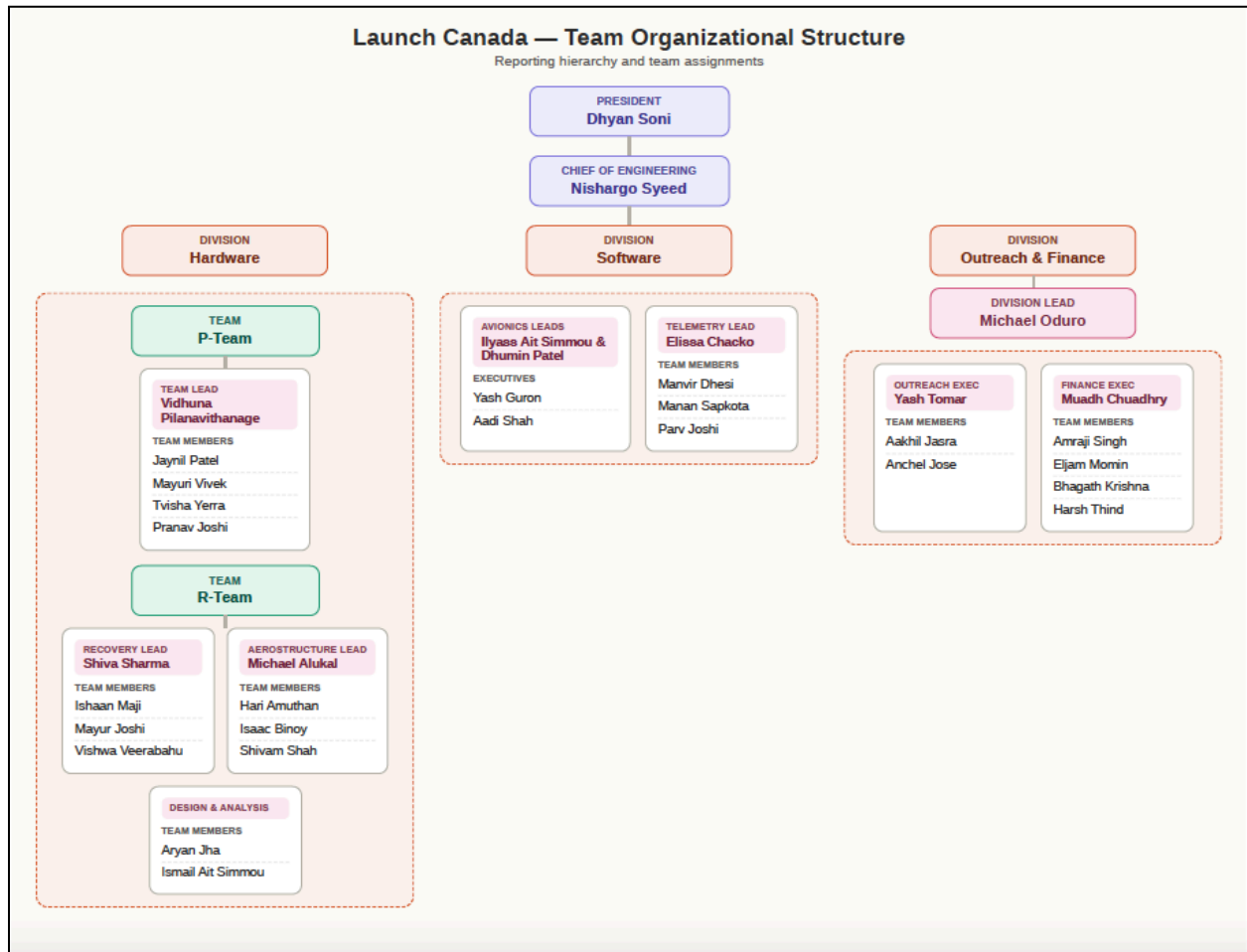


Figure 1: Technical Development Team Structure.

1.3 Team Development & Succession Planning

Turner Fenton Rocketry recognizes that long-term success depends as much on a strong people pipeline as on sound engineering. To ensure key roles are never left vacant, TFRA has built a structured recruitment and development process that identifies, trains, and retains capable members well before senior students graduate.

Recruitment begins early in the school year, with open positions advertised to underclassmen through club fairs, classroom presentations, and school-wide events. Rather than accepting members into the team at large, each sub-team (Aerostructures, Recovery, and Avionics/Software) conducts its own interviews to evaluate candidates' technical background and fit for that sub-team's specific needs, ensuring new members are placed where they can contribute and grow fastest.

Once selected, new members complete a structured onboarding project before joining an active design team, complete with deadlines, deliverables, and design reports, reviewed through frequent check-ins with senior members. Incoming Aerostructures members, for example, have designed and built a launch rail and a vacuum chamber, while incoming Avionics/Software members complete independent avionics projects to demonstrate coding and systems-integration skills. This gives recruits hands-on experience with the full design process, requirements, iteration, testing, documentation, in a lower-stakes environment before taking on responsibility within the main vehicle program. Afterward, new members are paired with senior teammates on live projects, so technical knowledge and team practices transfer directly through collaboration rather than documentation alone.

To prevent the loss of institutional knowledge as members graduate, TFRA maintains a centralized digital archive of design reports, CAD files, test data, and post-mortem write-ups organized by sub-team and year. New members are directed to this archive during onboarding, and outgoing leads update it with a handover document summarizing open problems, design rationale not captured elsewhere, and lessons learned.

Leadership transitions are staggered rather than abrupt: incoming sub-team leads begin shadowing outgoing leads at least one semester ahead of a full handover, participating in active decisions so authority transfers gradually. TFRA also maintains contact with graduated alumni, many of whom remain available for informal consultation during critical design or testing phases..

1.4 Student Outreach Program

For this year, Turner Fenton Rocketry's outreach strategy heavily focused on creating various events through our NOCSA (National Organisation of Canadian Student Aerospace) initiative.

Through NOCSA, we have created a fully automated outreach sequence involving the collection of expert engineers, designers, and more, and bringing them onto speaker events or other workshops for participants. This initiative, while brand new, has generated much traction already. Our first event of 2025 involved a speaker from MDA Space's satellite division, who spoke about his career growth and what it is like to work as an engineer in the aerospace field. As our inaugural event, we had 45 people from multiple different high schools join this event, marking it as a major success in the growth of this organization.

For the year of 2026, we hosted an additional speaker event with a propulsion systems engineer from Nordspace. This event was another successful event which involved our longest questioning session of over 30 minutes, on top of his explanation of his personal journey in the aerospace field and what Nordspace has been up to. This event had 60 people in attendance.

Aside from NOCSA, TFRA itself has been invited to numerous events as guest speakers to share the journey we have been through from building a high school rocketry team from the ground up. For starters, we were invited to the Moonshot 2025 event as keynote speakers and presented to a room of over 300 students. We met many students who got in touch with us and later gained our assistance in starting their own rocketry clubs.

Additionally, we were invited to one of the largest board wide events that is annually hosted: the Peel District School Board's Youth in Action Day, Career & Destination Fair for all SHSM and EHS students a part of the Peel District School Board and the Peel-Dufferin School Board. At this event of 3000 students, we were a keynote speaker and booth holder where we spoke about our journey as a club and how we aim to bring others on board our journey by aiding in their own aerospace endeavours in their schools.

TFRA was also invited to Morningside Public School, an elementary school in Malton, to speak to a third grade class about the Artemis II Mission, and the future of space in general. Following that success, Dhyan Soni was invited to the CBC in downtown Toronto, and got the opportunity to speak on a 15 minute national radio segment about Artemis II, TFRA, and the future of aerospace in Canada from the perspective of an aspiring young engineer.

2 BUSINESS CASE ANALYSIS

2.1 Financial Goals

Our financial goals this year were to diversify our revenue streams and avoid reliance on one method of funding. During the 2024-25 school year, roughly 60% of our funding came from the Peel District School Board's Leading Education Innovation Project (LEIP) grants, where we were granted \$4,500 to develop our rocket Spectre for Launch Canada 2025. Despite this and an additional \$3,025 from executive member fees, we still were about \$70 above budget by the time the school year ended in June 2025.

Coming into September of the 2025-26 school year, our finance team forecasted that our large rocket aspirations means larger material expenses while less enthusiasm from the Board to fund it. This forecast made the conclusion of revenue diversification evident. It was considered good fortune that the club was less than \$100 over budget in 2025, but the deficit would widen significantly if we hoped for funding from the PDSB again. Thus, these financial goals were the driver to our new sponsor outreach program. We increased the size of our outreach team by 2 spots, merged them with our finance team for smoother communication between the two, and centralized oversight to one head director over both teams, resulting in the formation of the Finance-Outreach team.

To diversify our revenue streams, we sought potential contacts within Canada (specifically Southern Ontario) who may donate to us (monetary or material), and attempted to reach out to them. Each member within the Finance-Outreach team deployed numerous different methods of communication: calls, in-person conversation at aerospace events, but most importantly, cold-email campaigns. Our Finance team researched potential contacts while the Outreach team tailored and sent emails. We experimented throughout the year debating whether the winning metric was the quality or quantity of the emails. We experimented with different campaigns aimed at different people who may be related to rocketry, whether they are an aerospace engineer, machiner, manufacturer, or works with computer hardware technology. Overall, from December to June, we sent over 325 emails and heard back from 7 contacts, highlighting the difficulties of finding and maintaining a mutual partnership between these companies and us.

Additionally, we aimed to amplify our credibility and recognizability by growing our online presence. Since the 2025-26 school year began, we made 7 posts, including 2 reels and one collaboration post with our school newspaper, on our Instagram, @tfss.rocketry, to increase our notability among younger audiences. The concluding result were 634 followers and an average of 93 likes between our 2 reels. Our logic was that our popularity among the youth meant that our program was proof that we can bring young minds into rocketry, serving as leverage in sponsorship negotiations with various companies. It worked well in conjunction with our NOCSA outreach program, providing an audience for various keynote speakers and a talent pool for inquiring aerospace companies. Furthermore, we created our own website at <https://tfssrocketry.com/> to give potential sponsors a look into the inner workings of Turner Fenton Rocketry, and connect them personally to our team members.

2.2 Value Proposition

TFRA's most significant contribution to the Canadian aerospace sector is a demonstrated one: we are the first ever high school team to launch at Launch Canada. This achievement alone establishes a new precedent, proving that motivated high school students, given the right support structure, can execute a serious rocketry project alongside university and professional teams. In doing so, TFRA has effectively lowered the age barrier for entry into Canadian rocketry, opening a pathway to hands-on aerospace experience that previously did not exist for students at our level.

Since our launch, we have worked deliberately to convert this achievement into long-term value for the industry's talent pipeline. We have visited middle schools across our region to introduce younger students to rocketry and engineering, aiming to spark interest well before students reach the point of choosing a career path or post-secondary program. We were featured on CBC News, using that platform to raise public awareness of student rocketry in Canada and to show that this kind of work is accessible outside of traditional university settings. We have also worked closely with the Peel Regional School Board, demonstrating firsthand what motivated students can accomplish when given adequate support, a case study the board can now point to when advocating for further investment in STEM programming.

Taken together, these efforts represent an educational value proposition: TFRA is actively building the next generation of aerospace talent for Canada, starting earlier than most existing programs do. By normalizing rocketry as an achievable pursuit for high schoolers, we are helping to widen and diversify the funnel of students who eventually enter Canadian aerospace and engineering programs, a benefit that compounds over time as more teams and schools follow the pathway we've helped establish. In this way, TFRA's contribution is not only technical, but structural: we have shown NOCSA, Launch Canada, and the broader community that high school teams are a viable and valuable part of the Canadian rocketry ecosystem.

2.3 Sponsor Outreach Program

In discussion about revenue diversification, we concluded that the revenue stream that we had the most control over was internal, prompting us to set our Rocketry fee at \$46.20 and have Executive members pay that fee. This self-investment into our club was the initial spark that would help us move throughout the year. Our funds helped kickstart our sponsor outreach program. As mentioned in 2.1, we primarily ran cold-email campaigns on a per-month basis and evaluated any struggles or strengths that we noticed throughout the month. The email campaigns sent three automatic emails to sponsors over a span of 5 days. We tracked email deliverability (maintained >90 throughout the year), email opens and reply rates (roughly 5% including automatic and out of office replies), and we tweaked email length and content..

All our emails emphasised our hope to lead the path for other high school rocketry clubs in both the Peel region and Canada more broadly. We envisioned a network of high school rocketry teams that bridged the gap between high schools and the Canadian aerospace industry. We clarified that our ambitions to achieve such goals only came from upscaling our operations in all aspects, and explained our intentions to develop larger, more advanced rockets to compete for the grand prize at Launch Canada. Additionally, we showed to sponsors our hope to foster a future for more high school rocketry clubs to comfortably sprout up and compete along with us through an expansion of our online presence to reach more students through extensive marketing campaigns, as mentioned in 2.1.

Our sponsor outreach program connected us with various companies over the year. Examples include the City of Brampton's \$500 donation through their Innovation District program that fostered homegrown technological ingenuity in Brampton, and the Canadian aerospace company Nordspace also granted us an additional \$500 dollars to help us with our operations too. In regards to the Peel District School Board, we re-applied to the LEIP grant but only received \$1,500 from them this year, marginally less than the year before, as predicted, but nonetheless supportive.

2.4 Financial Breakdown

As aforementioned in 2.1, the club tried many different ways to diversify their revenue streams. We aimed not to complicate how we tracked our finances and only ran a fairly simple spreadsheet that calculated the money we had in each account and our expenses that flowed out. Our school required of us to note the date, invoice #, company and cost. We had three accounts in total: the main account, the Board's LEIP funds, and the leftover funds from last year. An important limitation to mention is that this does not account for non-monetary forms of income, such as the donation of hardware or software materials.

ACCOUNT 1: TFRA MAIN ACCOUNT

The TFRA main account is the funds that came from Turner Fenton members such as students, staff, principals and any other external donations. It was required of the students in our club to pay \$46.20 in order to maintain the costs of our club. This was our primary account.

TFRA			
Budget	\$ 3,866.01	Remaining	\$13.31
Date	Invoice #	Company	Amount
October 15, 2025	95268153	DigiKey	\$178.86
October 15, 2025	3483-EFPV	EpicFPV	\$116.59
November 21, 2025	HD-9921405	Home Depot	\$149.83
December 19, 2025	96104832	Digikey	\$448.81
January 5, 2026	702-4921-385	Amazon	\$67.45
February 4, 2026	702-8813-904	Amazon	\$62.76
February 20, 2026	WR-2026-042	Wildman Rocketry	1082.69 USD
February 20, 2026	GLR-8841	Giant Leap Rocketry	\$148.00
February 20, 2026	97334109	Digikey	\$54.96
February 27, 2026	(N/A - Credit)	\$1000 School Funds	
March 4, 2026	SSC-22941	Sculpture Supply Canada	\$35.87

March 4, 2026	98001542	Digikey	\$92.35
March 4, 2026	RB-773952	Robotshop	\$40.72
March 4, 2026	ME-440921	Mouser Electronics	\$29.81
April 17, 2026	WR-2026-042R	Wildman Rocketry Refund of \$100 USD	
June 23, 2026	LC-2026-0623	Launch Canada Fee	\$1,020.00
June 23, 2026	HD-4412093	Home Depot & Assorted Purchases	\$998.08
June 24, 2026	CT-91301	Canadian Tire	\$324.72
June 24, 2026	3DPC-4092	3D Printing Canada	\$83.89
Total			\$3,852.70

Table 1: TFRA Funding Chart

ACCOUNT 2: 2026 LEIP FUNDS FROM PDSB

The 2026 LEIP Funds account was the donations from the Peel District School Board through their Leading Education Innovation Project grant. As forecasted, we received a 67% decrease in grants compared to last year.

LEIP FUNDS			
Budget	\$ 1,500.00	Remaining	\$24.46
Date	Invoice #	Company	Amount
June 26, 2026	AR-2026-0626	AllRockets	\$1,475.54
Total			\$1,475.54

Table 2: LEIP Funds

ACCOUNT 3: LC2025 REMAINING BALANCE

Our third and final account was the Disposable Revenue account from the Peel District School Board's donations last summer to prepare us for Launch Canada 2025. We used the remaining balance to help fund our operations this year.

TFRA - LC2025 REMAINING BALANCE			
Date	Invoice #	Company	Amount
November 29, 2025	HD-9902731	Home Depot	\$17.63
November 29, 2025	CT-77871	Canadian Tire	\$6.76
November 30, 2025	HD-9903213	Home Depot	\$61.62
December 5, 2025	96182012	Digikey	\$219.76
January 14, 2026	AR-2026-0114	AllRockets - Denau Hobby Supplies	\$713.71
March 10, 2026	CC-72124	Composites Canada	\$431.16
April 9, 2026	HD-9982176	Home Depot	\$18.03
April 9, 2026	98172642	Digikey	\$464.53
May 21, 2026	CT-87871	Canadian Tire	\$170.56
May 21, 2026	MT-100293	Metal Supermarket	\$163.85
May 21, 2026	711-2989-383	Amazon	\$114.00
May 21, 2026	CC-78212	Composites Canada	\$318.25
June 2, 2026	RC-2026-0602	Rigging Canada	\$87.45
June 2, 2026	98819235	Digikey	\$380.49
June 2, 2026	HD-10138202	Home Depot	\$32.23
June 2, 2026	HD-10139404	Home Depot	\$218.95
June 11, 2026	CT-89071	Canadian Tire	\$246.79
June 26, 2026	99122153	Digikey	\$445.32
Total			\$4,111.09

Table 3: History of Purchases from Disposable Revenue

3 CONCEPT OF OPERATIONS (CONOPS)

The CONOPS for Goldeneye follows an eight phase operation: final rocket integration, rail departure, powered flight, unpowered flight, drogue deployment, main deployment, and recovery operations. The procedures for each phase are detailed in the following section.

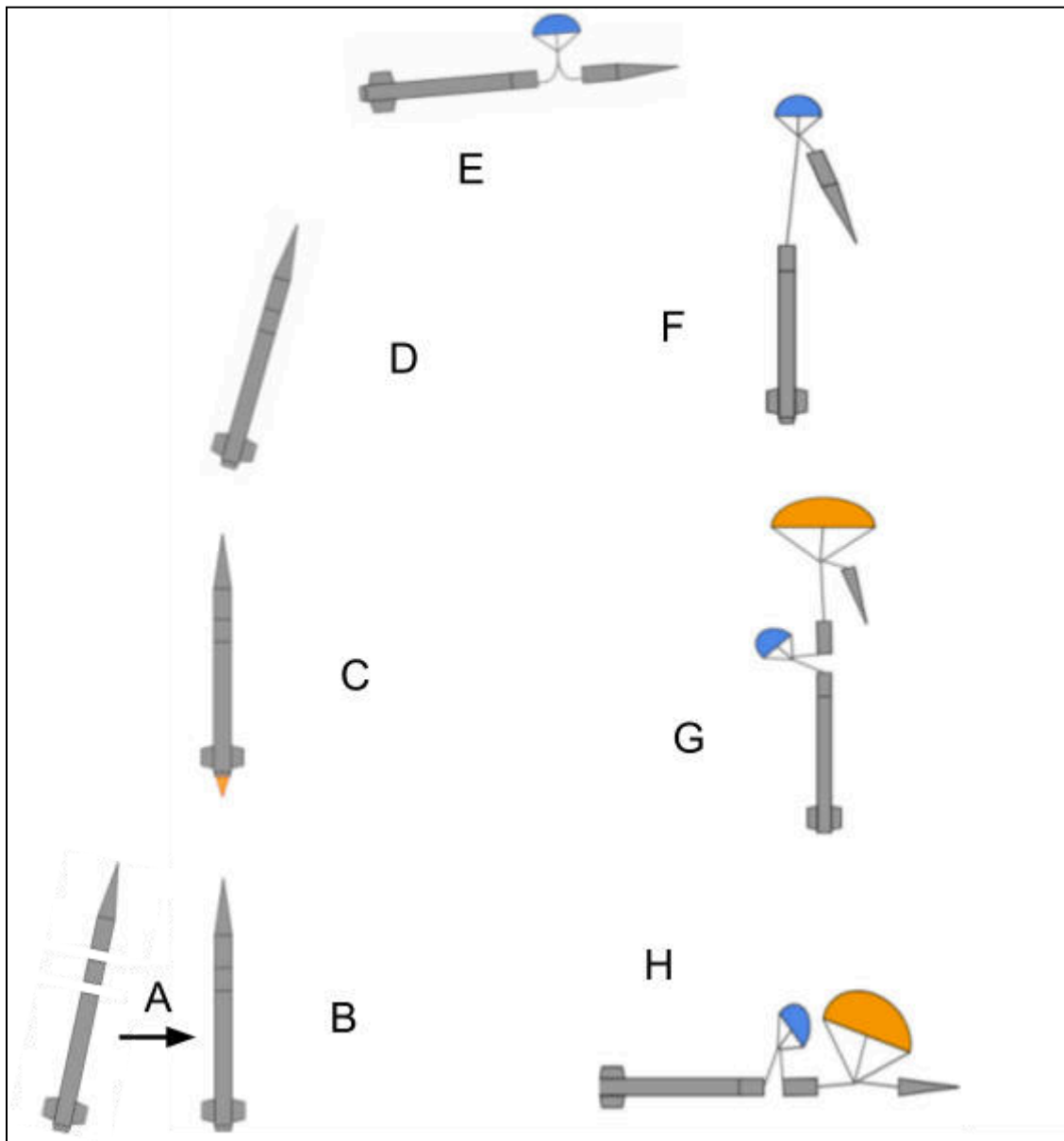


Figure 2: Visual Representation of Mission Operations.

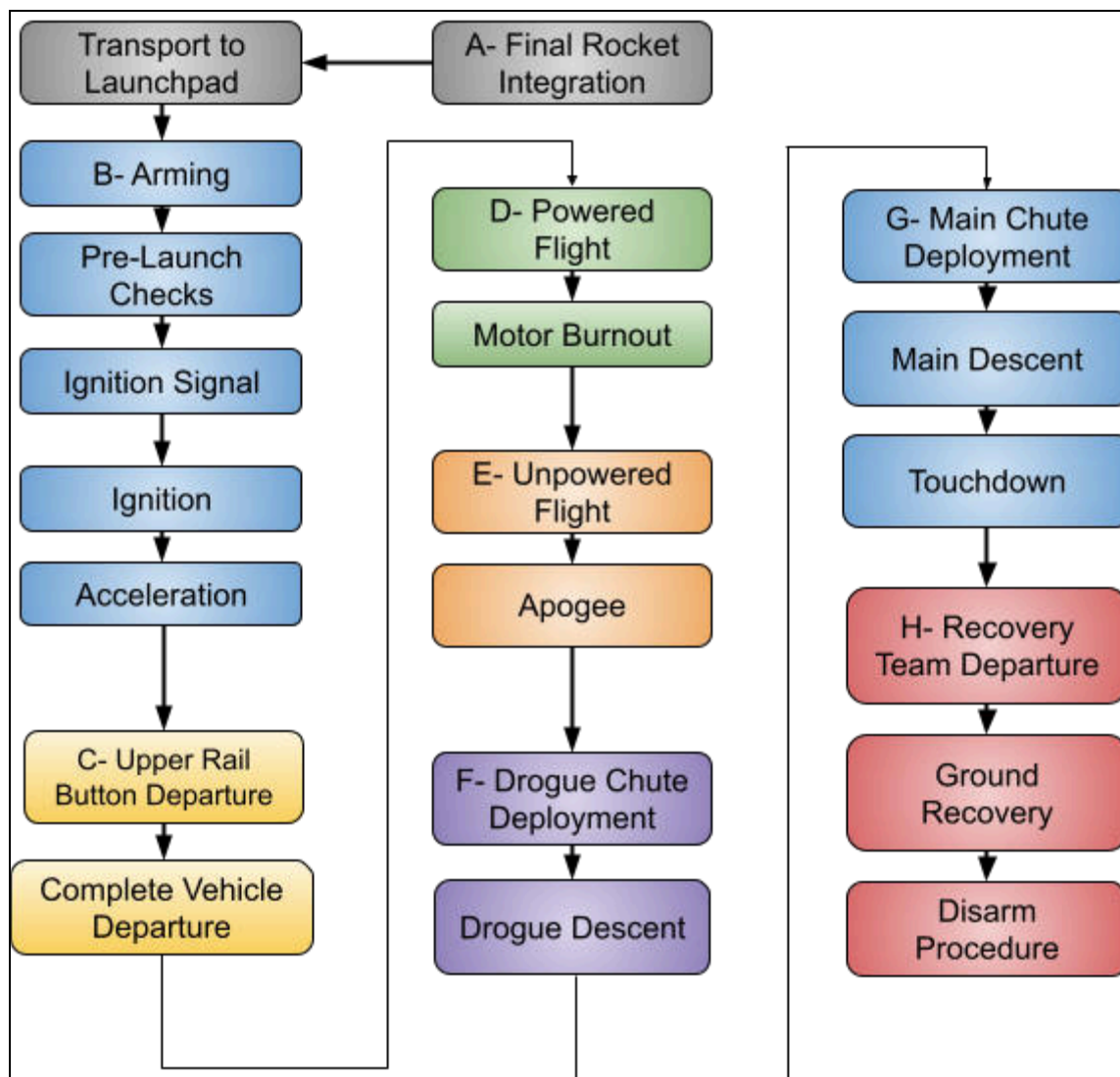


Figure 3: Mission Phase Flowchart.

A. Final Rocket Integration

Start: N/A

End: Rocket departs the assembly and integration area and is transported to the launchpad.

Final integration will be conducted in the main assembly area. The following is the final integration procedure:

- Secure the avbay panels with retention bolts.
- Thread the pitot tube into the nose cone and connect the airlines to the payload sled.
- Attach nose cone coupler to nose cone via retention bolts and ensure secure fit.
- Attach forward airframe to forward avbay coupler via retention bolts and ensure secure fit
- Attach the nose cone to the forward coupler and check to see if it can slide out freely with moderate force, then secure with shear pins.
- Attach the aft airframe to the aft avbay coupler and once again check to see if it can slide out. Secure with shear pins.
- Screw in the motor retainer.

Abort: Should an issue be encountered the rocket must be disassembled in the following sequence:

- Remove all shear pins and separate sliding points.
- Remove all retention bolts and disassembled fixed components.
- Disconnect pitot tube airlines.
- Conduct further disassembly if required for investigation.

B. Arming

Start: Avionics and payload components are powered on and armed.

End: Ignition signal is given.

Arm the electronics, following the pre-launch checklist.

Abort: Should a fault be detected, the arming switches will be turned to the “OFF” position. The vehicle will be disassembled, charges disconnected, and will be brought back to the assembly area for further investigation.

C. Rail Departure

Start: The rocket’s upper rail button leaves the launch rail.

End: The rocket’s lower rail button leaves the launch rail. The rocket fully departs the launch rail.

As the vehicle accelerates up the launch rail, the upper launch button will eventually clear the rail. This means that the rocket is free to pivot about the lower rail button. This puts the rocket in a position to weathercock into the wind, introducing a potential risk of destabilisation. However, as will be discussed in subsequent sections, the rocket is travelling fast enough, and with enough stability to prevent a weathercock into the wind.

D. Powered Flight

Start: The rocket has departed the launch rail.

End: The motor burns out.

After departing the launch rail, the rocket is in free flight. At the same time, the rocket motor is accelerating the rocket, and it is rapidly gaining velocity. The rocket will continue to gain velocity until the motor burns out approximately 1.9 seconds after ignition. At this point, the rocket is at its highest velocity, but will suddenly decelerate.

E. Unpowered Flight

Start: The rocket motor has burnt out.

End: The rocket reaches apogee.

The rocket will continue travelling upwards, whilst being decelerated by the force of aerodynamic drag and gravity. This eventually results in the rocket velocity decreasing to 0, and the rocket reaching apogee.

F. Drogue Deployment

Start: The drogue parachute deploys when the rocket reaches apogee.

End: Main parachute deployment.

At apogee (~12,590 ft), the primary flight computer (Blue Raven), will deploy the drogue parachute at apogee. Using a “+1 second” timer, the secondary flight computer (RRC3+), will deploy its black powder charge, in case the primary charge fails. The rocket will continue descending using the drogue parachute at approximately 27 m/s. This phase of flight continues until the rocket reaches 1500 ft AGL, at which point, the main parachute will deploy.

G. Main Deployment

Start: The main parachute deploys.

End: Touchdown.

At 1500 ft AGL, the primary flight computer, will fire its ejection charge which will deploy the main parachute. Again, with a “+1 second” timer, the secondary flight computer will deploy its own black powder charge, as a backup. According to the OpenRocket simulation, the rocket will descend at a velocity around 5 m/s. This phase of flight concludes when the rocket touches down.

H. Recovery Operations

Start: Confirmation from Launch Canada recovery personnel.

End: Disarm procedure.

After confirmation of touchdown from the telemetry systems, the team will refer to authorized LC personnel and staff for further direction. When the permission is acquired, the team will venture into the launch zone with reference to the coordinates from the GPS telemetry. Upon locating the rocket, in the event that the black powder charges did not fire nominally, they will be emptied and neutralised immediately. The rocket will be transported to the assembly area where the avbay and nosecone will be dissembled and all electronics powered off.

4 REQUIREMENTS

This section condenses some of the **key** criteria from the Launch Canada DTEG, as well as TFRA's internal vehicle requirements. Please note that in the interest of brevity, not all criteria have been listed.

4.1 Mission Requirements

4.1.1 Top-Level Mission Requirements

- **MISS-01 (Performance):** The launch vehicle shall depart the launch rail safely with a launch rod/rail departure velocity sufficient to ensure aerodynamic stability (minimum 30 ft/s) under maximum wind conditions.
- **MISS-02 (Performance):** The vehicle shall reach a maximum velocity exceeding Mach 1.2 and maintain supersonic flight for a minimum duration of 0.5 seconds during the boost phase. *(Source: TFRA.R.03)*
- **MISS-03 (Functional):** The vehicle shall execute a dual-event recovery sequence, deploying a drogue parachute at or near apogee and a main parachute no higher than 1,500 ft AGL. *(Source: LC-R4.1.1 / LC-R4.3.1)*
- **MISS-04 (Functional):** Upon touchdown, the vehicle shall retain complete structural integrity such that access to internal avionics, recorded data, and the experimental payload bay remains unobstructed and undamaged. *(Source: LC-DTEG)*

4.1.2 Subsystem Requirements Mission Requirements

Aerostructures & Airframe System

- **SYS-AERO-01:** The integrated airframe shall utilize a standardized 4-inch (102 mm) outer diameter across all primary body tubes and payload sections. *(Source: TFRA.R.01)*
- **SYS-AERO-02:** The vehicle shall be statically stable (1.5 body calibers minimum) at rail exit and throughout the entire powered flight envelope *(Source: LC-R7.2.1)*
- **SYS-AERO-03:** The vehicle shall incorporate external, reinforced hard-points for two rail guides capable of supporting 100% of the fully fueled vehicle launch weight while vertical *(Source: LC-R8.8.10 / LC-R8.8.11)*

Propulsion System (Solid Motor)

- **SYS-PROP-01:** The vehicle shall utilize a commercially available (COTS) solid rocket motor utilizing non-toxic propellants, with a total impulse not exceeding 40,960 Ns (Class O or lower) for Basic Category eligibility. *(Source: LC-R2.1.1 / LC-R2.1.3)*
- **SYS-PROP-02:** The propulsion system shall not utilize propellants containing large-particle metals designed to leave the combustion chamber unreacted to generate spark showers. *(Source: LC-R2.1.4)*

Recovery System

- **SYS-REC-01:** The recovery system shall achieve a descent rate under drogue parachute of approximately 100 ft/s (30.5 m/s) and shall not exceed 120 ft/s (36.6 m/s) or fall below 50 ft/s (15.2 m/s). *(Source: TFRA.R.04 / LC-R4.2.2)*
- **SYS-REC-02:** The recovery system shall achieve a terminal descent rate under main parachute of approximately 20 ft/s (6.1 m/s) and shall not exceed 30 ft/s (9.1 m/s). *(Source: TFRA.R.05 / LC-R4.3.1)*
- **SYS-REC-03:** Deployment energetics (black powder ejection charges) shall be controlled by completely independent, redundant electrical architectures (primary and backup) that share no common components or power supplies. *(Source: LC-R4.8.1 / LC-R4.8.2)*
- **SYS-REC-04:** All energetics arming features shall be externally accessible and configurable only from a safe remote distance or via two-step mechanical/electronic safing protocols. *(Source: LC-R5.2.1 – LC-R5.2.4)*

Avionics, Telemetry & Active Flight Controls

- **SYS-AV-01:** The primary avionics bay shall host at least one flight-proven COTS altimeter/flight computer dedicated to recovery deployment, paired with at least one redundant secondary system (COTS). *(Source: LC-R4.8.3)*
- **SYS-AV-02:** The vehicle shall continuously transmit real-time GPS coordinates, barometric altitude, and flight telemetry to the ground station at a minimum transmission rate of 0.5 Hz (every 2 seconds). *(Source: LC-R6.4.1 / LC-R6.10.2)*

4.2 Subsystem Requirements

4.2.1 Nosecone & Forward Payload Section

- **SUB-NC-01 (Performance):** The nosecone profile shall utilize an optimized von Kármán profile to minimize wave drag and aerodynamic heating across transonic and supersonic flight regimes
- **SUB-NC-02 (Functional):** The forward nosecone section shall integrate a metallic tip capable of housing the dynamic pressure inlet for the experimental pitot-tube payload without deforming under aerodynamic stagnation temperatures

4.2.2 Airframe Tubes, Couplers & Joints

- **SUB-AF-01(Mechanical):** All internal sliding couplers and permanent airframe joints shall extend into the mating body tube by a minimum of one (1.0) body caliber (4 inches / 102 mm) on either side of the joint interface to prevent bending under aerodynamic flight loads. *(Source: LC-R8.7.5 / LC-R8.7.7)*
- **SUB-AF-25 (Environmental):** All sealed airframe sections shall incorporate static vent holes sized to prevent internal over-pressurization during rapid ascent. *(Source: LC-R8.8.7)*

4.2.3 Fin Assembly & Aerodynamics

- **SUB-FIN-01 (Structural):** The fin-to-airframe root joints shall be assembled using a precision alignment jig and reinforced with internal root tabs extending through the airframe to center-tube or motor-mount centering rings.
- **SUB-FIN-02 (Performance):** Fin thickness, span, and composite layup schedule shall be verified via aeroelastic flutter analysis to ensure the divergence and flutter velocities exceed the maximum predicted flight velocity (*Source: LC-R8.7.1*)

4.2.4 Motor Retention & Aft Hardware

- **SUB-MOT-01 (Mechanical):** The solid rocket motor case shall be positively retained within the airframe utilizing a mechanical aft-retention system (*Source: LC-R8.7.2*)
- **SUB-MOT-02 (Structural):** The motor mounting assembly shall transfer 100% of thrust loads directly through structural centering rings into the primary body tube skin, bypassing non-structural internal components. (*Source: LC-R8.7.3*)
- **SUB-MOT-03 (Thermal):** All centering rings, retention hardware, and adhesives adjacent to the motor tube shall be rated to withstand continuous operational temperatures exceeding 150°C without mechanical degradation. (*Source: LC-R8.7.4*)
- **SUB-MOT-04 (Mechanical):** External launch rail buttons shall be mechanically fastened to reinforced internal blocks independent of the motor tube, ensuring mounting screws do not impinge upon or transfer shear loads to the COTS motor casing.

4.2.5 Recovery Hardware & Parachute Subsystem

- **SUB-REC-01 (Hardware):** All load-bearing attachment points (eyebolts, U-bolts) shall be closed-loop forged or machined steel/aluminum rated for the peak shock load; open-wire or bent-eye bolts are strictly prohibited. (*Source: LC-R8.8.1*)
- **SUB-REC-02 (Hardware):** All shock cords and harness connections shall incorporate heavy-duty ball-bearing swivel links to prevent harness twisting and unthreading during descent spinning. (*Source: LC-R4.5.1*)
- **SUB-REC-03 (Thermal Protection):** Recovery charge wells shall be capped with Nomex blankets, baffles, or vulcanized silicone blankets to shield nylon canopies and Kevlar harnesses from direct exposure to hot black powder combustion gases. (*Source: LC-R4.4.1*)

5 SYSTEMS ARCHITECTURE

Goldeneye is a single stage rocket composed of 5 active subsystems: aerostructures, recovery, propulsion, avionics, and payload technologies.

Goldeneye is a 4 inch diameter rocket. Airframes are constructed with COTS carbon fibre. The nose cone is COTS fibreglass and is of the Von Karman profile. The motor is mounted in a COTS carbon fibre motor mount tube. Additionally, four 1/4-inch fibreglass fins will be integrated with through-the-wall mounting. An SRAD 6061 aluminum boat tail is integrated at the aft end of the rocket for drag reduction and motor retention.

Goldeneye uses a dual separation, dual deploy recovery system. The drogue parachute deploys at apogee which is approximately 12,590 ft. The main parachute deploys at 1,500 ft. Both drogue and main parachutes will be ejected with black powder charges. Nylon shear pins hold together the airframe at the separation points which will break upon ejection.

Goldeneye uses a CTI Pro75 4864L2375-P rocket motor. The rocket will reach a maximum velocity of 429 m/s (Mach 1.25) and apogee at 12,502 ft. The average stability of flight is 2.40 cal, and the stability off-the-rail is 1.84 cal. The vehicle has a thrust to weight ratio of 21.71:1.

The avionics system consists of 2 COTS flight computers, 2 COTS GPS trackers, and a passive SRAD electronics suite. The primary flight computer is the Blue Raven. It offers robust sensors, intuitive software, and reliable performance while remaining simple and easy to use. The secondary flight computer is the RRC3+ which is a simple, but reliable flight computer.

The payload includes the pitot tube constructed with 6061 aluminium and carbon fiber tubing. The pitot tube measures the vehicle's airspeed.

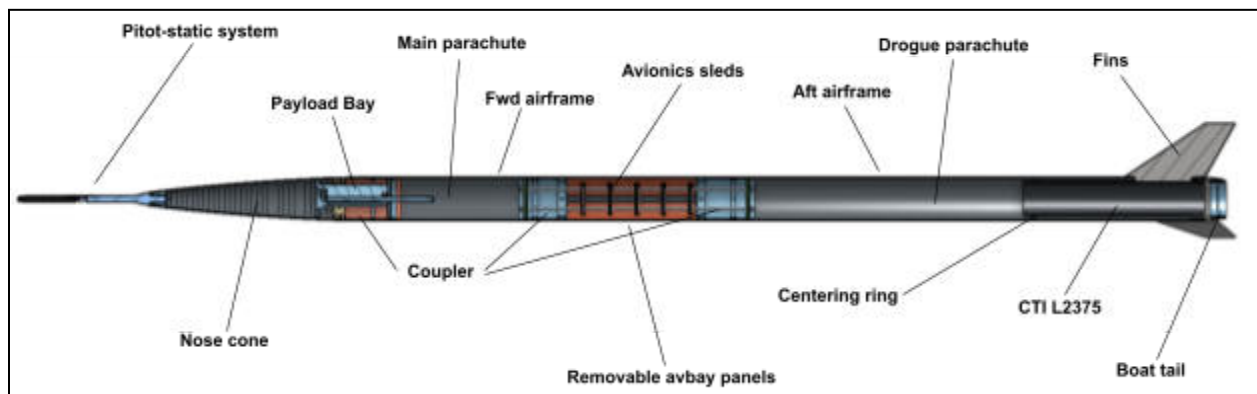


Figure 4: Cross Section of Rocket with Major Components Labelled.

Rocket Name	Goldeneye
Configuration Type	Single Stage
Propulsion Type	COTS
Total Impulse	4864 Ns
Average Thrust	2375 N
Stability Calibers	2.73 cal
Rocket Length	98.85 in
Dry Mass	8934 g
Wet Mass	11256 g

Table 4: General Rocket Configuration and Geometry

5.1 Aerostructures Summary

Component	Qty	Material	COTS/SRAD
Pitot tube fwd section	1	6061-T6 Aluminium	SRAD
Pitot tube mid section	1	Carbon fibre	COTS
Pitot tube aft section	1	6061-T6 Aluminium	SRAD
Threaded insert	1	6061-T6 Aluminium	SRAD
Nose cone	1	Fibreglass	COTS
Nose cone coupler	1	Fibreglass	COTS
Retaining ring	1	Maple wood	SRAD
Payload sled	1	PLA	SRAD
Shear pin ring	2	PLA	SRAD
Nose cone bulkhead	1	Fibreglass	SRAD
Locking ring	1	Fibreglass	COTS
Forward airframe	1	Fibreglass	COTS
Avbay bulkhead	3	Fibreglass	SRAD

Component	Qty	Material	COTS/SRAD
Avbay coupler	2	Fibreglass	COTS
Avbay switchband	2	Fibreglass	COTS
Avbay bracket	2	Fibreglass	COTS
Avbay panel	2	Fibreglass	COTS
Avbay sled	6	PLA	SRAD
Centering ring	1	Fibreglass	SRAD
Motor mount tube	1	Carbon fibre	COTS
Rail button	2	Delrin	COTS
Fin	3	Fibreglass	SRAD
Boat tail	1	6061-T6 Aluminium	SRAD
Motor retainer	1	6061-T6 Aluminium	COTS

Table 5: Summary of Aerostructures Components

5.2 Mass Budget

Each component in the following table has been classified according to the following estimate types:

- E = Estimate
- M1 = Physical mass measured on balance
- M2 = Mass from manufacturer
- M3 = Mass calculated by OnShape

Component	Estimate Type	Qty	Unit Mass (g)	Total Mass (g)	Relative Mass (%)
Aerostructures			Subtotal	4082	36.27%
Nose cone	M1	1	384	384	3.41%
Nose cone coupler	M1	1	213	213	1.89%
Nose cone bulkhead	M1	1	88	88	0.78%
Forward airframe	M1	1	302	302	2.68%
Avbay bulkhead	M1	3	87	261	2.32%
Avbay forward coupler	M3	1	119	119	1.06%
Avbay aft coupler	M3	1	140	140	1.24%

Avbay switchband	M3	2	11	22	0.20%
Avbay panel	M3	2	137	274	2.43%
Avbay bracket	M3	2	59	118	1.05%
Aft airframe	M1	1	928	928	8.24%
Centering ring	M1	1	30	30	0.27%
Motor mount tube	M1	1	233	233	2.07%
Fin	M3	4	150	600	5.33%
Rail button	M1	2	10	20	0.18%
Boat tail	M3	1	279	279	2.48%
Slim line motor retainer	M1	1	71	71	0.63%
Avionics			Subtotal	288	2.56%
Avbay sled	M1	6	20	120	1.07%
Blue Raven	M2	1	6.8	7	0.06%
RRC3+	M2	1	14	14	0.12%
Featherweight GPS	M2	2	17	34	0.30%
DAQ PCB	E	1	30	30	0.27%
Pitot tube PCB	E	1	30	30	0.27%
Payload bay sled	M1	1	53	53	0.47%
Recovery			Subtotal	1248	11.09%
Drogue parachute	M2	1	76	76	0.68%
Main parachute	M2	1	357	357	3.17%
Main chute protector	E	1	50	50	0.44%
Drogue chute protector	E	1	25	25	0.22%
Shock chord	M2	2	338	676	6.01%
Quicklinks	M1	2	32	64	0.57%
Payload			Subtotal	249	2.21%
Pitot tube forward section	M3	1	55	55	0.49%
Pitot tube mid section	M3	1	6	6	0.05%
Pitot tube aft section	M3	1	112	112	1.00%
Threaded insert	M3	1	66	66	0.59%
Pitot tube hardware	E	1	10	10	0.09%
Propulsion			Subtotal	4161	36.97%

L2375	M2	1	4161	4161	36.97%
Miscellaneous			Subtotal	1228	10.91%
Nose cone bulkhead hardware	M1	1	176	176	1.56%
Retaining ring	M1	4	54	216	1.92%
Shear pin ring	M1	2	10	20	0.18%
Avbay bulkhead hardware	M1	2	91	182	1.62%
Avbay allthread	M1	4	96	384	3.41%
Fin fillets	E	1	250	250	2.22%
TOTAL				11256	100.00%

Table 6: Vehicle Mass Budget

6 PERFORMANCE ANALYSIS

6.1 Flight/Demo Performance

The flight performance of the vehicle was simulated using OpenRocket and RASAero II. The following section will provide a comparison of the analyses performed by both softwares. RasAero and OpenRocket simulations were performed in accordance with launch site conditions.

6.1.1 Simulation Parameters

Software	OpenRocket
Windspeed	3 m/s
Wind Direction	90°
Latitude	48.4758°N
Longitude	81.3°E
Altitude	967 ft
Launch Rod Length	18.5 ft
Launch Angle	0°

Table 7: OpenRocket Simulation Parameters

Software	RasAero II
Windspeed	3 m/s
Altitude	967 ft
Launch Rod Length	18.5 ft
Launch Angle	0°

Table 8: RasAero II Simulation Parameters

6.1.2 Flight Analysis

Software	OpenRocket
Max Altitude	12406 ft
Max Velocity	421 m/s
Max Mach Number	1.247
Max Acceleration	232 m/s ²
Average Stability	2.73 cal
Stability OTR	1.90 cal
CG	65.54 in
CP	76.50 in
Velocity OTR	49.28 m/s
Velocity at Drogue Deploy	2.46 m/s
Velocity at Main Deploy	23.77 m/s
Thrust to Weight Ratio	21.15:1

Table 9: OpenRocket Performance Analysis Summary

Software	RasAero II
Max Altitude	15570 ft
Max Velocity	434 m/s
Max Mach Number	1.265
Max Acceleration	237 m/s ²
Average Stability	3.98 cal
Stability OTR	2.66 cal
CG	65.54 in
CP	76.23 in

Table 10: RasAero II Performance Analysis Summary

The analyses revealed discrepancies between the two softwares. Specifically, the predicted maximum altitude differed by 25.5%, and the calculated average stability and off-the-rail stability differed by 46% and 40%, respectively. Interestingly, all other common flight metrics presented above do not vary more than about 3%.

The discrepancies observed between OpenRocket and RasAero II are primarily due to differences in their aerodynamic modelling approaches, particularly in the transonic and supersonic flight regimes. RasAero II incorporates more advanced modelling of transonic drag rise, wave drag, base drag, and boat-tail effects, whereas OpenRocket relies on simplified aerodynamic corrections derived from the Barrowman method. Since the Goldeneye reaches roughly Mach 1.25 and incorporates a boat tail, RasAero II is expected to provide a more accurate representation of the vehicle's aerodynamic behaviour at supersonic speeds. As such, the team believes the higher altitude and stability predictions from RasAero II are considered more reasonable. RasAero II was therefore selected as the primary source for flight analysis while OpenRocket remained an additional source to validate the vehicle's flight characteristics and was more convenient for locating metrics during specific flight events (such as rod clearance and recovery device deployment) which occurred in the subsonic regime. Additionally, OpenRocket was used to calculate the vehicle's CG.

Upon recovery of the flight data, the team plans to compare the results with the simulations from both softwares. This will allow the team to evaluate which software was more accurate in predicting supersonic flight performance.

Primary Analysis Software	<u>RasAero II</u> ■ Improved accuracy in transonic /supersonic regimes ■ Improved accuracy in modelling boat tail effects
Secondary Analysis Software	<u>OpenRocket</u> ■ Used in initial design phase ■ Calculating CG ■ Locating specific flight events

Table 11: Primary and Secondary Software Analysis

6.1.3 Graphical Analysis

The following stability vs. time graphs confirm that the vehicle maintains stability greater than 1.5 cal throughout powered and unpowered flight.

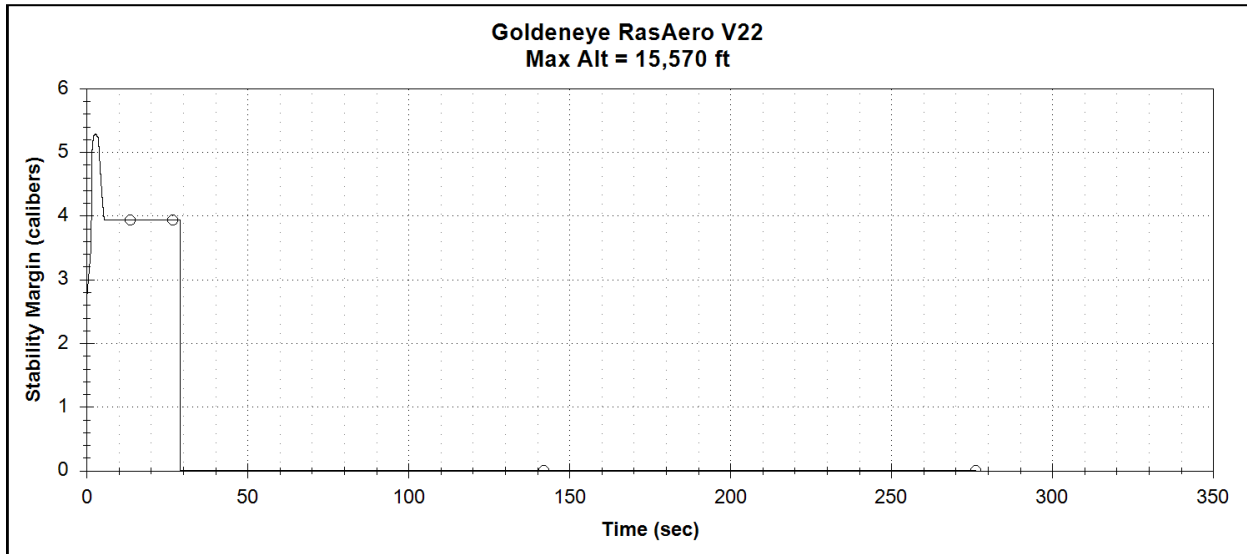


Figure 5: RasAero II Stability vs. Time Graph

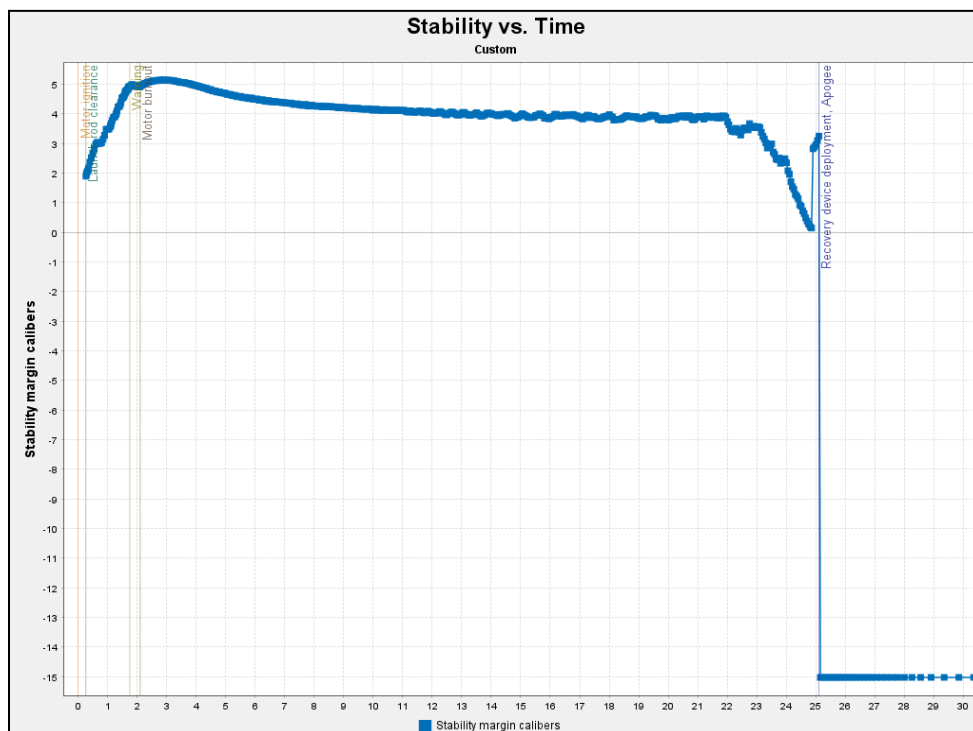


Figure 6: OpenRocket Stability vs. Time Graph

6.2 Recovery Dispersion Analysis

To evaluate the recovery footprint of the Goldeneye, a high-fidelity randomized simulation was conducted. Instead of relying solely on a nominal 3D trajectory, a 600-iteration Monte Carlo analysis was executed to model real-world variations in manufacturing, launch operations, and weather patterns.

The analysis was performed using RocketPy.

The flight vehicle was subjected to normal distributions across three primary risk categories: structural tolerances, launch rail alignment, and intense weather patterns. The standard deviations (σ) were chosen somewhat arbitrarily. Without every single subsystem being built to completion, it became difficult to estimate these standard deviations. The standard deviations for wind components were determined using historical wind data for the Matagami / Timmins area during mid-August, in the late morning and early afternoon.

Parameter	Baseline Mean (μ)	Standard Deviation (σ)	Standard Deviation Justification
Vehicle Mass	11.256 kg	± 0.250 kg	This magnitude accounts for discrepancies from hardware masses and paint.
Center of Gravity Position	1.65 m (from nose tip)	± 0.020 m (2 cm)	This accounts for shifting center of gravity from objects (parachute, recovery hardware, etc.) shifting during flight under acceleration.
Rail Inclination	84.0°	$\pm 2.0^\circ$	Models mechanical play in the launch rail.
Rail Heading	0.0°	$\pm 2.0^\circ$	Models mechanical play in the launch rail.
Easterly Wind Component	7.0 m/s (~25 km/h)	± 3.5 m/s	Evaluates vehicle response in high to severe wind conditions.
Northerly Wind Component	4.0 m/s (~14 km/h)	± 3.5 m/s	Evaluates vehicle response in high to severe wind conditions.

Table 12: Parameters of the Monte Carlo analysis

For each flight profile, the simulator solved the rigid-body equations of motion across two critical phases:

1. **Boost Phase & Unpowered Coast (Power-On/Power-Off):** The rocket accelerates under the thrust profile of the Cesaroni Pro75 L2375 motor. Structural mass variations alter the thrust-to-weight ratio, while the randomized launch rail angles directly dictate the initial velocity vector at rail exit. The interaction between the intense, randomized wind gusts and the vehicle's aerodynamic center introduces instantaneous angles of attack, capturing realistic wind-cocking behavior.
2. **Parachute Descent:** Upon reaching apogee, the dual-recovery sequence triggers parachute deployment. Because the rocket transitions to a high-drag drift state, the randomized wind fields (Easterly, Northerly Wind Components) completely dominate this phase, transporting the vehicle downwind until ground touchdown.

The output coordinates of all successful ground impacts were compiled to construct a statistically sound landing footprint mapped around the launch origin (0,0).

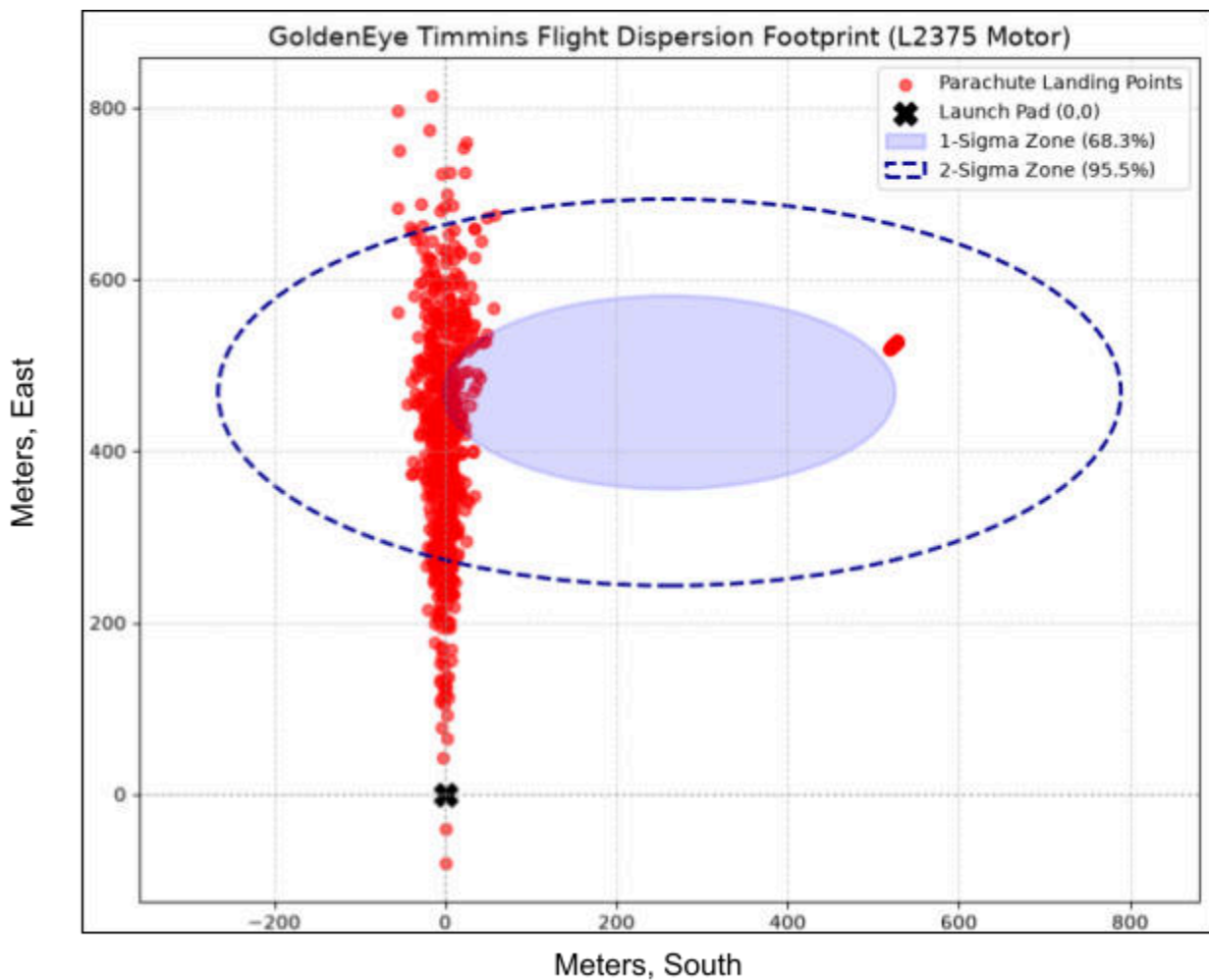


Figure 7: RocketPy Dispersion Estimate

- **1-Sigma Zone (68.3% Confidence Interval):** The inner solid blue ellipse defines the core landing zone where the bulk of nominal flights cluster.
- **2-Sigma Zone (95.5% Confidence Interval):** The outer dashed blue ellipse represents our primary safety buffer. This boundary accounts for severe compounded anomalies (e.g., a heavy rocket launching off a significantly tilted rail directly into a maximum-velocity wind gust).

In order to compare this data, the OpenRocket estimation was overlaid with the RocketPy estimation. Both approaches appear to be in agreement with each other, as the OpenRocket simulation's estimated landing position coincides closely with the estimated landing position from the RocketPy script.

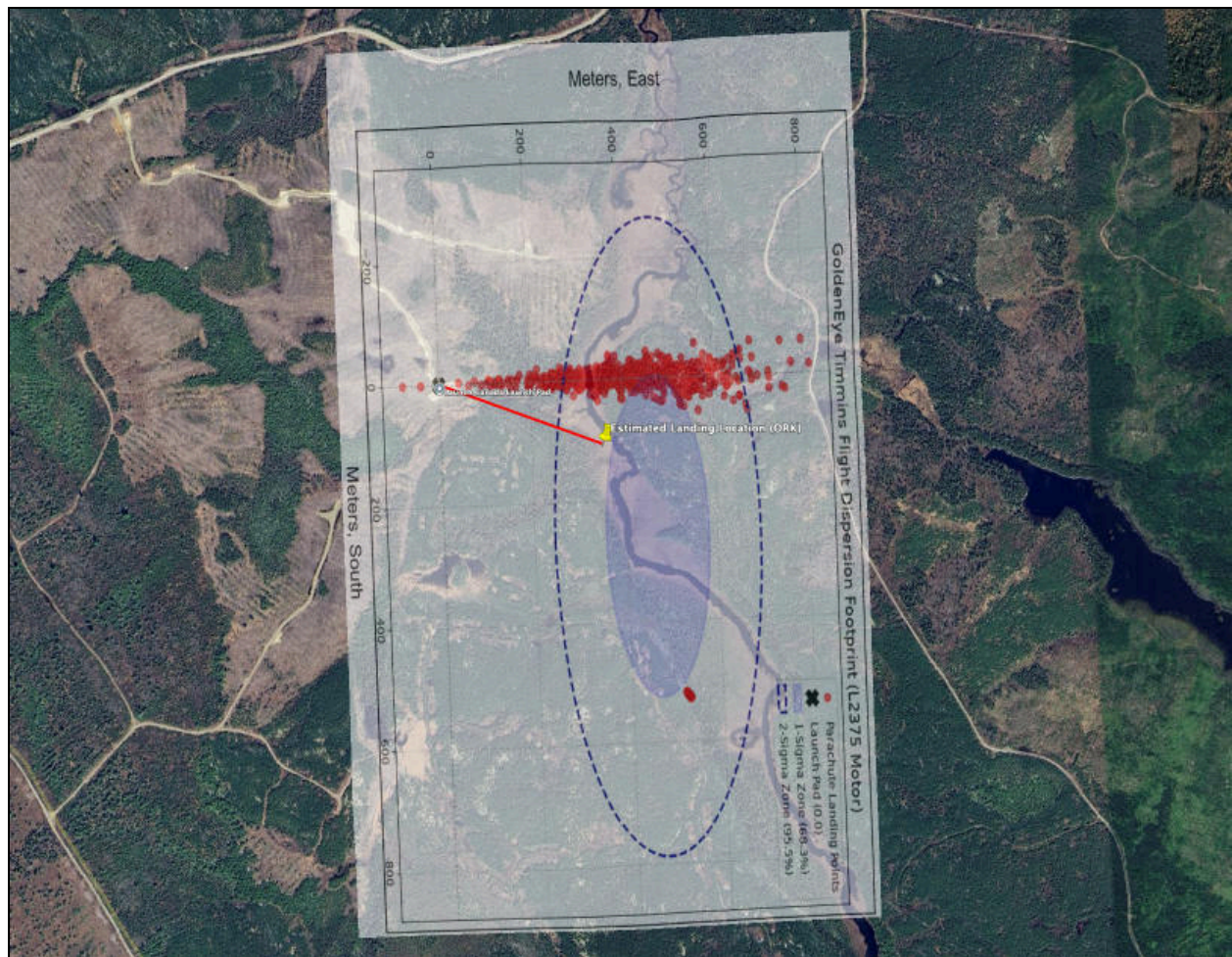


Figure 8: RocketPy / OpenRocket Dispersion Estimate

7 NOSE CONE

7.1 Detailed Design

The nose cone is COTS 23 inch fiberglass with a Von Karman profile. It was selected for its availability, RF transparency, and modularity with previous vehicles. This section houses the payload, including the pitot tube, which extends outward through the tip, and its associated PCB, which is mounted on an internal sled. The nose cone shoulder functions as a sliding-point coupler, and has been designed to allow easy access to these internal components.

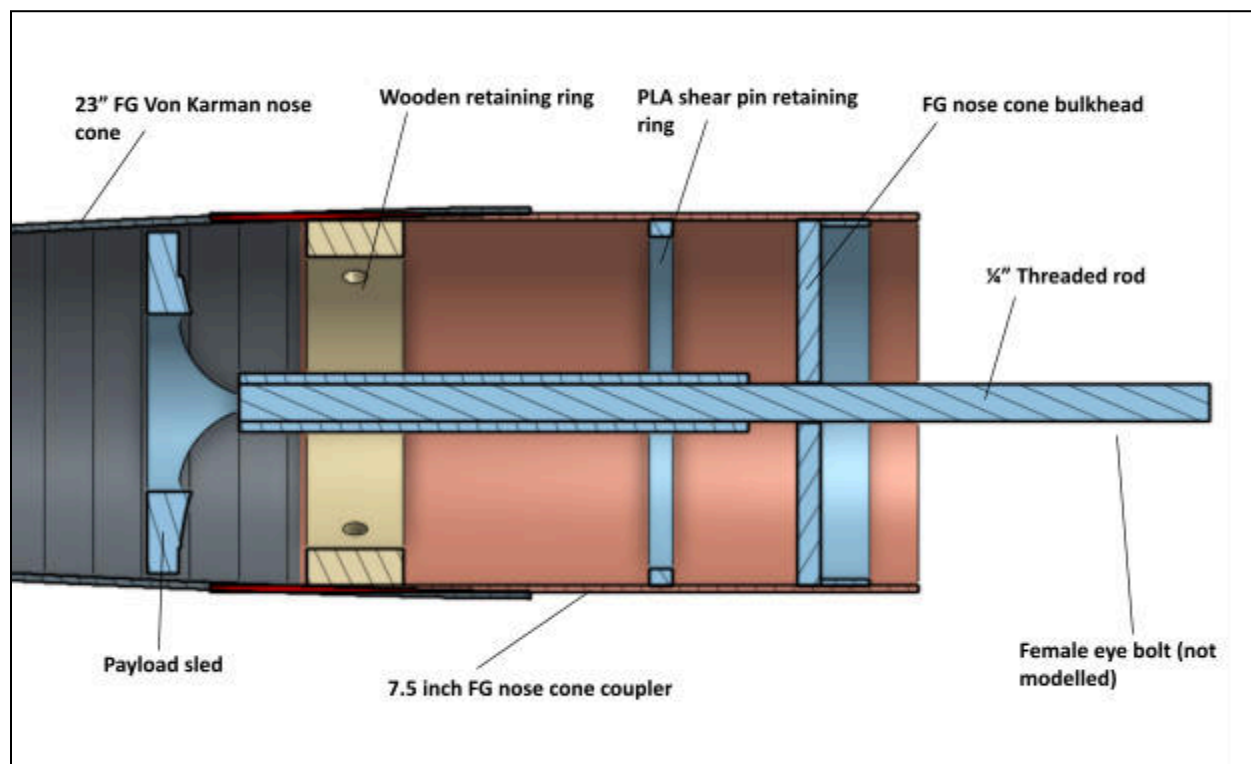


Figure 9: Nose Cone Assembly Diagram.

Eight (8x) 3mm retaining bolts secure the coupler to the nose cone, anchoring into a hardwood backing ring. Brass threaded inserts are epoxied directly into this wooden ring, providing robust anchor points for the bolts. This assembly creates a highly secure attachment between the shoulder and the nose cone that is fully capable of withstanding anticipated bending and recovery loads. Furthermore, the system remains removable, ensuring that internal components like the pitot tube PCB are easily accessible. While previous TFRA projects utilized a similar bolted retention system relying on self-clinching nuts, this year's iteration transitions to the wooden retention ring with brass inserts. This updated design was chosen for its increased reliability and stronger pull-out resistance and thread engagement.

The aft end of the coupler serves as the vehicle's forward separation point. Here, the coupler is secured to the forward airframe using three shear pins. To optimize retention and prevent flight vibrations from dislodging the pins (which could trigger premature separation), the shear pins are reinforced by a custom 3D-printed PLA backing ring.

Finally, a $\frac{1}{4}$ inch fiberglass bulkhead is integrated into the bottom of the coupler. It is permanently fixed in place using high-strength epoxy fillets and a thin fiberglass locking ring. This reinforced construction renders the bulkhead exceptionally resilient to recovery loads, ensuring structural integrity even in the event of a premature ejection.

7.2 Manufacturing Process

Although the nose cone was not manufactured in-house, the overall requirement for the nose cone volume to be usable, and accessible, made the entire subsystem more complicated. The manufacturing process for unique components, as well as overall system integration is detailed below.

7.2.1 Nose Cone Bulkhead

As mentioned prior, the nose cone bulkhead is manufactured from a $\frac{1}{4}$ " G10 Fiberglass plate. First, the bulkhead is cut using a 4" Milwaukee Hole Dozer. The diameter of the off-cut from this hole saw closely matches the internal diameter of the coupler section. Thereafter, using a belt-sander, the diameter is further reduced until a friction-fit is established between the inner diameter of the coupler, and outer diameter of the bulkhead.

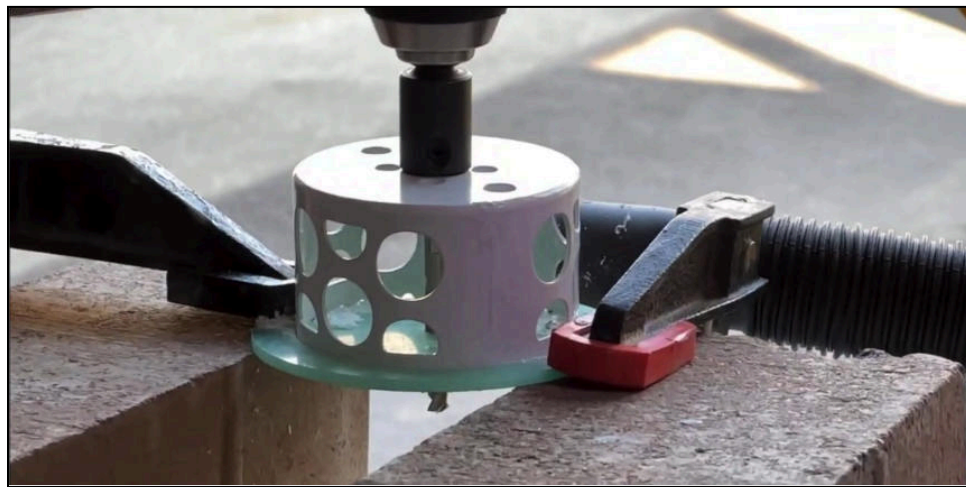


Figure 10: Nose Cone Bulkhead Manufacturing Process

7.2.2 Wooden Retaining Ring

The wooden retaining ring provides a backing plate into which the retaining bolts are threaded into. Therefore, the retaining ring is a critical component. As such, the decision was made to manufacture the retaining ring from 1" thick maple plank. Sequentially using a drill press and 3" and 4" hole saws, a toroidal ring was created. This component was then sanded until the outer diameter matched the inner diameter of the coupler (~ 3.755 ").



Figure 11: Machining the Retaining Ring on the Drill Press

Following this step, eight radial holes of diameter $13/64$ " were drilled using a 3D-printed jig, and a standard drill. This created a bore large enough to insert metric threaded inserts. Traditionally, these brass inserts are heat-set in 3D-printed components. However, with a thin layer of West System Epoxy (105 Resin, 206 Hardener), as well as an appropriately-sized bore, this insert is correctly-sized and incredibly strong. These epoxy-coated inserts are press-fitted into the retaining ring using a vise as a press. Finally, using a belt sander, the outer edge of this component is sanded until it fits smoothly into the coupler.



Figure 12: Completed Retaining Ring

The most challenging stage of the assembly process involves using a 90° Dremel attachment to bore holes into the coupler, and the nose cone from the retaining ring. This process requires precise placement of holes to ensure alignment through all three components, while ensuring no damage to the threads on the brass inserts. Eventually, the holes align, and eight (x8) 3mm bolts are threaded from the nose cone, into the coupler, and into the brass inserts.

The ¼" bulkhead, the 3D-printed shear pin ring, and the wooden ring are then epoxied into the coupler

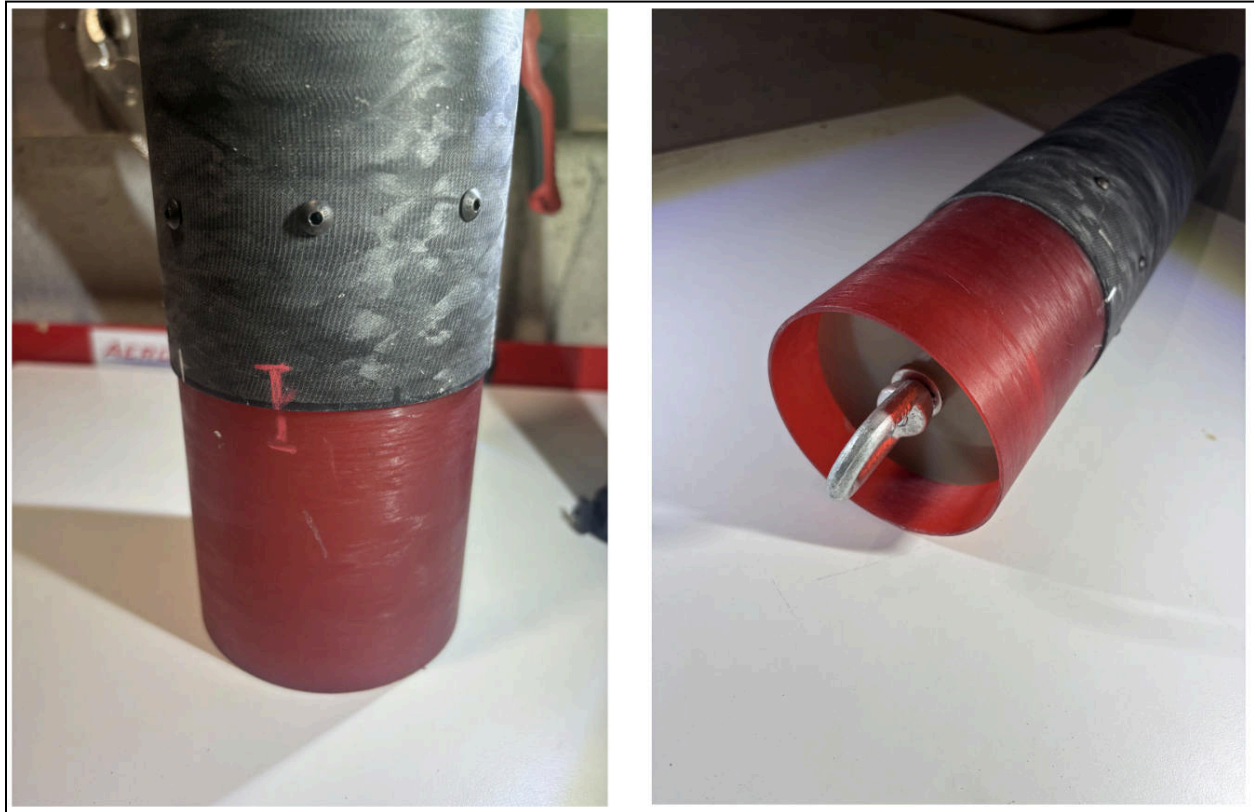


Figure 13: Completed Nose Cone Assembly

section using a series of 3D-printed jigs. This results in a strong, yet removable joint that achieves the required parameters for the Goldeneye flight vehicle.

8 COUPLERS

8.1 Separation Points

The rocket includes both fixed, and sliding couplers, and therefore, permanent joints and impermanent separation points. The separation points are held by #2-56 nylon shear pins, while the permanent joints are held by retaining. The specifications of each fastener is given in the recovery, and AvBay section respectively.

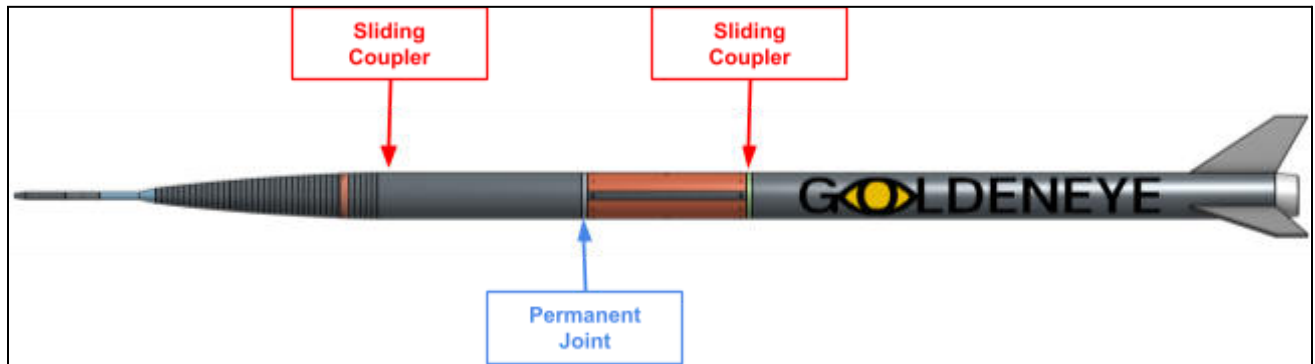


Figure 14: Separation Points Diagram

The first sliding coupler is between the nose cone and the forward airframe. This separation occurs at 1500' AGL, during Main Deployment. The shear pins are broken by the force of the nose cone actuating upwards, as a result of hot gasses from black powder ignition.

The second sliding coupler is between the aft of the AvBay and the aft airframe. This separation occurs at apogee, during Drogue Deployment. The shear pins are broken by the force of the nose cone actuating upwards, as a result of hot gasses from black powder ignition.

The permanent joint is between the forward airframe and the forward section of the AvBay. The bolts in this region prevent the forward airframe from separating from the AvBay during the firing of the main parachute.

9 RAIL BUTTON

The rail buttons serve the important role of providing stability for the launch vehicle while it is placed on the launch pad, and during the initial phases after ignition. These rail buttons are critical structural components that support significant load, namely the fully loaded vehicle sitting on the pad.

9.1 Detailed Design

This year, the team opted to use Giant Leap Rocketry's 1515 Airfoiled Rail Button. These components are machined from Delrin, and have an "airfoiled" profile, thereby reducing the drag. Given that Goldeneye's primary objective is to achieve high-fidelity supersonic air pressure data, it was critical to reduce drag to maximize airspeed. As such, the airfoil shape of these rail buttons were highly desired, especially compared to the standard circular profile of most rail buttons. Compared to standard circular buttons, the airfoil shape reduces the "wake drag", acting as a miniature boattail by reducing the low-pressure zone formed at the rear of the component. The reduction in drag, alignment with overall mission profile, and relatively low cost made this rail button a useful change.

From the previous year, and the Spectre rocket, the team found that a simple, but reliable rail button system involves the use of a wooden backing plate. This wooden block serves as a "hardpoint", wherein the rail button can be attached. This distributes the forces being exerted along to a much larger area of the thin-walled airframe, thereby greatly reducing the risk of "tear-out", where the airframe fails due to highly-concentrated, localized stress.

Furthermore, the wooden backing plate eliminates the need for a dedicated threaded insert, as it provides a material for the threads to secure onto. Thereafter, epoxy can be used to fill the voids and strengthen the joint even further.

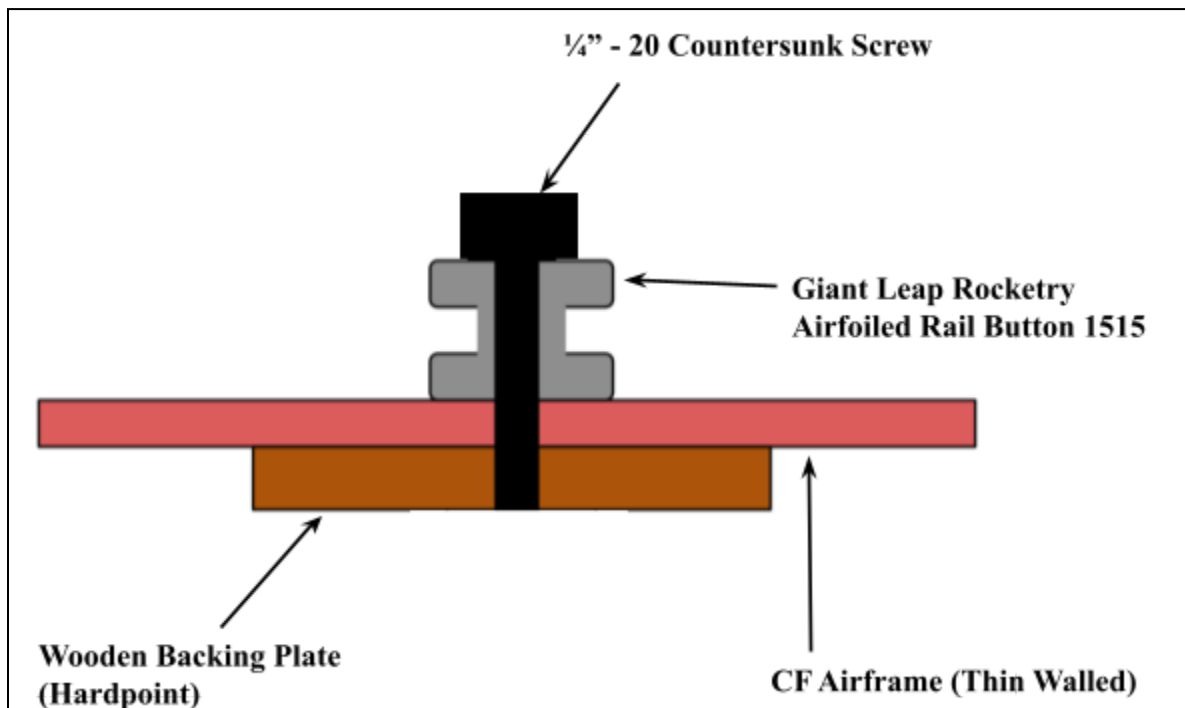


Figure 15: Cross-Sectional Illustration of the Rail-Button and Backing Plate

The axial placement of the rail button was a critical design decision. Placing both rail buttons closer to the aft lessens the time period that the vehicle can freely pivot after the first rail button clears the 18.5' launch rail. However, this placement can make the vehicle prone to bending, as the lever arm through which force on the rail button is applied is increased. In contrast, spacing the rail buttons farther apart improves structural integrity on the launch-pad, but can make the vehicle more prone to weather-cocking as it departs the launch rail (at a comparatively lower velocity).

To find a balance between these two conditions the decision was made to place the forward rail button on the center of gravity, and the aft rail button just forward of the fins. Thus, the aft rail button provided the maximum possible time in contact with the rail, while the forward rail button found a balance between both aforementioned conditions.

9.2 Manufacturing Process

The manufacturing process for the rail button is detailed in the following section.

1. **Manufacture backing plate:** The wooden hardpoint is made from maple wood. It was selected due to its strength, and the fact that this same wood was used in all other wooden components of the vehicle. The 1" thick plank was shortened using a jigsaw. Thereafter, using a belt sander, the outer edge of the wooden block was shaped to match the inner diameter of the airframe (~3.898").
2. **Surface preparation:** The inner wall of the airframe is sanded using 120-grit sandpaper and cleaned with isopropyl alcohol. This is to ensure that the epoxy is properly bonded to the material, which is critical for a structural joint.
3. **Surface-to-surface bond:** Using a Q-tip, the West System Epoxy (105 Resin, 206 Hardener), was applied to the wooden block, which was then placed in the airframe.
4. **Fillet:** Normally, on a surface-to-surface adhesive bond, clamping the surfaces together can improve the strength of the joint. Due to the inaccessible nature of this bond, the surface-to-surface bond is not clamped, and therefore may be structurally weak. As such, a fillet is shaped around the backing plate. This uses the 105/206 WSE, with 406 colloidal-silica filler, to prevent sagging or dripping during application.
5. **Bore:** Using a 1/4" drill bit, and a 3D-Printed alignment jig, the bore through the backing plate and the airframe is made.
6. **Epoxy Filler:** After the screw is inserted and seated within the backing plate, 105/206 WSE is applied using a Q-tip, in order to fill any voids, as well as to lock the screw in place.

This entire process is repeated for the forward and aft rail buttons. Alignment between both rail buttons is verified using a 1515 aluminum extrusion (launch rail).



Figure 16: Cross-Sectional Image of 1515 Rail Button Installed



Figure 17: Image of the Airfoiled 1515 Rail Button after Installation

10 AVIONICS BAY

10.1 Detailed Design

The Goldeneye's avionics bay was designed primarily with concern for ease of access to the electronics within. In previous TFRA flight vehicles, namely Spectre, troubleshooting electronic faults required partial disassembly of the vehicle. This had the adverse effect of delaying launch operations by one day during Launch Canada 2025.

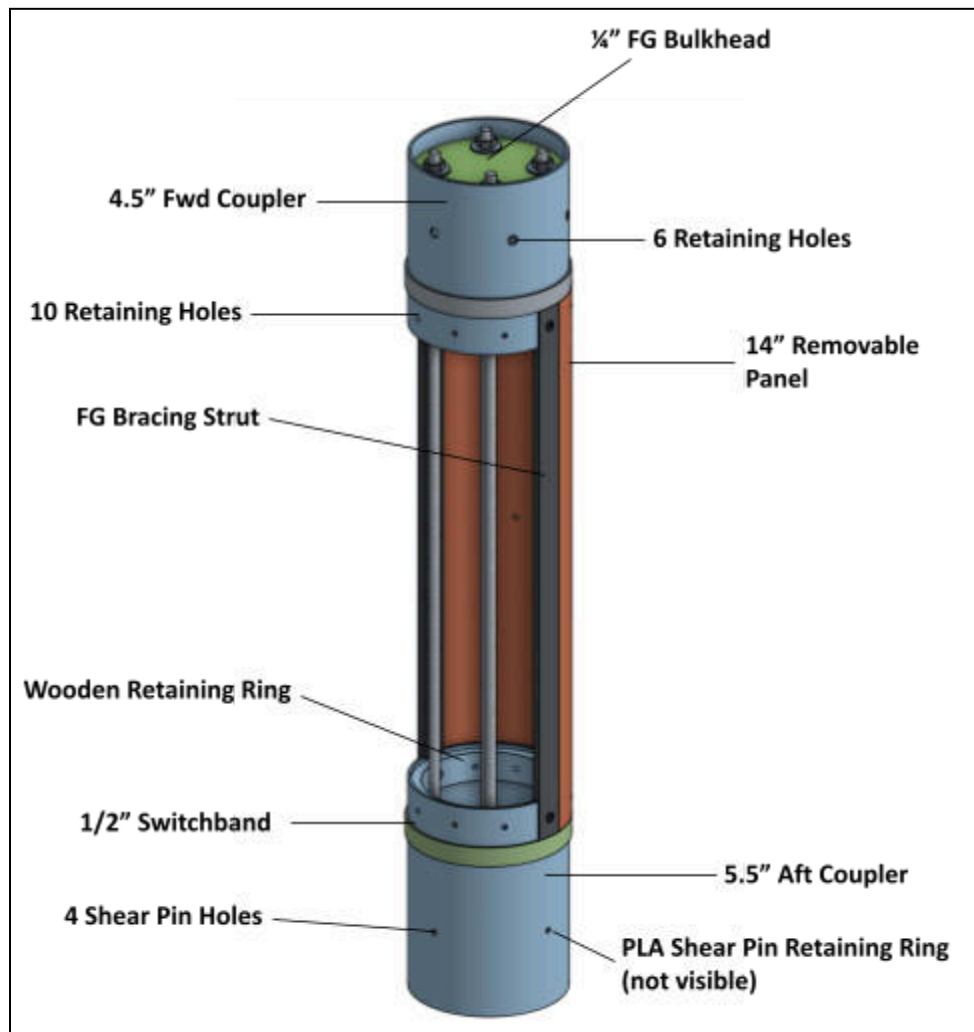


Figure 18: Nose Cone Assembly Diagram.

10.1.1 14" Removable Panels & Bracing Struts

The Avbay fulfills the design goal of easy accessibility using two 14" tall, removable fibreglass panels built into the Avbay section of the airframe, as shown in orange. These panels easily can be secured and removed by screwing and unscrewing their retaining bolts.

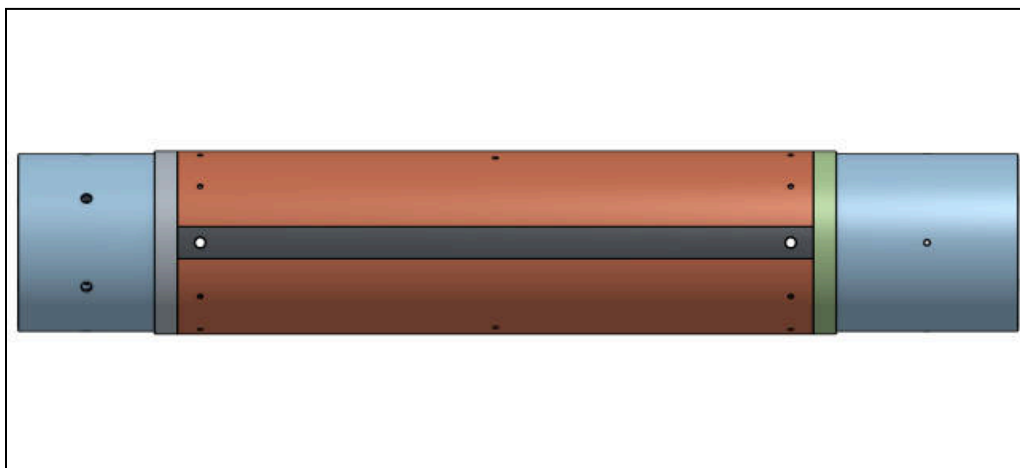


Figure 19: Avionics Bay Sideview

These removable panels are attached via eight (8x) 3mm bolts, both on the top and bottom couplers. When removed, the top and bottom AvBay couplers are held together by two bracing struts, shown in grey. These bracing struts connect to the couplers via $\frac{1}{4}$ " bolts.

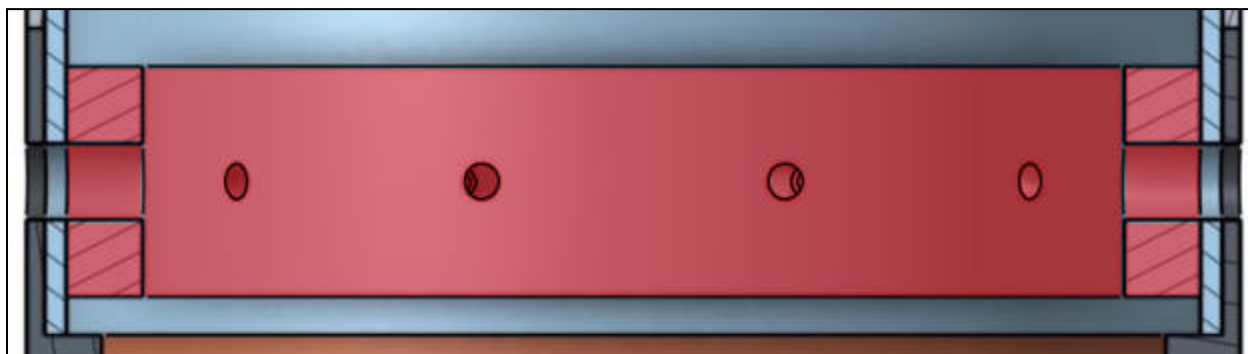


Figure 20: Panel Retainer & Coupler Cross Section

The ten retaining bolts on each coupler go through three layers: the removable panel/FG bracing strut (shown in grey/orange), the coupler (light blue), and the panel retainer (red). Both the aft and fwd panel retainers are made out of wood and include brass inserts in all retaining holes for the bolts to thread into.

10.1.2 Sleds & Threaded Rods

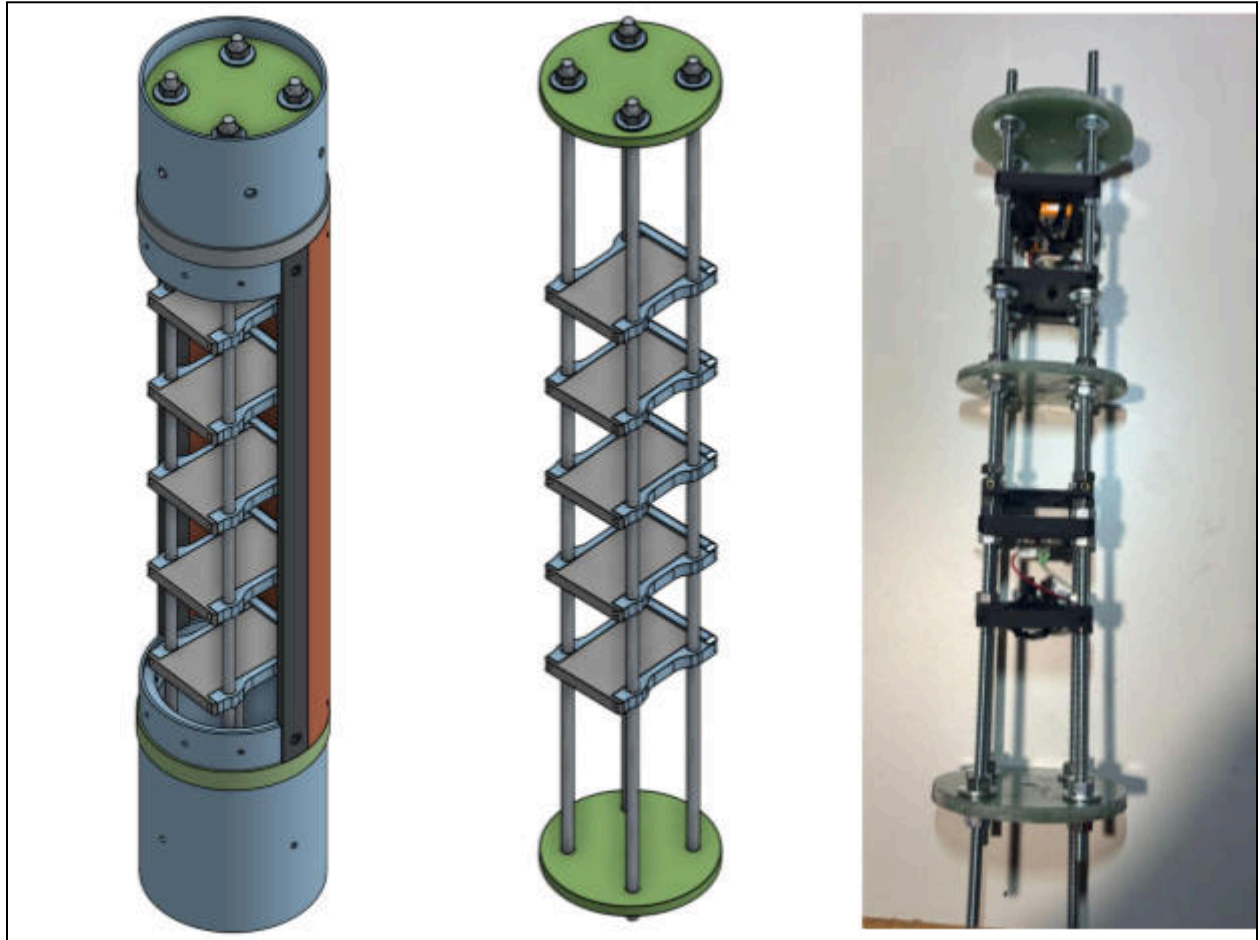


Figure 21: AvBay Open Assembly & Internals

Within the avionics bay are multiple sleds held up via four $\frac{1}{4}$ " steel, threaded rods running through the Avbay, from bulkhead to bulkhead. These sleds are all 3d printed, and made out of PLA.

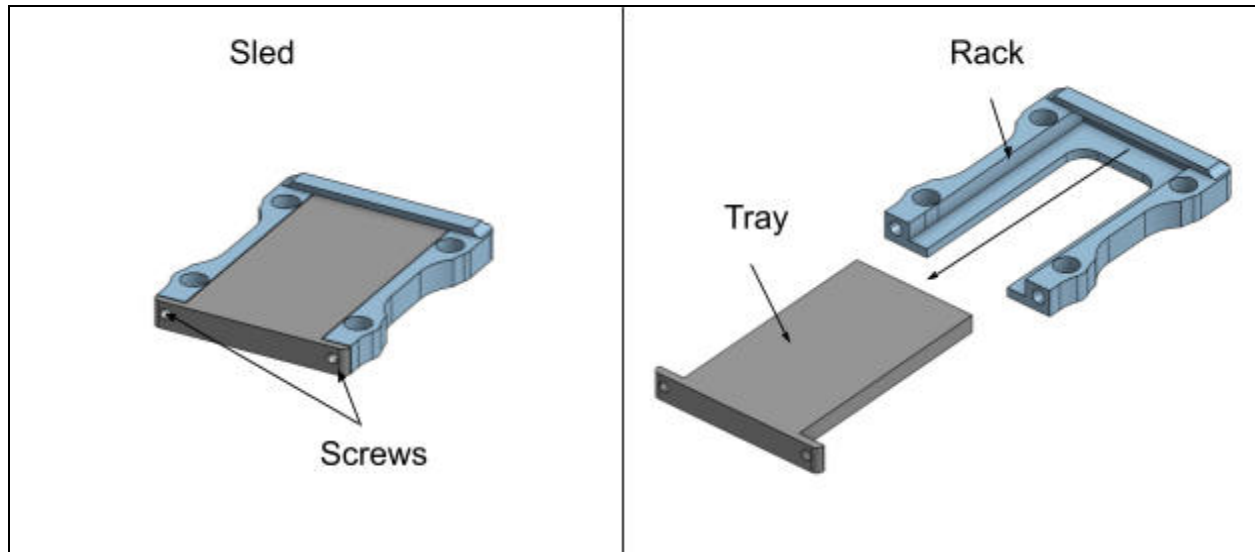


Figure 22: Sled CAD

The sleds, like the Avbay, are designed with ease of access in mind. Since removing sleds from the threaded rods would be highly time intensive, the sleds employ a sliding design. Each sled is composed of a rack, and a tray. The frame features four holes for the 1/4" threaded rods, and holds the tray. The tray holds the electronic components, and slides into the frame. The sled is secured to the frame using 3 mm metric bolts, which thread into heat-set brass inserts in the rack.

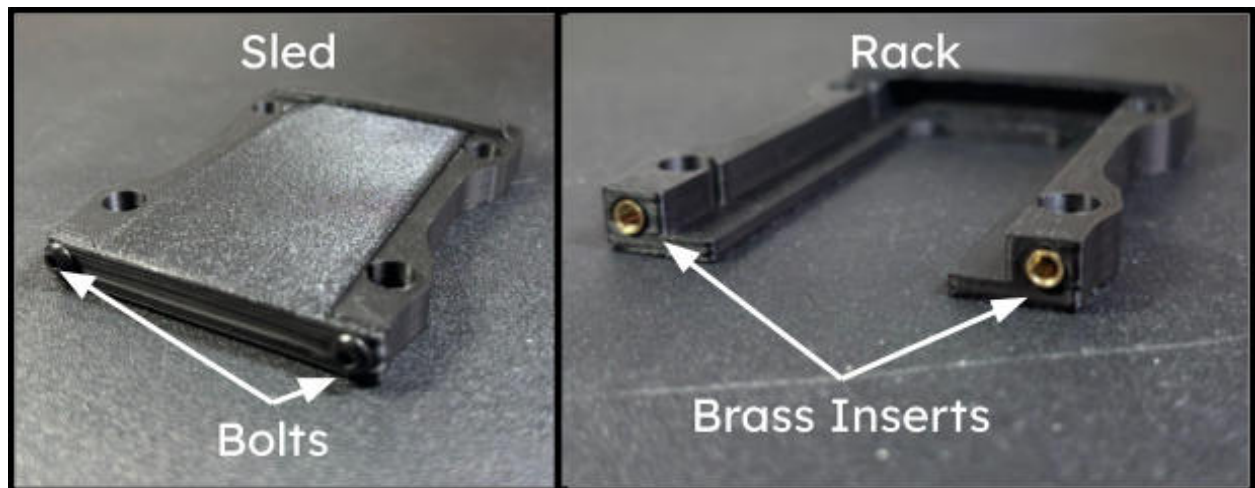


Figure 23: Sled

Overall, by having the electronics on a removable sled, electronics can easily be taken in and out of the avionics bay without disassembling it vertically.

10.1.3 Static Vent Holes

The vent holes are placed on both removable AvBay panels. These are critical to ensure that all the avionics systems, particularly the COTS altimeters that manage recovery, are able to accurately sense altitude changes, and therefore fire recovery charges.

There are a total of four vent holes, each of 0.125" diameter. These were manufactured with an alignment jig, and a standard power drill.

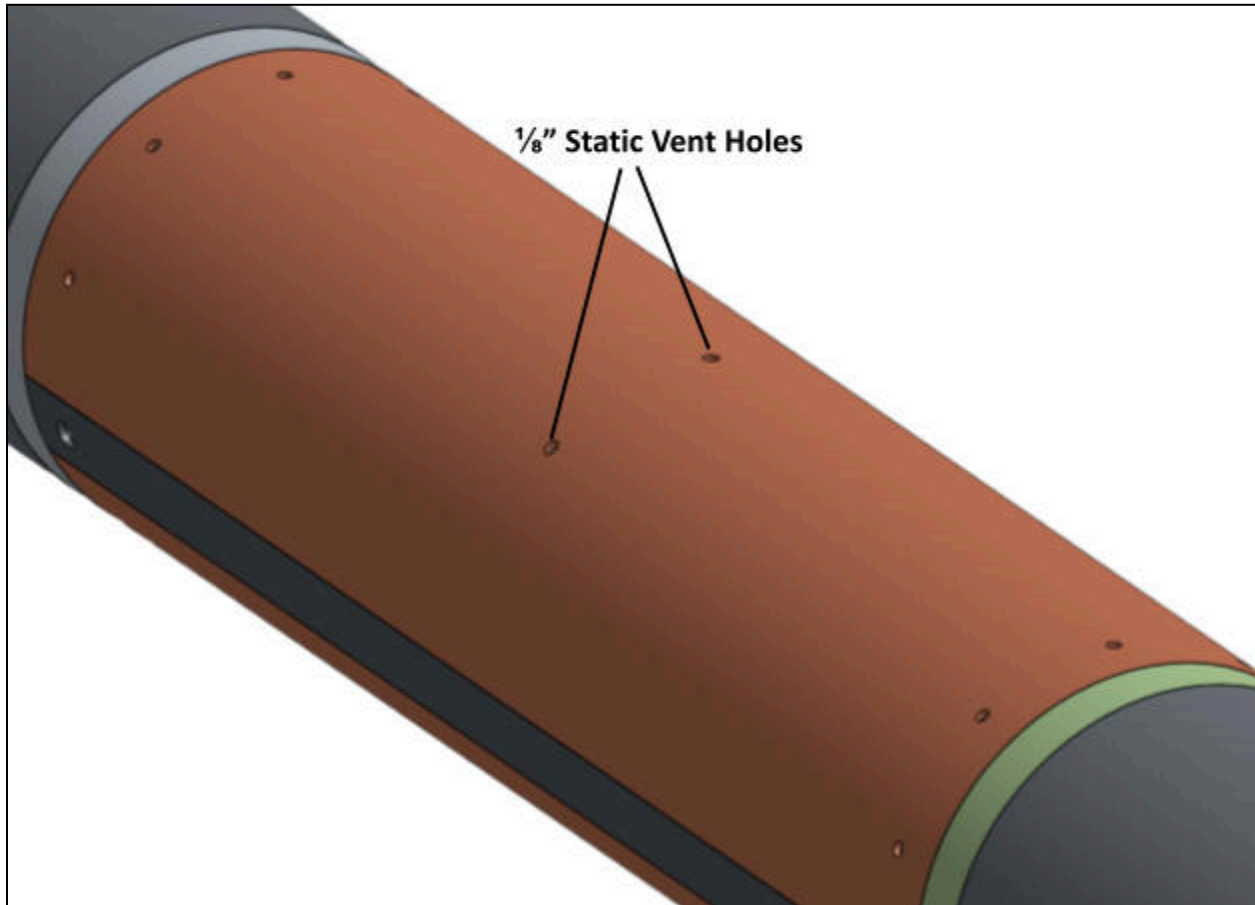


Figure 24: AvBay Vent Hole Placement

10.1.4 Couplers, Bulkheads, and Switchbands

The avionics bay features two fibreglass couplers: an aft coupler, and forward coupler. Each coupler comes with a $\frac{1}{4}$ " thick fibreglass bulkhead, and $\frac{1}{2}$ " tall fibreglass switchband.

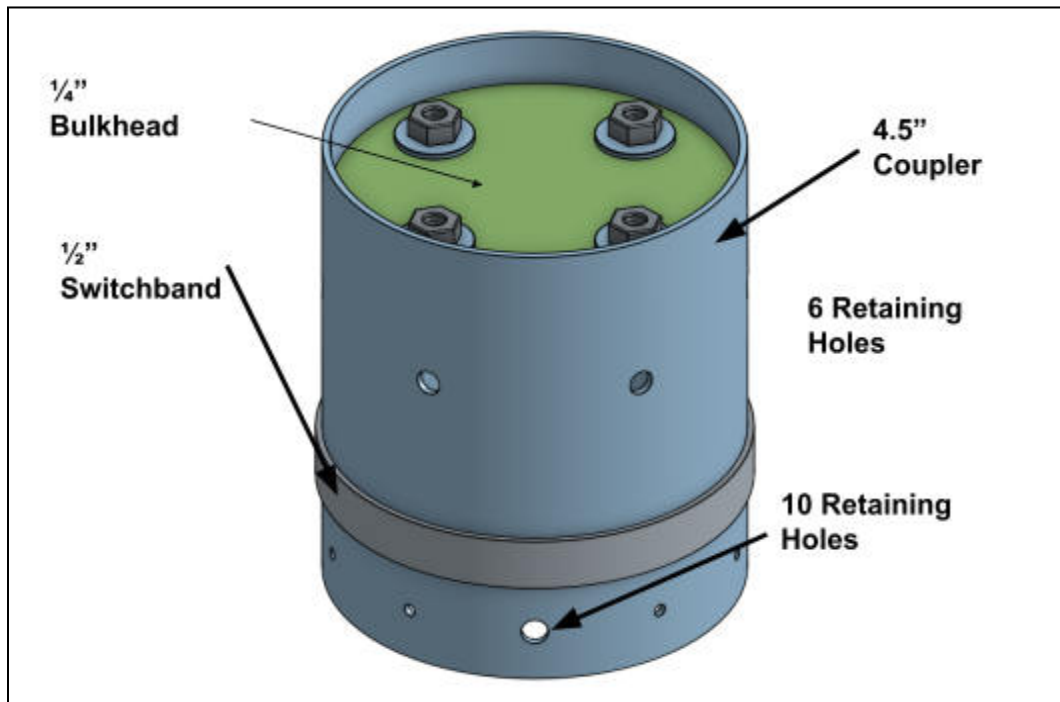


Figure 25: FWD Coupler

The avionics bay features a 4.5" fwd fibreglass coupler, with a $\frac{1}{2}$ " fibreglass switchband separating its upper and lower shoulders. The lower shoulder extends 1" below the switchband, and holds the ten retaining holes for the removable panels and FG struts— of which eight are 3 mm, and two are $\frac{1}{4}$ " for the FG struts. Meanwhile, the upper shoulder extends 3" above the switchband, and includes six 4mm retaining holes for the fwd Airframe.

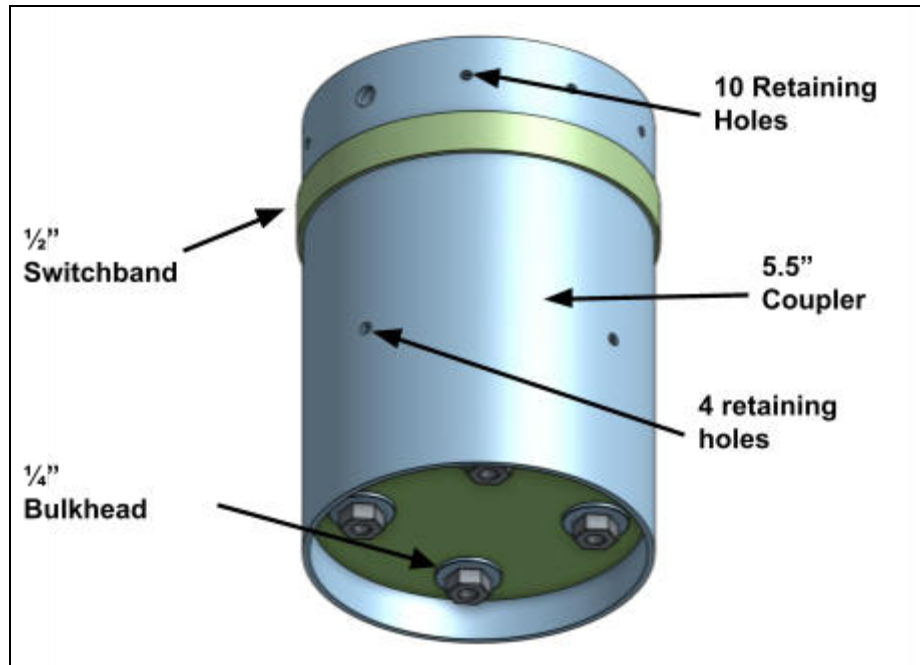


Figure 26: AFT Coupler

The avionics bay also includes a 5.5" aft fibreglass coupler connecting to the aft airframe. Its bulkhead and switchband are the same as the fwd coupler's. For the aft coupler, the switchband is located 1" below the top of the coupler. The upper shoulder extends 1" above the switchband, and the lower shoulder extends 4" below the switchband. The top shoulder has ten retaining holes: eight 3mm holes, and two 1/4" holes. The lower shoulder has four 1.5 mm retaining holes for four shearpins.

There are two 1/4" fibreglass bulkheads in the avionics bay. Each bulkhead has four 1/4" holes for the steel threaded rods running through the avionics bay. The fwd bulkhead is located 1/2" below the top of the fwd coupler, and the aft bulkhead is located 1/2" above the bottom of the aft coupler.

10.1.5 Power Switches

Although not illustrated in the CAD models, the power-on switches are placed on the bracing struts. Given that these struts are fixed, they are an ideal choice for placing the switches, as it removes the need for rewiring switches each time the AvBay is opened. The team found great success in using key switches on the Spectre launch vehicle. As such, the team decided to use key switches on this vehicle as well. These switches are mounted vertically on the bracing strut.



Figure 27: Key Switch Listing

Each switch is wired to an independent system, resulting in four separate key switches (two on either bracket).

11 RECOVERY

Goldeneye features a standard dual-separation, dual deployment recovery scheme. This recovery system ensures a reliable deployment of the main and drogue parachutes to safely recover the vehicle. Furthermore, TFRA has developed a total of three flight vehicles with this form of recovery system, and has seen successful flights from each vehicle. Thus, the simplicity and previous experience with the dual-separation, dual-deployment recovery setup were crucial factors in the decision to use this recovery scheme.

11.1 Recovery Load Estimation

In order to accurately select recovery hardware, such that mass is minimized, whilst sufficient factors of safeties are implemented, it was critical to estimate the peak recovery loads. The peak loading that the recovery system faces is upon the maximum extension of the shock cord, resulting in a “snap” of all recovery components, and imparting hundreds of pounds of force in an instant. Another significant moment is upon the inflation of the parachutes, where significant loads are imparted on the parachute, and the risers/suspension lines.

The primary method in order to estimate the peak shock loads during recovery involved creating a Python script to model a one-dimensional, 2-body spring-mass-damper system. This Python script was adapted from the University of Victoria [7]. The spring in this case, is the 1” tubular nylon shock cord. The damper is the parachute, which is simplified to produce drag according to the following equation:

$$F_D = 0.5 \cdot \rho \cdot C_D \cdot A_{eff}$$

Further, by overlaying the respective performance characteristics of both the main and drogue parachutes alongside environmental conditions in OpenRocket, this Python script yields an estimate on peak shock loading. This process was iterated using variations of density, drag coefficient, effective parachute area, amongst other variables, and the highest reasonable shock expected is 455 lbf, or 2.02 kN. This figure was the highest output in simulations considering both the main and drogue parachutes. Assuming the worst case scenario, this figure was used to size and select recovery hardware.

In addition, Launch Canada’s recovery loads estimator was also used to verify the validity of the Python script [8]. After inputting the various data points required within this calculator, and accounting for worst-case scenarios (i.e. low apogee resulting in denser air during drogue deployment), the “Instantaneous Opening Force” was found to be 425.8 lbf. This represents a 6.6% variance between the two figures, which increased the team’s confidence within the methodology used to calculate recovery loads.

11.2 Parachutes

The following table details the parachutes selected for Goldeneye. These parachutes were chosen based on performance, accessibility, and affordability factors. Two conspicuous, but different colour parachutes were also chosen, in order to clearly discern distinct flight events (Drogue versus Main opening).

Main Parachute			
Designation	Toroidal/Annular	Size	72" Diameter
Manufacturer	Rocketman	Material	Ripstop Nylon
Weight (grams)	357	Drag Coefficient	2.20
Descent Rate (ft/s) / (m/s)	17 / 5	Deployment (ft AGL)	1,500
Drogue Parachute			
Designation	N/A	Size	24" Diameter
Manufacturer	Rocketman	Material	Ripstop Nylon
Weight (grams)	76	Drag Coefficient	0.97
Descent Rate (ft/s) / (m/s)	90 / 26	Deployment (ft AGL)	12,590 (Apogee)

Table 13: Main Parachute and Drogue Parachute Key Information

11.3 Recovery Hardware

The following table details the associated hardware to the recovery system on Goldeneye. Components were chosen to optimize strength with mass.

As mentioned prior, the expected shock loading is 455 lbf / 2.02 kN.

Component	Vendor	Load Rating (lbf / kN)	Factor of Safety	Location
3/8-16 Eyebolt	Rigging Canada	7750 / 34.5	17.0	Motor Fwd Closure
3/8-16 Eyenut	Rigging Canada	6250 / 27.8	13.7	Nose Cone Bulkhead
1/4" Quicklink	Rocketman Parachutes	5000 / 22.2	11.0	Assorted
3000 lbs Stainless Steel Swivel	Rocketman Parachutes	3000 / 13.3	6.6	Main / Drogue
1" Tubular Nylon	Mountain Equipment Company	4000 / 17.8	8.8	Assorted
3/8-16 U-Bolt	McMaster Carr	5375 / 23.9	11.8	AvBay Bulkheads

Table 14: Goldeneye Recovery Hardware

11.4 Lines Diagram

The following diagrams depict the connections within both the main and drogue recovery systems. For the sake of conserving detail and clarity, the two respective systems are depicted in two separate diagrams. Additionally, in order to prevent the respective vehicle sections from impacting each other upon separation and descent under parachutes, the shock cords are “staggered”. In other words, the parachute does not sit at the midpoint of the shock cord segment (5m / 5m), but rather at a $\frac{2}{3}$ to $\frac{1}{3}$ distance. This stagger reduces the probability of rocket segments colliding while descending.

Drogue Parachute

Please note that the diagram is NOT to scale.

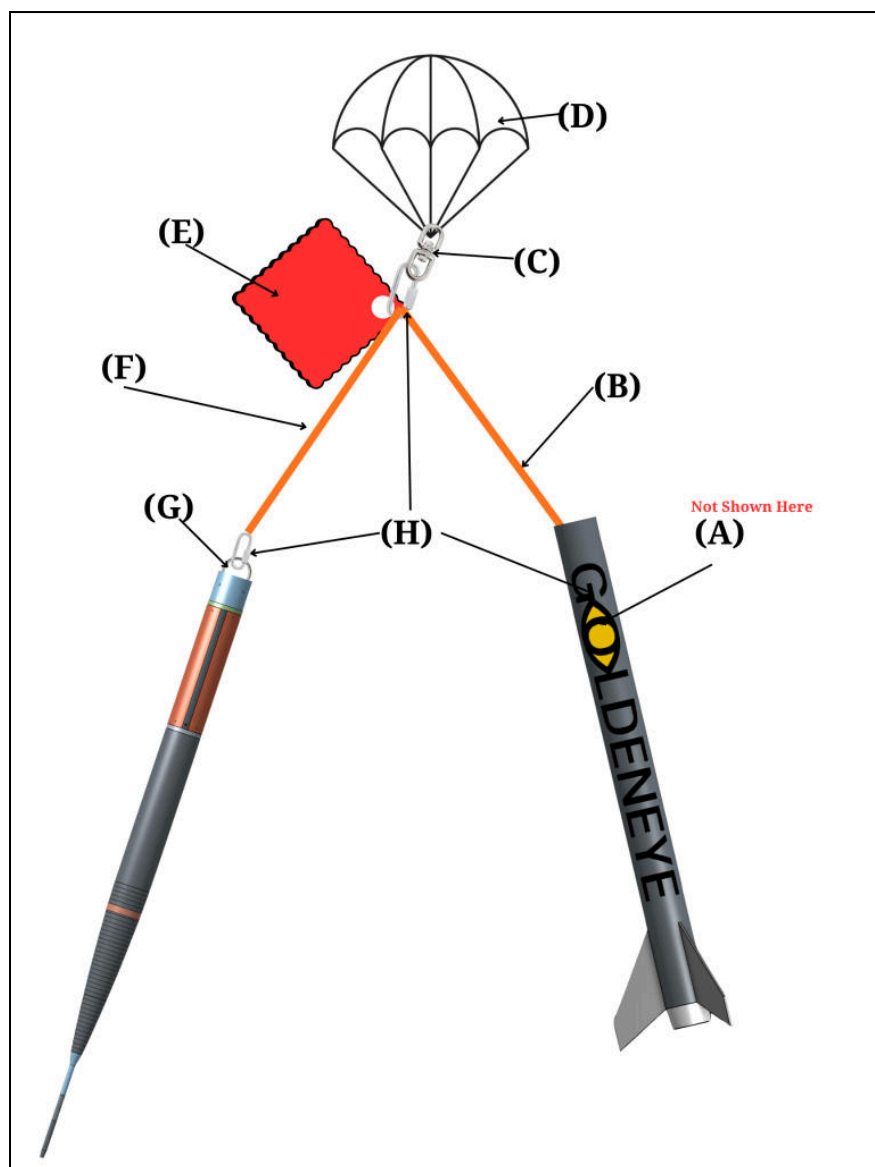


Figure 28: Drogue Parachute Lines Diagram

Letter	Component	Location
A	$\frac{3}{8}$ - 16 Eyebolt	Not illustrated. Threaded into the forward motor closure. Attached to (H).
B	1" Tubular Nylon	6.6 Meter Segment. Attached to quicklinks (H) that connect to the swivel, and eventually the drogue parachute.
C	3000 lbs Stainless Steel Swivel	Attached to (H) and (D). Prevents lines from tangling.
D	24" Drogue Parachute	Attached to (C).
E	9" x 9" Nomex Chute Protector	Attached to (H) and protects the chute (D) from ejection gasses.
F	1" Tubular Nylon	3.3 Meter Segment. Attached to (H). Connects avionics bay (G) to the chute (D).
G	$\frac{3}{8}$ - 16 U-Bolt	Bolted into the avionics bay. Attached to (H).
H	$\frac{1}{4}$ " Quicklinks	Connects (A) and (G) to (C).

Table 15: Letter key to the diagram in Figure 28

Main Parachute

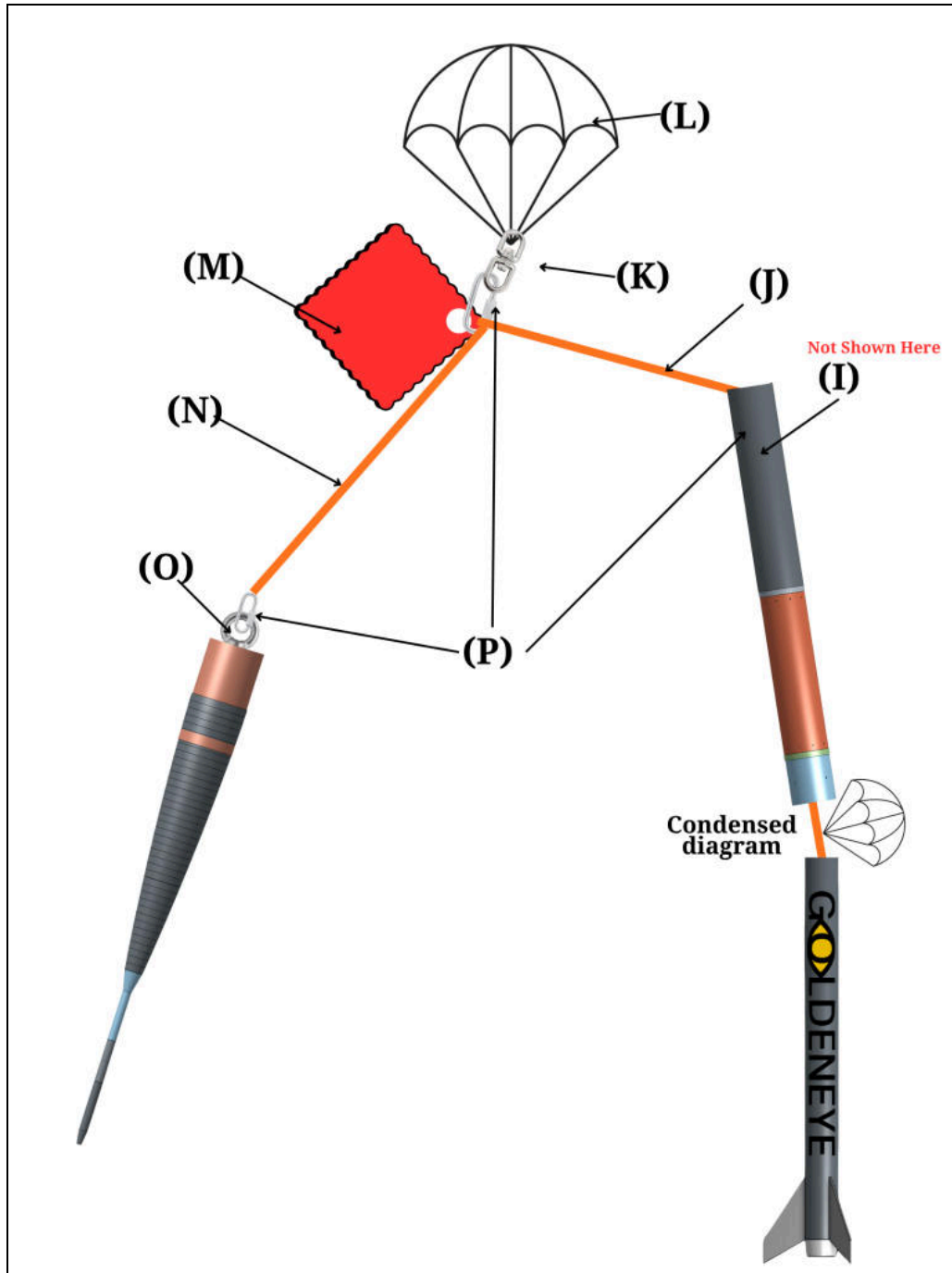


Figure 29: Main Parachute Lines Diagram

Letter	Component	Location
I	$\frac{3}{8}$ - 16 Eyebolt	Not illustrated. Threaded into the forward motor closure
J	1" Tubular Nylon	6.6 Meter Segment. Attached to swivels (P) that connect the chute (L) and the avionics bay..
K	3000 lbs Stainless Steel Swivel	Attached to (L) and (P). Prevents lines from tangling.
L	72" Main Chute	Attached to (K).
M	18" x 18" Nomex Chute Protector	Attached to (P) and protects the chute (L) from burn marks.
N	1" Tubular Nylon	3.3 Meter Segment. Attached to (P). Connects nose cone (O) to the chute (L).
O	Female Eyunut	Bolted into the nose cone bulkhead (O). Attached to (H).
P	$\frac{1}{4}$ " Quick Links	Connects (A) and (G) to (C)

Table 16: Letter key to the diagram in Figure 29.

An image of the recovery hardware is provided below. Once again, for the sake of clarity, two separate photos were taken, depicting the drogue and main parachute systems.



Figure 30: Drogue Parachute Hardware

Please note that the left hand side depicts the AvBay aft bulkhead. The right hand side features the $\frac{3}{8}$ "-16 eyebolt, which threads into the motor forward closure.



Figure 31: Main Parachute Hardware

Please note that the left hand side depicts the nose cone. The right hand side features the forward AvBay bulkhead.

11.5 Black Powder

The parachutes will be deployed using the combustion of FFFF Black powder. This method was chosen simply because it is lightweight, reliable, simple, and TFRA has had experience with pyrotechnic systems.

	PRIMARY	DEPLOYMENT	SECONDARY	DEPLOYMENT
MAIN PARACHUTE	2.0 grams	1500 FT AGL	2.7 grams	1500 FT AGL +1 Second
DROGUE PARACHUTE	1.7 grams	APOGEE	2.7 grams	APOGEE +1 Second

Table 17: Black powder key details in Main and Drouge parachutes.

The BP charges will be secured within a nitrile glove finger, and secured with rubber bands. A J-TEK Igniter is also used to ignite the black powder. The BP Charges are held within the charge wells, which are made from $\frac{3}{8}$ " Copper end caps.

Component	Usage
Black Powder	Used to generate the hot gasses for parachute ejection.
J-TEK Igniter	Used to ignite the black powder charges.
Nitrile Glove Fingertip	Used to house BP.
$\frac{3}{4}$ " Copper Charge Well	Used to house complete BP Charge (within nitrile fingertip).

Table 18: Pyrotechnic Components List.



Figure 32: Low-Resolution Image of BP Charge.

11.6 Pressure Equalization Holes

The pressure equalization holes play a crucial role in preventing premature vehicle separation by preventing overpressurization of the airframe. Briefly, without these pressure equalization holes, the inside of the airframe will remain at a higher pressure than the ambient air during its ascent. As such, the rocket will be inclined to separate. This is particularly dangerous when coupled with the risk of premature drag separation. As such, the pressure equalization holes are included in both the aft and forward airframes to prevent overpressurization.

An important decision is the axial placement of the pressure equalization holes. Under acceleration, the parachute and most recovery hardware will be pushed towards the aft of the vehicle. Thus, if the pressure equalization holes are included in the aft of the airframe, it can be rendered useless by sliding hardware. As such, the pressure equalization holes are drilled 1" behind the most forward section of each airframe.

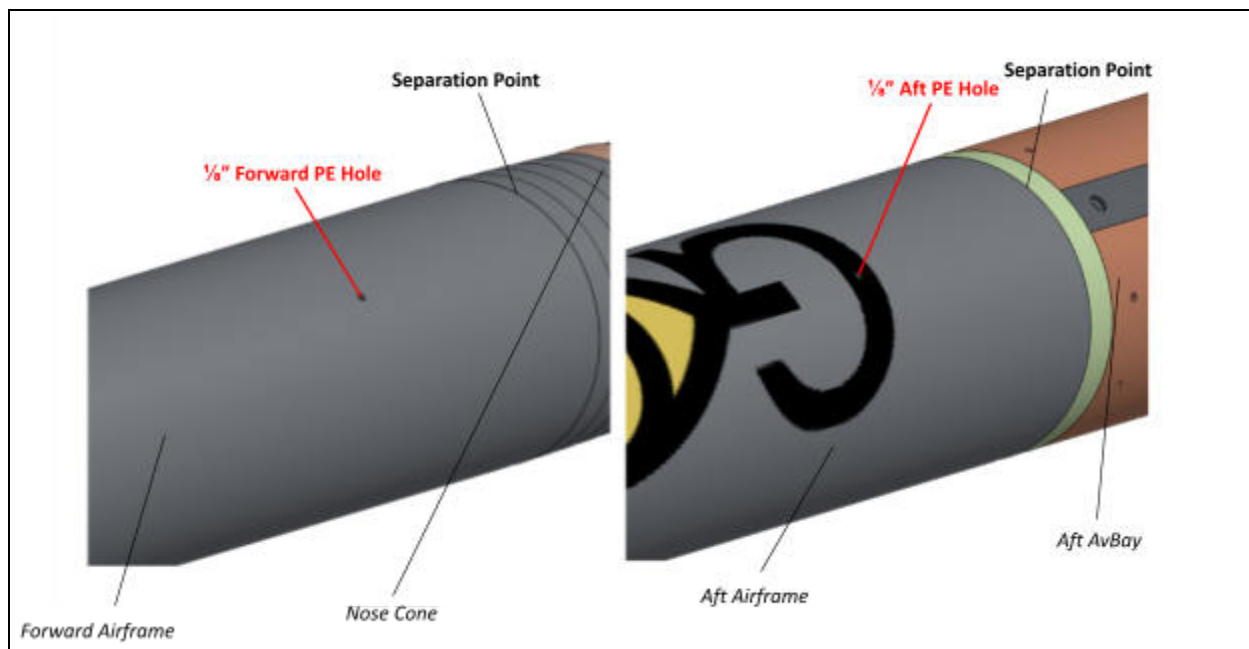


Figure 33: Illustration of Pressure Equalization Holes

11.7 Testing

In order to determine if the recovery system and its components function properly, there is no substitute for all-up testing. As such, TFRA placed emphasis on conducting recovery testing early in the development cycle in order to incorporate any necessary changes into the novel design.

This approach yielded important insights and design changes. Initially, Goldeneye was designed to have two (x2) 4-40 nylon shear pins. In previous rockets, the nylon shear pins that were used (#2-56), were unable to thread into material, as the threads themselves were far too small and delicate. As such, using #4-40 shear pins instead was suggested. However, after the initial ground tests, it became apparent these thicker shear pins required far more force than expected to break. This would introduce a great deal of unnecessary stress on the recovery system. Thereafter, the decision was made to revert to #2-56 threads. To alleviate the lack of threads issue, 3D printed retaining rings were used, as detailed in prior sections.

TFRA's typical ground test procedure is to remotely activate the black powder charge while the rocket is resting at a $\sim 30^\circ$ inclination on a custom test stand.



Figure 34: Ejection Testing of Goldeneye.

An image of the ejection test is provided above. Quite evidently, the camera and lighting conditions were not ideal. Nevertheless, the ground testing provided valuable data and insight into design elements that needed to change, that could stay the same, and the improvements that could be made.

12 BULKHEADS

The bulkheads on this vehicle serve to provide recovery hardpoints, isolate internal volume, protect against recovery gasses, and provide structural integrity in critical locations. There are three key bulkheads within the Goldeneye flight vehicle. Each bulkhead is made of $\frac{1}{4}$ " G10 fiberglass. This material was chosen mainly due to its strength-to-weight ratio, and workability.

The bulkhead also serves critical function in terms of recovery, as it houses lever nuts, ejection wells, and black powder charges.



Figure 35: Goldeneye Bulkhead, Isometric View



Figure 36: Goldeneye Bulkhead, Head-on View

The pictures above depict the Goldeneye AvBay bulkheads, which are placed in the forward and aft of the assembly. They are identical. The bulkhead features the following components:

- **$\frac{3}{8}$ " - 16 Eyebolt:** This is the recovery hardpoint, where the quicklink is tethered to. This component is secured to the bulkhead with four (4x) $\frac{3}{8}$ " stainless steel hexnuts.
- **$\frac{3}{4}$ " Ejection Wells:** The $\frac{3}{4}$ " ejection wells are epoxied onto the bulkhead and serve as a container for the black powder charge (inside the nitrile glove tip).
- **5/32" Bore:** The bulkhead also features four (4x) 9/32 bores to allow for $\frac{1}{4}$ " threaded rods
- **Lever Nuts:** These lever nuts are substitutes for typical screw terminals for black powder igniters. These lever nuts hold the wires securely, resist vibration, and are far more convenient than typical screw terminals.

12.1 Manufacturing Process

The nose cone bulkhead presides in the coupler within the nose cone. It serves as a hardpoint for the recovery system. It has a central $13/32$ " bore in order to accommodate a $3/8$ -16 threaded rod. This bulkhead was created by using a 4" hole saw, and sanding the outer circumference using a belt sander, until it was reduced to approximately 3.725", and fit securely in the coupler.

The remaining two bulkheads preside on the forward and aft ends of the AvBay. They are made in an identical fashion to the nose cone bulkhead. However, these AvBay bulkheads differ from the nose cone bulkheads, as the four AvBay bulkheads have four equidistant $9/32$ " holes to accommodate four $1/4$ " threaded rods. Further, in order to isolate the ejection gasses from the delicate electronics inside the AvBay, the central bore made by the hole saw arbor is epoxied using West System Epoxy.



Figure 37: Bulkhead Sanding Process

13 FINS

13.1 Detailed Design

The primary design objective of the fin system was to minimize drag and mass while maintaining sufficient stability and structural integrity throughout the flight. Since the vehicle was designed to exceed Mach 1.2, reducing mass at the aft end of the rocket was particularly important to maximize the achievable altitude and velocity while maintaining a stability margin greater than 1.5 cal. Consequently, the fins were designed to optimise performance while still providing the required stability.

The fins were manufactured from fibreglass composite for its excellent strength-to-weight ratio and availability. Although carbon fibre was initially considered because of its superior stiffness and lower mass, procurement constraints prevented its use. Fibreglass nevertheless provided sufficient structural performance while remaining lightweight and manufacturable.

Number of Fins	3
Root Chord	7.55 in
Tip Chord	2.8 in
Height	5 in
Sweep Length	7.6 in
Sweep Angle	56.7°
Thickness	0.25 in

Table 19: Overview of Fin Geometry.

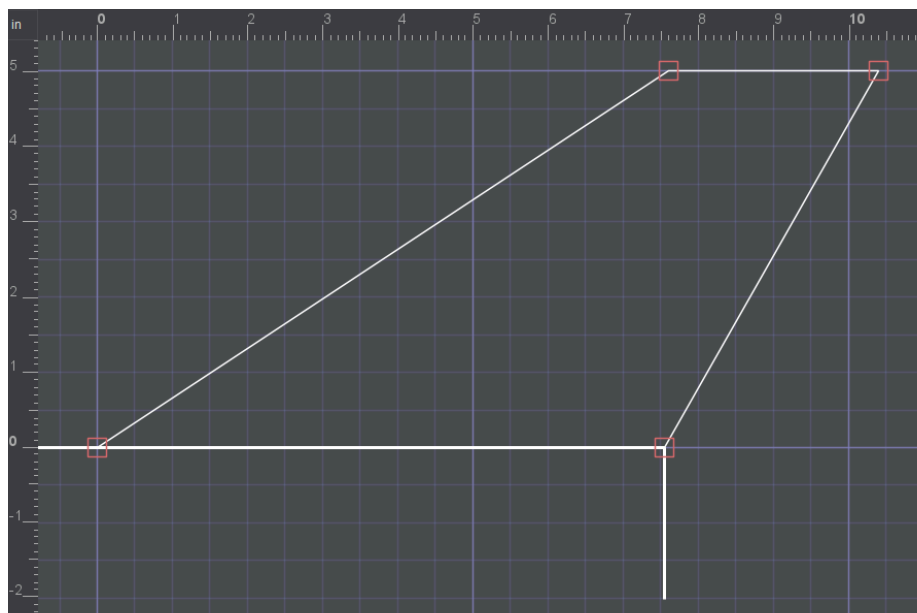


Figure 38: Fin Freeform Diagram

The final vehicle utilizes a three-fin configuration. Earlier design iterations employed a four-fin arrangement, consistent with previous TFRA vehicles; however, OpenRocket simulations indicated that this configuration produced excessive stability while introducing unnecessary mass and increased drag. Transitioning to a three-fin configuration reduced both drag and mass improving performance while maintaining the required stability margin.

The fin geometry also evolved throughout the design process. Initial designs featured a shorter fin span of 3.8 inches. Following discussions with the Launch Canada technical team, concerns were raised regarding the fins effectiveness during supersonic flight. Specifically, it was identified that the boundary layer and wake generated at transonic and supersonic speeds could reduce the effectiveness of shorter fins if they did not sufficiently extend into undisturbed airflow. The fin span was therefore increased to 5 inches as suggested, to ensure adequate stability throughout flight.

Structurally, the fins utilize through-the-wall mounting to efficiently transfer aerodynamic loads into the motor mount tube. Both internal and external epoxy fillets are incorporated to improve load distribution and increase resistance to bending and shear loads experienced during powered ascent. Finally, the fins were designed to achieve a 29.3% fin flutter safety margin, ensuring that they will not experience flutter within the expected flight regime.

13.2 Manufacturing Process

The fins were cut via water jet, provided by our partners at the Ontario Tech Space and Rocketry team (OTSR). This process allowed high-precision cuts and ensured excellent accuracy to the desired fin geometry such that aerodynamic performance remains consistent across all three fins.

The fins' leading edges were chamfered to approximately 45° using a belt sander. The fin slots were cut using a custom 3D printed jig to ensure accurate positioning around the airframe. The base of the fins tabs were epoxied to the motor mount tube and epoxy fillets were applied along the root chord of the fins to ensure a secure attachment. Finally, internal fillets were applied using a syringe and surgical tubing between the fin tabs and the motor mount tube to act as an additional hardpoint. During curing processes the fins were held in alignment using a 3D printed fin jig.

13.3 Analysis

Fin flutter analysis was conducted using a pre-made calculator by John K. Bennett. This calculator provides a simple though limited approximation of the fin flutter velocity. This year the team wanted to better understand the equations governing the determination of the fin flutter velocity as well as the limitation involved with this calculator. Following a report published by Bennett it was found that the following equation is cited as the governing equation for fin flutter velocity.

$$V_f = a \times \sqrt{\frac{G_E}{\frac{39.3 \times A^3}{(\frac{t}{c})^3 \times (A+2)} \times (\frac{\lambda+1}{2}) \times (\frac{p}{p_0})}}$$

Figure 39: Fin Flutter Equation [20]

The denominator term features the number 39.3, representing the “denominator constant”. However, the value presented here is not an accurate approximation for highly-swept back fins such as the ones on Goldeneye. As such, another equation was applied to calculate a more appropriate denominator constant.

$$DN = \frac{(24 \times \varepsilon \times \kappa \times p_0)}{\pi}$$

Figure 40: Denominator Constant Equation [20]

The calculator used accounts for this and calculates DN separately. The equation was thus revised as such:

$$V_f = a \times \sqrt{\frac{G_E}{\frac{DN \times A^3}{(\frac{t}{c})^3 \times (A+2)} \times (\frac{\lambda+1}{2}) \times (\frac{p}{p_0})}}$$

Figure 41: Fin Flutter Revised Equation

The largest assumption made by the use of the calculator is shear modulus of the fin material, in this case G10 fibreglass. Given its anisotropic properties the team struggles to find a reliable value for this variable, denoted G_E in the equation above. Upon accessing the manufacturer's TDS on the G10 material the Young's Modulus and Poisson's Ratio of the material was found as 3100000 psi and 0.12 respectively. Applying the equation:

$$G_E = \frac{E}{2 \times (1 + \nu)}$$

Figure 42: Shear Modulus Equation [20]

where E is the Young's modulus and ν is Poisson's ratio, the shear modulus was found to be 1383929 psi. As an extra precaution a 15% safety factor was applied, reducing the shear modulus to the final value of 1176339 psi. Applying this value to the fin geometry and launch conditions produced a final value for fin flutter at 545.29 m/s, giving a safety margin of 123.69 m/s or 29.3% (1.29 FOS).

Critical Fin Flutter Velocity	545.29 m/s
Vehicle Max Velocity	422 m/s
Margin of Safety	123.69 m/s
Margin of Safety (%)	29.3%

Table 20: Summary of Fin Flutter Analysis Results

According to Bennet, any value for fin flutter with a safety margin $\geq 25\%$ is considered safe for flight. Thus, the results indicate that the fins will not experience flutter within the expected flight regime. It should be noted however, that the calculator does not account for the effects of a through-the-wall fin attachment on fin flutter, therefore this aspect of the design could not be evaluated.

13.4 Testing and Validation

13.4.1 Objective

The objective of this testing campaign was to determine the optimal surface preparation procedure for the fin-to-airframe epoxy bond. Specifically, the effects of sandpaper grit size and isopropyl alcohol (IPA) evaporation time prior to epoxy application were evaluated to determine their influence on bond quality and failure behaviour.

Bond performance was assessed through destructive mechanical testing, with failure mode classification serving as the primary indicator of bond integrity. A cohesive failure mode was considered the desired outcome, as it indicates that the adhesive bond exceeded the internal strength of the cured epoxy.

13.4.2 Methodology

A total of six representative bonded test specimens were fabricated using the same fiberglass materials and epoxy system selected for the flight vehicle fin-to-airframe joints. Each specimen was prepared using a controlled combination of:

- Sandpaper grit: 60, 80, and 120 grit
- IPA evaporation time prior to epoxy application: approximately 10 s and 30 s

All other variables, including material geometry, adhesive application procedure, cure time (24 hours), and ambient curing conditions, were maintained constant throughout testing.

Sample photograph of the prepared test samples are provided in section B.2.1.

Following curing, each specimen was subjected to destructive mechanical testing using a custom lever-based testing apparatus. The apparatus applied an increasing load directly to the fin joint until structural failure occurred.

Due to the limited availability of instrumentation capable of accurately measuring ultimate failure loads, bond performance was evaluated using failure mode classification, a widely accepted qualitative method for assessing adhesive joint performance during prototype development and aerospace validation.

Failure modes were defined as follows:

- Adhesive Failure: Separation occurred at the epoxy-substrate interface, indicating insufficient adhesion.
- Mixed Mode Failure: Failure occurred through a combination of adhesive separation and epoxy fracture, indicating partial bond integrity.
- Cohesive Failure: Fracture occurred within the epoxy while both bonded interfaces remained intact, indicating that the adhesive bond exceeded the internal strength of the cured epoxy.

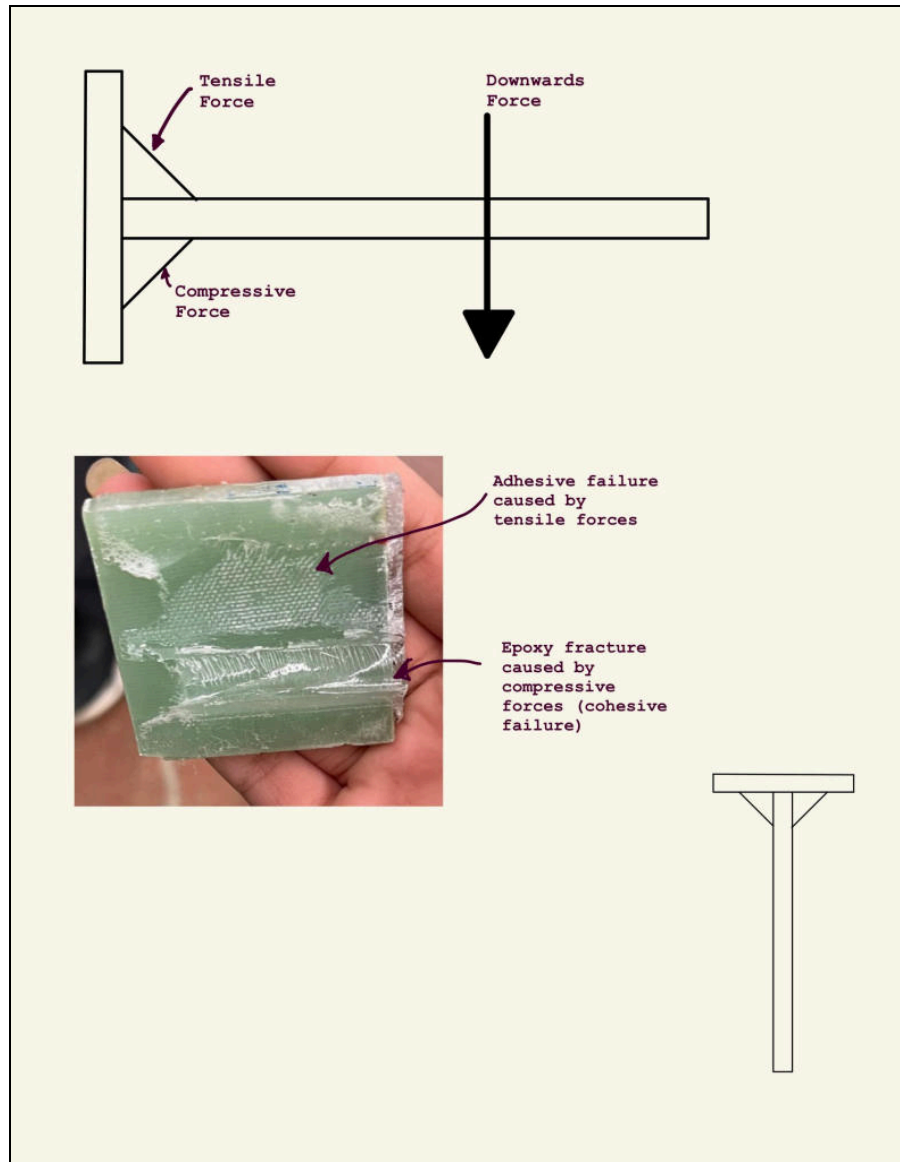


Figure 43: Testing Apparatus & Adhesive Failure Mode Classification

Representative photographs of the testing apparatus during mechanical testing are provided in Appendix B.2.2

13.4.3 Results

Batch #	Sandpaper Grit	IPA Application Time	Observed Failure Mode	Qualitative Observation
1	60	10s	Adhesive Failure	Minimal bonding observed; fillets detached at one of the surfaces.
2	60	30s	Mixed Mode Failure	Slight improvement in adhesion, but inconsistent bonding remained.
3	80	10s	Adhesive Failure	No effective bonding; fillets detached from one substrate face.
4	80	30s	Mixed Mode Failure	Partial bonding achieved; inconsistent adhesion between fillets.
5	120	10s	Cohesive Failure	Epoxy fractured internally; bonded interface remained intact.
6	120	30s	Mixed Mode Failure	Reduced performance compared to shorter IPA application time.

Table 21: Summary of surface preparation testing results.

Representative examples of observed failure modes provided in Appendix B.2.3.

13.4.4 Discussion

The experimental results demonstrated that surface preparation had a significant influence on epoxy bond performance, with sandpaper grit exhibiting a greater effect than IPA evaporation time.

Effect of Surface Roughness

Specimens prepared using 60-grit and 80-grit sandpaper consistently exhibited adhesive failure when the IPA evaporation time was limited to 10 seconds, indicating inadequate bonding between the epoxy and substrate. Increasing the IPA evaporation time to 30 seconds improved performance to mixed-mode failure for both grit levels, suggesting that additional solvent evaporation reduced residual surface contamination and promoted more consistent adhesion. Representative adhesive and mixed-mode failures shown in Appendix B.2.2.

In contrast, specimens prepared using 120-grit sandpaper produced the highest bond quality observed during testing. The 120-grit specimen with a 10-second IPA evaporation period was the only configuration to exhibit cohesive failure, indicating that the adhesive interface was stronger than the cured epoxy itself.

Effect of IPA Evaporation Time

IPA evaporation time produced a secondary but measurable influence on bond performance. At both 60-grit and 80-grit surface finishes, increasing the evaporation time from 10 seconds to 30 seconds improved the observed failure mode from adhesive to mixed-mode failure.

However, when the surface was prepared using 120-grit sandpaper, extending the IPA evaporation period from 10 seconds to 30 seconds resulted in a reduction in bond quality from cohesive to mixed-mode failure. Although the exact mechanism requires further investigation, these results suggest that surface preparation had a greater influence on bond strength than IPA evaporation time within the range tested.

Overall, the transition from adhesive failure to cohesive failure demonstrates that optimizing surface preparation substantially improved the structural integrity of the fin-to-airframe bond.

13.4.5 Conclusion

Based on the results of the destructive bond testing, the following surface preparation procedure was selected for all flight-critical fin-to-airframe joints:

- Surface preparation using 120-grit sandpaper
- Cleaning with isopropyl alcohol (IPA)
- Approximately 10 seconds of IPA evaporation prior to epoxy application
- Standard epoxy application followed by a 24-hour room-temperature cure

This procedure was selected because it was the only tested configuration to achieve cohesive failure, demonstrating that the adhesive interface exceeded the internal strength of the cured epoxy. These results provide confidence that the selected bonding procedure will produce a robust fin-to-airframe joint capable of withstanding the aerodynamic and landing loads expected during flight.

14 PROPULSION

Goldeneye is propelled by Cesaroni Technologies 4864L2375-P motor, in a Pro75 4G casing.
The motor specifications and thrust curve can be found below.

Common Name	CTI L2375
Length	24.45 in
Total Weight	4161 g
Propellant Weight	2322 g
Average Thrust	2451 N
Peak Thrust	2798 N
Total Motor Impulse	4864 Ns
Specific Impulse (ISP)	214 s
Burn Time	1.90 s

Table 22: Technical Specifications of CTI L2375.

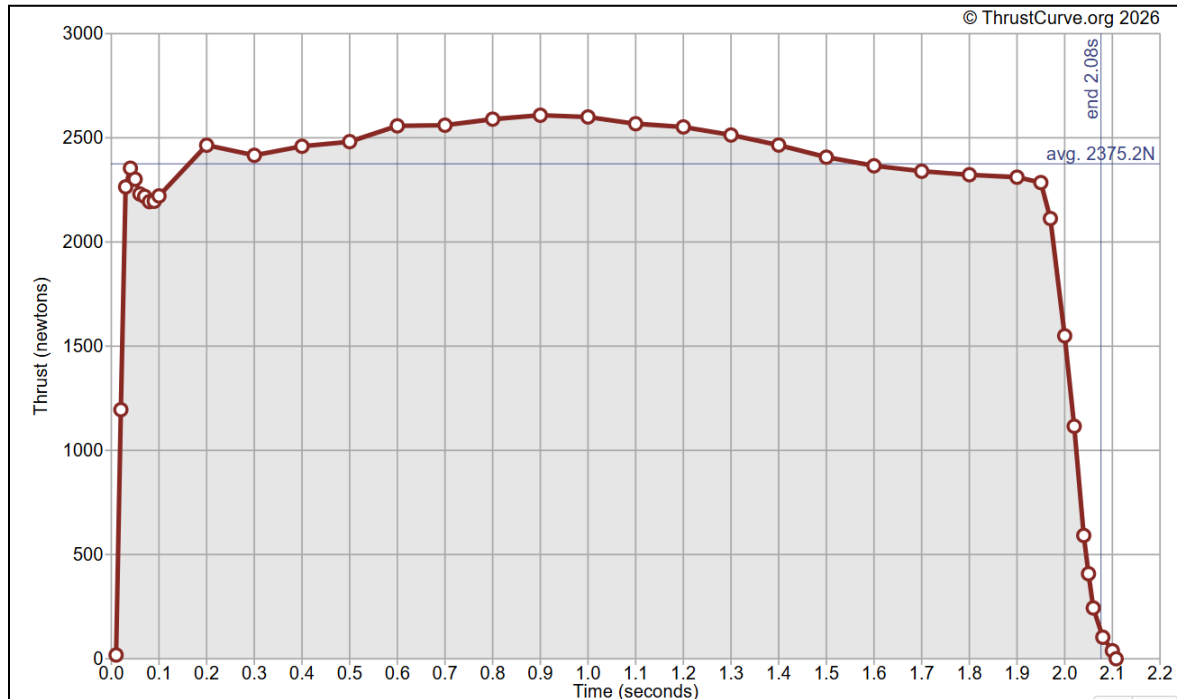


Figure 44: CTI L2375 Thrust Curve.

During the IDP stage of the project, the team intended to power the vehicle using a CTI M3400 motor. This configuration was not approved by competition officials due to the team's status as a high school team. The propulsion system was changed to the L2375 motor resulting in a major rework to the entire vehicle.

This motor was chosen for its short burn time (1.9 seconds) and relatively high thrust (2798 N). These factors were critical in ensuring that the vehicle has sufficient thrust to overcome wave drag and enter the supersonic flight regime.

15 THRUST STRUCTURES

The primary objective of the thrust structure is to securely transfer the forces imparted by expelled exhaust onto the airframe. Furthermore, considering its critical role in safety, the thrust structure features redundancy through separate primary and secondary thrust transfer mechanisms. Finally, to increase the vehicle's aerodynamic efficiency, a boat tail was integrated into the thrust structure.

The boat tail functions by reducing the wake behind the rocket, and thereby reducing the low-pressure zone that forms behind the vehicle. As such, it reduces the “base drag” and improves aerodynamic performance [43].

15.1 Detailed Design

The primary thrust transfer mechanism is the aluminum boat tail. It was designed such that thrust could reliably be transferred from the boat tail into the airframe, whilst also providing aerodynamic efficiency increases.

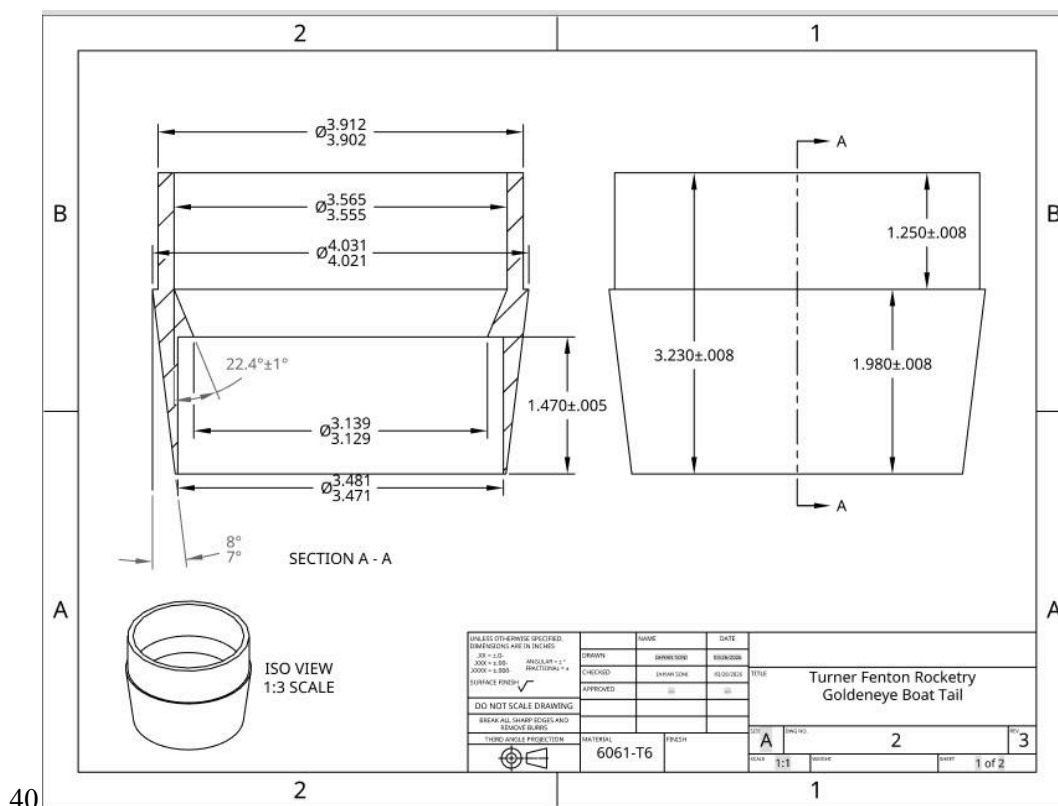


Figure 45: Boattail Engineering Drawing.

The boat tail, fabricated from 6061-T6 round stock, was manufactured to act as an aerodynamic and structural component. The thrust is transferred from the rocket motor, to the motor casing, to the Slimline Motor Retainer, to the boat tail, and finally into the airframe. This is the primary thrust transfer mechanism, and is designed with a factor of safety of 9.7, with detailed calculations present in Section 14.3.

The thrust transfer mechanism is designed such that the primary thrust path is sufficient to sustain all propulsion loads intended during flight. The secondary thrust path is designed purely for redundancy, as well as fulfilling the crucial role of centering the thrust, ensuring that the axis of thrust is aligned with the longitudinal axis of the rocket.

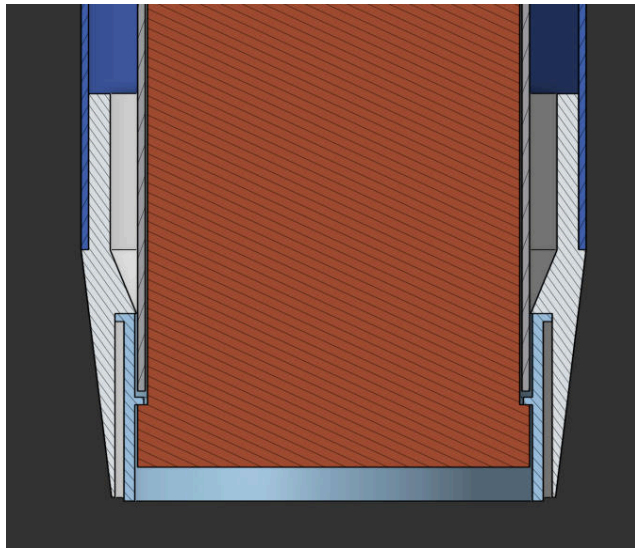


Figure 46: Boattail Cross Section

Primary Thrust Transfer Mechanism

Orange: Motor & Motor Casing
Light Blue: 75mm Slimline Motor Retainer
White: SRAD Boattail
Grey: Motor Mount Tube

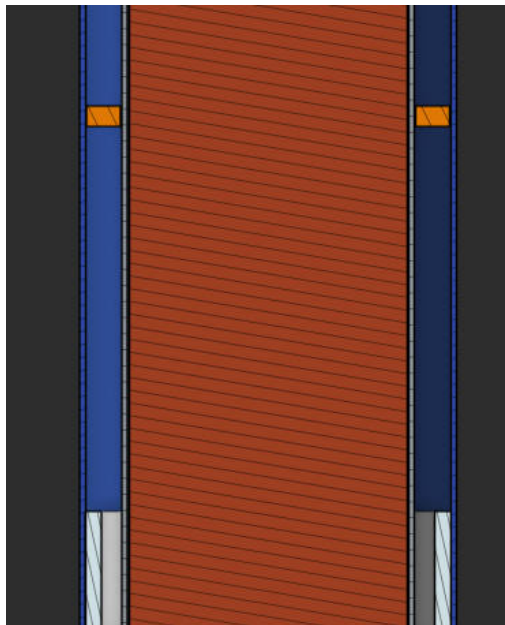


Figure 47: Motor Mount Cross Section

Secondary Thrust Transfer Mechanism

Orange: Motor & Motor Casing
Light Blue: 75mm Slimline Motor Retainer
Bright Orange: 1/4" Fiberglass Centering Ring
Grey: Motor Mount Tube

The various components within the thrust transfer mechanism are joined together with West System 105/206 structural epoxy. This adhesive was chosen due to its relatively high compressive, tensile, and shear strengths, making it ideal for high strength applications.

However, the joint between the Slimline Motor retainer, and the motor mount tube is made with JB-Weld. This is due to the fact that JB-Weld is more suitable for high-temperature environments. The motor casing is expected to be heated during the motor burn, and thus requires an adhesive that is capable of withstanding high heat.

15.2 Manufacturing Process

The boat tail is machined from 6061-T6, 4.500" round stock. This material was selected due to its machinability, availability, and relatively low cost. The boattail was machined at Arbalest Rocketry's machine shop.

The machining process, in a simplified format, is presented below. (All dimensions presented in inches)

1. **Rough Material Removal:** The 4.500 round stock of aluminum was mounted in a lathe. Using a standard cutting insert, the stock was reduced to 4.050
2. **Fine Material Removal:** The stock's outer diameter was thereafter reduced to 4.026 ± 0.005 and 3.907 ± 0.005
3. **Center Bore:** The center bore is made with a $\frac{1}{4}$ " pilot drill, and enlarged using progressively larger drill bits, until a $1\frac{1}{2}$ " center bore is created
4. **Inner Diameter:** Using an internal boring bar, the respective inner diameters (3.555, 3.129, 3.471) are made
5. **Internal Taper:** Using the compound slide, a 22.4° inner taper is made
6. **External Taper:** Using the compound slide, a $7\text{--}8^\circ$ outer taper is made

This component was machined with assistance from the team members at Arbalest Rocketry.

The motor mount tube is a 3" Carbon Fiber tube from Wildman Rocketry. The $\frac{1}{4}$ " Fiberglass centering ring is made from a $\frac{1}{4}$ " G10 Fiberglass plate. Using a hole saw and various sanders, the team reduced the diameters until the centering ring was of sufficient inner and outer diameter.

15.3 Analysis

Considering the significant loads that are expected upon the boattail, and the overall thrust structure, careful design assessment was conducted in order to maintain a high factor of safety.

To ensure structural survivability without adding unnecessary mass to the vehicle, a rigorous structural integrity verification was conducted. The evaluation methodology models maximum transverse aerodynamic bending moments and peak motor thrust forces. To capture the most severe structural loading environment experienced during the flight profile, all aerodynamic parameters were evaluated at maximum dynamic pressure (Max-Q). This aerodynamic milestone occurs at an altitude of 1,616 ft AGL and an ascent velocity of 443.3 m/s, where the free-stream dynamic pressure reaches a peak value of 111.48 kPa (with a calculated air density at Max-Q of 1.13 kg/m^3).

During ascent, a significant magnitude of aerodynamic bending loads may occur at non-zero angles of attack (α). To quantify the worst-case bending moments experienced across the boattail section, a modified Barrowman analysis was implemented. While traditional Barrowman analysis assumes subsonic flow regimes, using data from OpenRocket, this model was adapted using guidance from Nakka Rocketry [44].

The vehicle was modeled under a severe, conservative maximum angle of attack of 6° (0.1048 rad) across a reference frontal area of 0.00817 m^2 (4.02 in airframe reference diameter). Bending loads were resolved by calculating the normal forces generated by the forward nosecone and aft fin assembly, and subsequently multiplying them by their respective moment arms relative to the vehicle's center of gravity (located at 1.592 m from the nose tip):

- **Fin Normal Force:** Evaluated at 847.57 N acting at the fin center of pressure (2.084 m from the nose tip). This generates a lever arm of 0.492 m relative to the Center of Gravity, producing a local fin bending moment of 416.91 N·m.
- **Nosecone Normal Force:** Evaluated at 190.99 N acting at the nose center of pressure (0.292 m from the nose tip). This results in a lever arm of 1.300 m relative to the Center of Gravity generating a forward bending moment of 304.06 N·m.

Summing these individual moments provides the cumulative peak bending moment (M) acting along the vehicle body axis:

$$M = M_{Fin} + M_{Nose} = 416.91 \text{ N} \cdot \text{m} + 304.06 \text{ N} \cdot \text{m} = 720.98 \text{ N} \cdot \text{m}$$

Another key detail in selecting 6061-T6 for the boat tail is its excellent strength-to-weight ratio. Furthermore, aluminum's isotropic properties make the analysis of components far more convenient and accurate. To prevent structural failure, stress concentrations were analyzed at the thinnest machined cross-section of the boattail, defined by an outer radius (r_{outer}) of 0.04955 m (3.902 in diameter) and an inner radius (r_{inner}) of 0.04520 m (3.56 in diameter). This geometry yields a cross-sectional area (A_c) of 0.001293 m^2 and a section modulus (Z) of $3.221 \times 10^{-5} \text{ m}^3$, derived via the circular annulus relationship:

$$Z = \frac{\pi(r_{outer}^4 - r_{inner}^4)}{4r_{outer}}$$

Under worst-case flight loading, this thin-walled section must simultaneously withstand two primary stress components: bending stress (σ_b) induced by aerodynamic loading, and axial compressive thrust stress (σ_t) generated by the propulsion system. Propulsion loading was evaluated using the Cesaroni 4864L2375-P solid rocket motor, which delivers a baseline peak thrust of 2,798 N. To account for manufacturer performance tolerances and ignition pressure spikes, a 10% safety margin was incorporated, establishing a design thrust force (F_{thrust}) of 3,077.8 N.

The individual stress components acting on the cross-section are defined as follows:

$$\sigma_b = \frac{M}{Z} = \frac{720.98 \text{ N} \cdot \text{m}}{3.2215 \times 10^{-5} \text{ m}^3} = 22.38 \text{ MPa}$$

$$\sigma_t = \frac{F_{thrust}}{A_c} = \frac{3077.8 \text{ N}}{0.001293 \text{ m}^2} = 2.38 \text{ MPa}$$

This yields the maximum cumulative stress (σ_{total}) experienced by the material:

$$\sigma_{total} = \sigma_b + \sigma_t = 24.76 \text{ MPa}$$

To establish the Factor of Safety (FoS), this peak cumulative stress was evaluated against the compressive yield strength of Aluminum 6061-T6 (240 MPa), which is lower than its tensile yield strength (260 MPa).

$$FoS = \frac{240 \text{ MPa}}{\sigma_{total}} = 9.69$$

A structural Factor of Safety of 9.69 is deemed sufficient for this application.

Beyond structural yielding of the aluminum body of the boat tail, the mechanical joint interfacing the machined boattail with the primary carbon fiber airframe was verified. Although the majority of the thrust loads are designed to be transferred through the “lip” on the boat tail, the overlapping “coupler” of the boattail-airframe interface is designed to be sufficient to transfer expected loads into the airframe. This internal shoulder joint is permanently bonded using West Systems Epoxy (WSE), which provides a rated shear strength of 1,300 psi (8.97 MPa based on manufacturer technical data) [45].

To model shear load transfer across the bonding interface, the combined cross-sectional area of the mating shoulder was evaluated using an outer radius of 0.05105 m (4.02 in diameter) and an inner radius of 0.04521 m (3.56 in diameter). This establishes a total interfacial cross-sectional area (A_{shear}) of 0.001767 m². The maximum longitudinal force exerted across this shear plane (F_{max}) is driven by the peak cumulative stress acting over the joint cross-section:

$$F_{max} = \sigma_{total} \cdot A_{shear} = 24.76 \text{ MPa} \cdot 0.001767 \text{ m}^2 = 43.75 \text{ kN}$$

This figure was used to determine the required bond area between the boattail’s coupler section and the airframe, and therefore, the length that the boattail would extend into the airframe. Further, a Factor of Safety of 2.0 was applied to this surface joint. In other words, even without the presence of the crucial boattail “lip”, or even the redundant thrust path, the epoxied joint between the boattail and the aft of the airframe would be sufficient to transfer all potential flight loads. Thus, this joint would have to sustain a shear force of 87.49 kN

The minimum required bonding surface area (A_{req}) was calculated via:

$$A_{req} = \frac{F_{design}}{\tau_{wse}} = \frac{87.49 \text{ kN}}{8960 \text{ kPa}} = 0.00967 \text{ m}^2$$

Given the circular geometry of the boattail shoulder, this required surface area translates directly to a minimum shoulder insertion height (h_{req}):

$$h_{req} = 3.13 \text{ cm (1.23 in)}$$

This 3.13 cm insertion height requirement directly governed the final machining dimensions of the boattail mating shoulder. By integrating this insertion depth into the physical part geometry, the design guarantees secure load transfer from the propulsion assembly into the airframe without risking shear failure or joint delamination during flight.

16 PAYLOAD

16.1 Mechanical Design

Pitot tube design is dictated by the speed regime anticipated during the flight. A pitot has two sets of orifices, one which is referred to as the total port which collects data on the total pressure, and another set of orifices called the static ports which aim to capture the ambient or ‘static’ air pressure of the atmosphere during the flight. Refer to the diagram below:

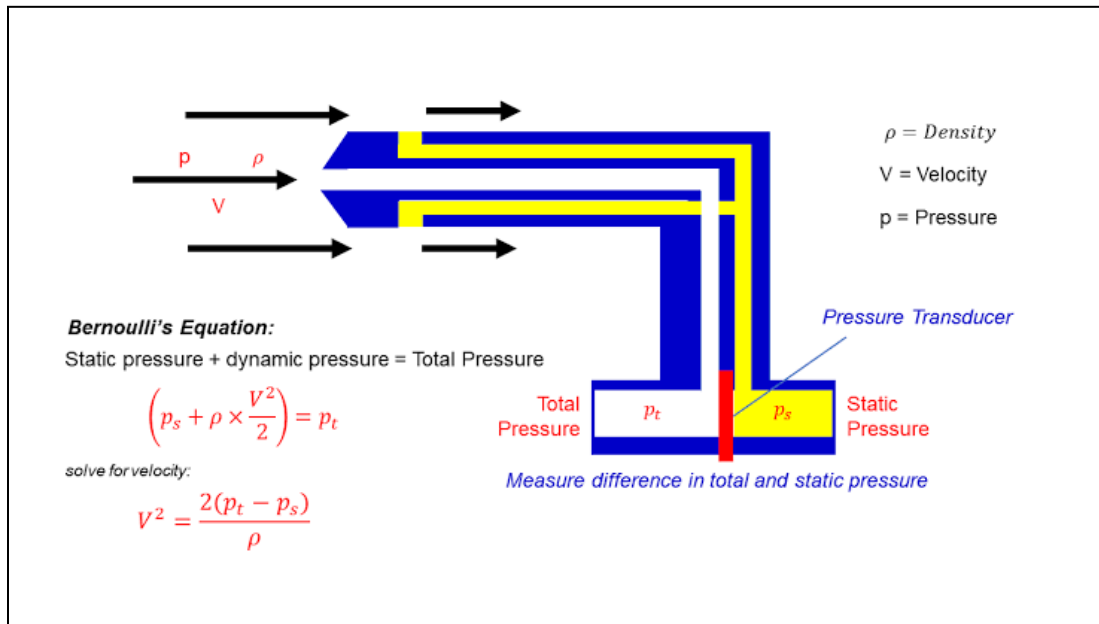


Figure 48: Pitot Tube Diagram.

In the supersonic regime, a bow shock forms ahead of the tip of the pitot tube. As the air passes through the bow shock, it decelerates to a subsonic speed resulting in a loss of total pressure. The flow is then brought to rest at the stagnation point located at the tip of the pitot tube. The bow shock and the resulting flow field disturb the local static pressure in the vicinity of the pitot tube. Therefore, the static ports must be positioned sufficiently far behind the tip, where the influence of the shock and the flow disturbance is negligible, allowing the measured static pressure to closely approximate true ambient air. Finding the placement of the static ports was therefore a key design choice that the team researched. Additionally, factors such as the length of the pitot tube, the inner and outer diameter, the shape of the total port opening, and the overall assembly were substantiated design decisions shaped by the team's research, and manufacturability constraints. The following table summarises the design choices made and how each were substantiated.

System Component	Design Choice	Substantiation
Pitot tube probe (body) diameter	0.56 inches	- Improved manufacturability achieved with more internal space. - A thicker pitot tube will be stronger against increased bending loads in supersonic flight.
Pitot tube opening diameter	0.336 inches	- Creates a d/D ratio of 0.60 with the probe diameter, where d = opening diameter, D = probe diameter. 0.60 ratio minimizes error in the total port pressure readings when AoA is present. More specifically, a ratio of ~ 0.60 shows no error in the pressure reading up to an AoA of $\sim 8^\circ$. It is anticipated that an AoA no greater than 6° will occur during flight. See the graph below.

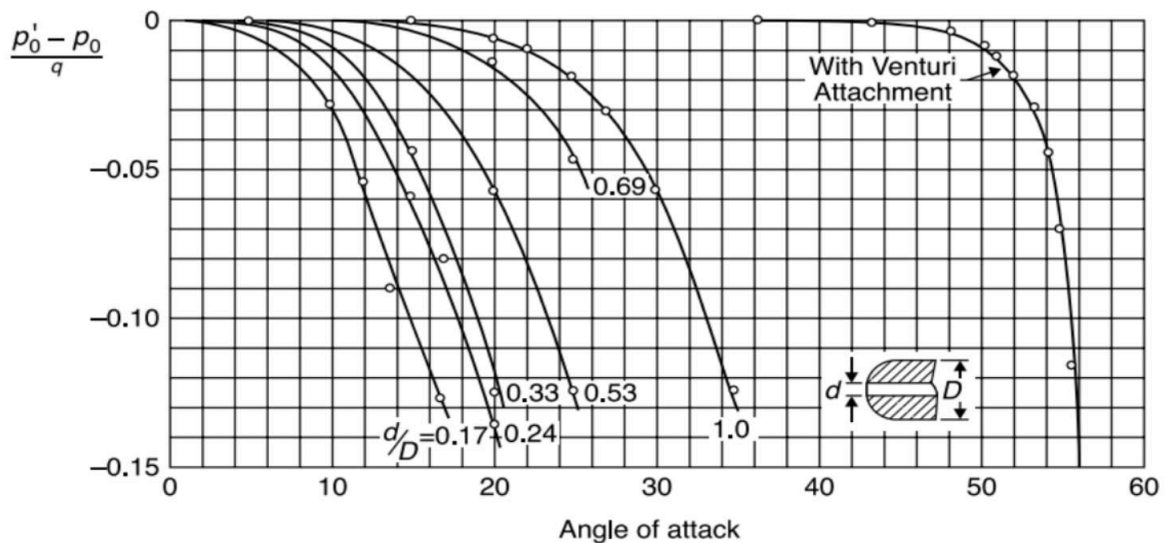


Figure 49: Diameter Ratio vs. Angle of Attack vs. Total Port Pressure Error [46].

Position of static ports	5 inches downstream from the tip of the pitot tube.	- The static ports must be positioned in undisturbed air far enough from the stagnation point in order to obtain true static pressure conditions. - Multiple reports suggest that the static ports should be positioned 10-20D (probe diameters) downstream from the stagnation point for a supersonic pitot tube [19] [23]. Given that the probe diameter is 0.56 inches, the team has chosen to place the static ports 5 inches from tip.
System Component	Design Choice	Substantiation

Opening shape	Tapered inwards to create larger opening diameter but smaller internal diameter of the tube	■ The shape chosen limits AoA error up to $\sim 20^\circ$ according to the graph below. ■ The inwards taper creates a large opening diameter, which respects $d/D = 0.60$, and a small internal channel diameter. ■ A small internal diameter was desirable as it makes the pitot tube more rigid and thus stronger against bending loads. ■ Furthermore, a conversation with a member of the Aerospace Team Graz (ASTG) revealed that a small internal diameter for airflow reduces lag in the pressure readings.
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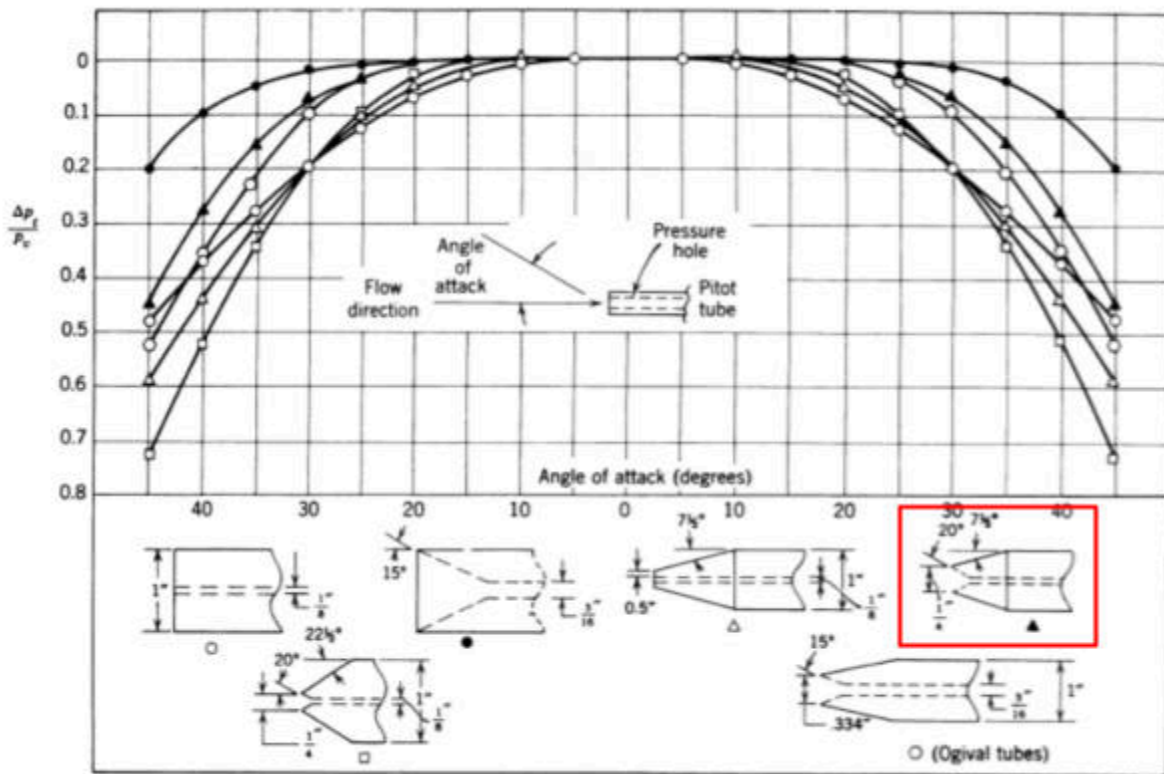


Figure 50: Pitot Tube Opening Shape Designs [22].

System Component	Design Choice	Substantiation
Length of pitot tube (exposed to freestream)	$10D + 10D = 20D = 10$ inches	- There are two points of the pitot tube that cause pressure disturbances along the stream. As discussed before, the stagnation point creates pressure disturbances referred to as head effects. Second the point at which the pitot tube transitions to the nose cone, the 'stem' also creates pressure disturbances. These are called stem effects. Observing the graph below shows the importance that the static ports are positioned to limit head and stem effects. As such the pitot tube itself must such that the static ports are far enough from the head and from the stem. These parameters thus determine the total length of the pitot tube. - The first 10D of the pitot tube are for the static ports placement, this reduces head effects. - The second 10D after static ports reduces stem effects, once again allowing for more accurate measurement of static pressure [46].

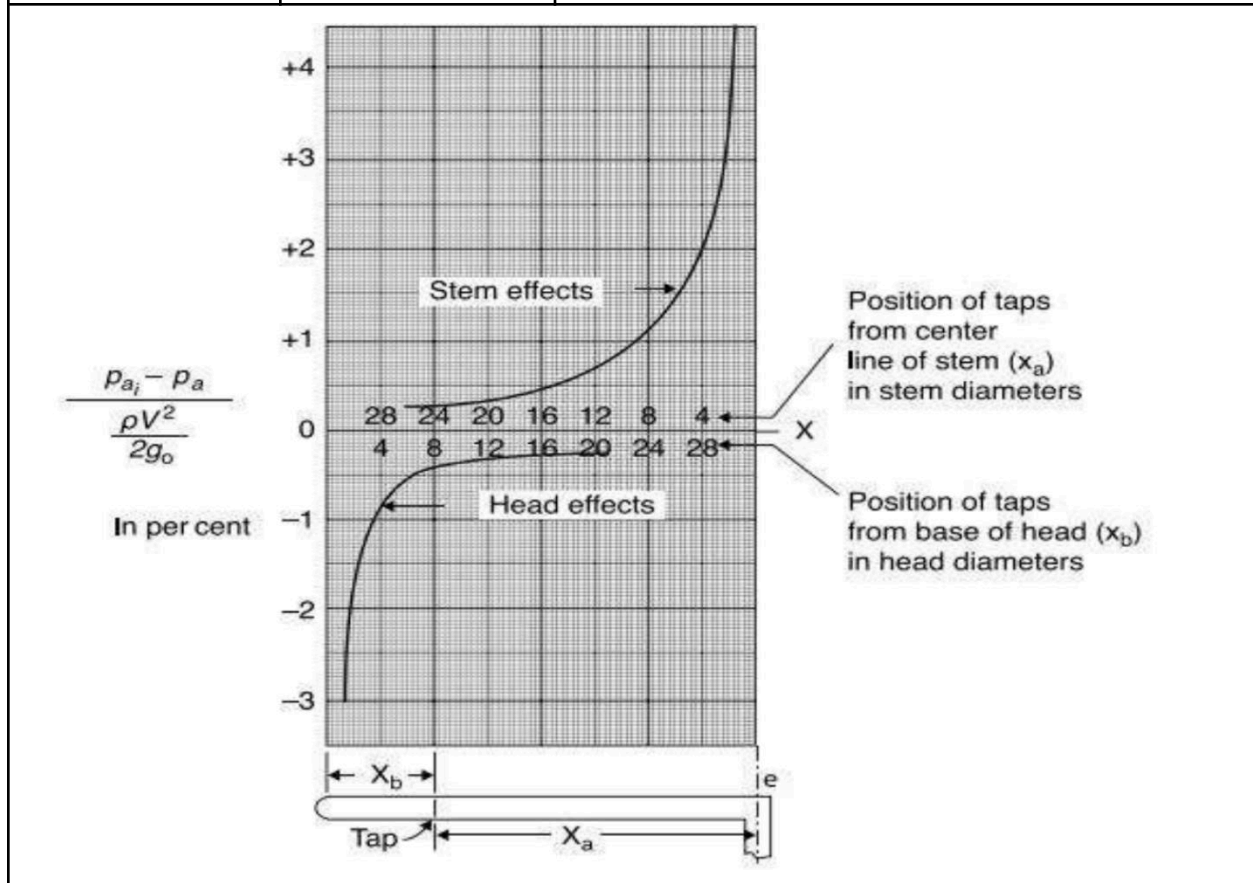


Figure 51: Pitot Tube Head and Stem Effects [19].		
System Component	Design Choice	Substantiation
Static port diameter	0.040 inches	■ As recommended on NASA 4184 [23].

Table 23: Pitot Tube Design Development.

16.2 System Design

The core obstacle in manufacturing the pitot tube was the inability to machine internal channels which would allow the respective ram and static airflows to reach the correct sensors. Further, the internal volume of the “static chamber” had to be minimized to prevent lag in the pitot tube data. As such, the design required several iterations. In other words, the most time-consuming aspect of this payload was designing for manufacturability.

Instead of manufacturing a single component with intricate internal channels, an alternate approach using three distinct parts was created. This split the assembly into the Pitot Tube Fwd Section, Pitot Tube Midsection, and the Pitot Tube Aft Section.

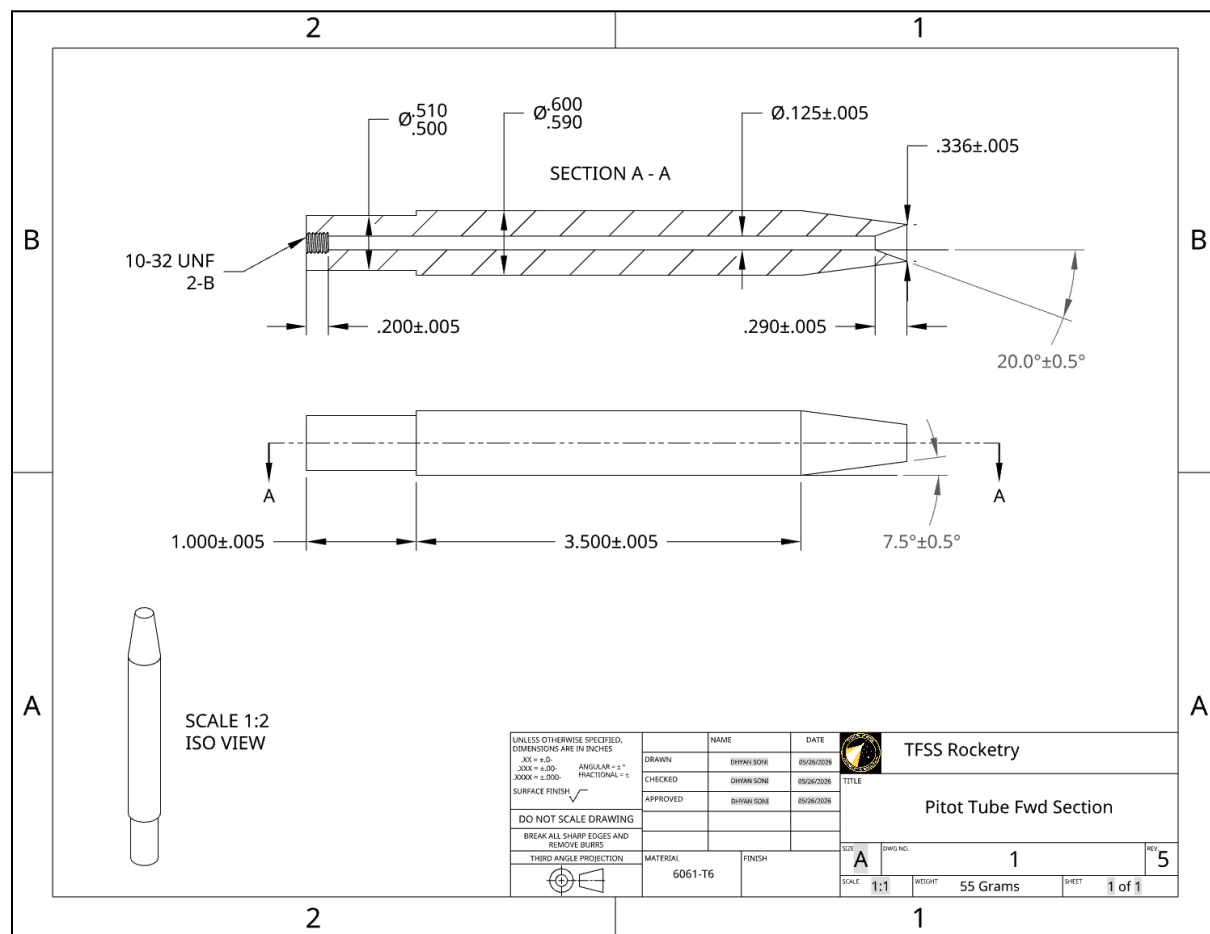


Figure 52: Pitot Tube Fwd Section Drawing.

The forward section features a 20-degree tapered opening and 1/8" inch internal channel for the flow of the ram air. The ram air is directed through the internal channel of the Forward Section. This component terminates in a #10-32 x 1/8" plastic hose barb. This barb connects to a section of Tygon tubing that bypasses the static chamber and delivers the ram air directly to the appropriate sensing element.

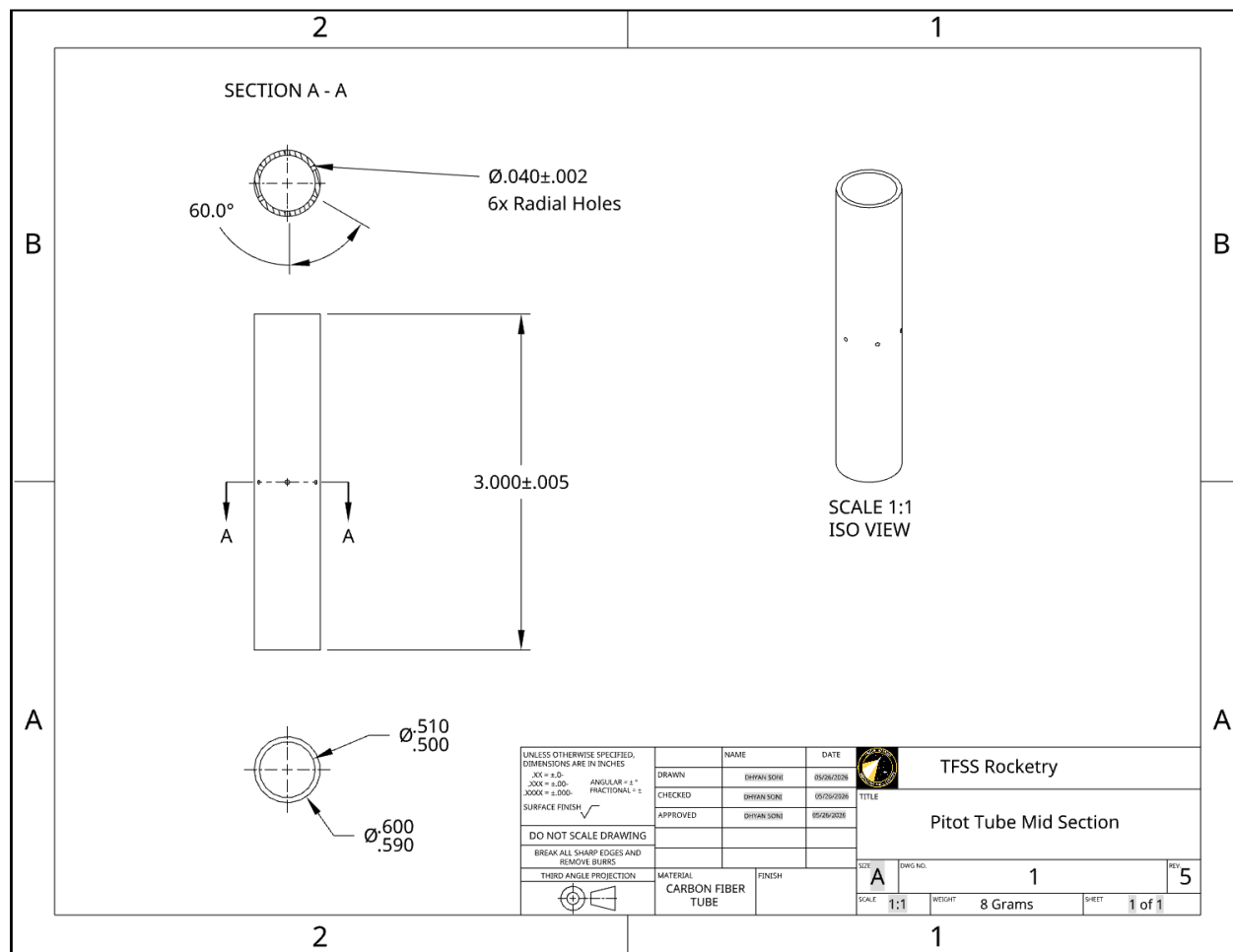


Figure 53: Pitot Tube Mid Section Drawing.

This component is comparatively simpler than the other two pitot tube sections, and as such, was manufactured in-house. Using a simple, 3D-printed jig, six (6x) radial holes were made in the carbon fiber tube. After the drilling operation, deburring the bores was critical, in order to prevent obstructions in airflow.



Figure 54: Static Port Drilling Operation

Thermocouple Placement

Although not included in the engineering drawing, a K-type thermocouple was included in the design. This thermocouple is critical for measuring ambient air pressure, which is important to convert between Mach number and true airspeed, as explained prior.

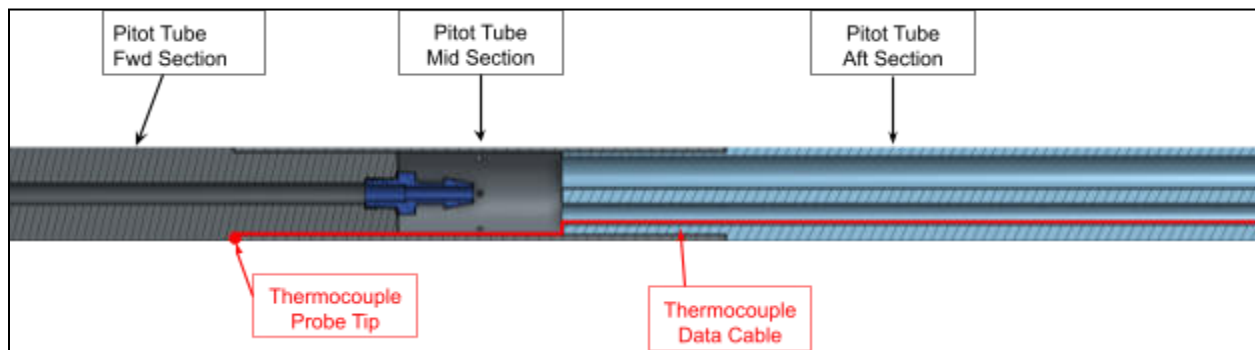


Figure 55: Thermocouple Placement

A small notch is filed into the carbon fiber tube (pitot tube midsection). The pitot tube thermocouple is placed in the overlap between the pitot tube midsection and pitot tube forward section, such that it is exposed to ambient air. The wires of the pitot tube are then routed through the $\frac{1}{8}$ " bore, and into the pitot tube computer.

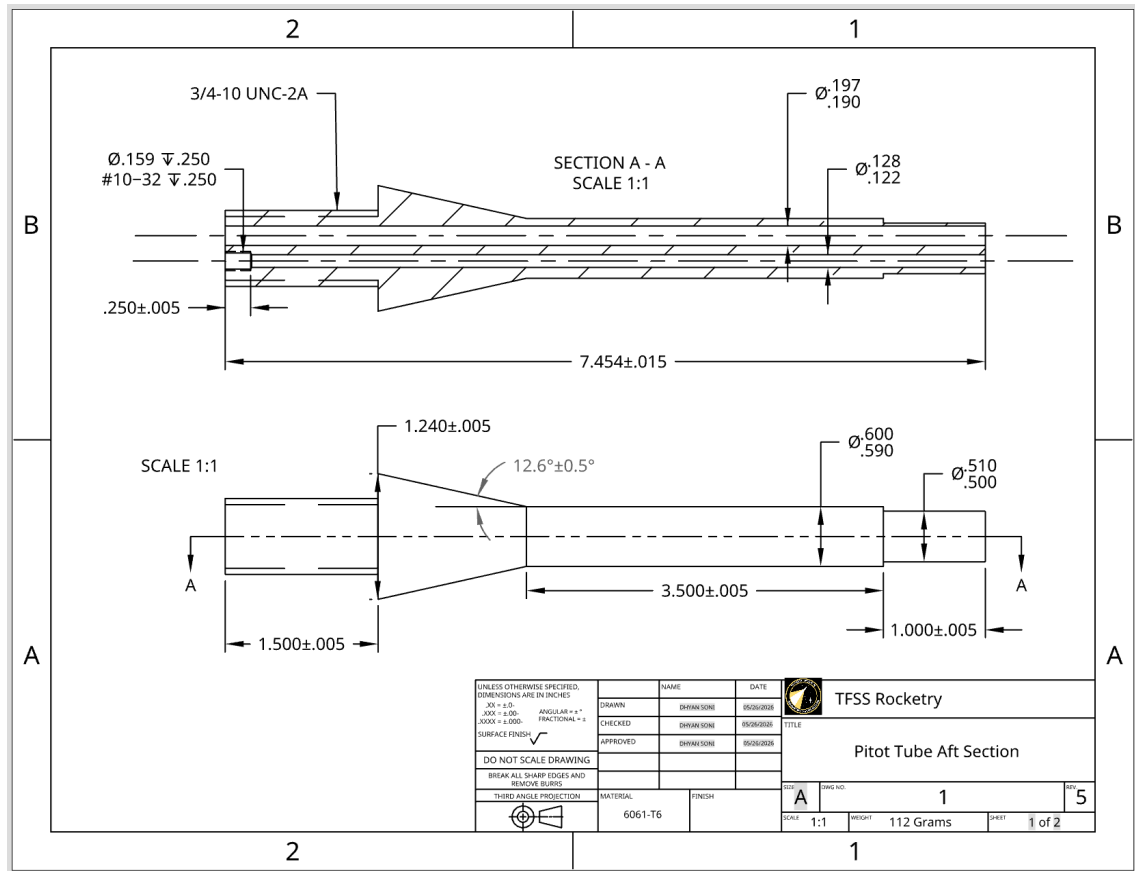


Figure 56: Pitot Tube Aft Section Drawing.

The pitot tube aft section was the most difficult component to manufacture. It requires two thin internal bores in a narrow component. This results in thin walls, the potential for chatter, and other manufacturing challenges. As a result, we collaborated with AOT Solutions to manufacture this part.

This component was machined from 1.375" 6061-T6 round stock. Using a standard manual lathe, the rough geometry was machined. Thereafter, fine finishing passes were made to arrive on the final outer geometry. Single point threading was used to manufacture the 3/4-10 UNC-2A thread. The most challenging phase of the design as mentioned prior, was the internal channels. To manufacture those features, a sinker EDM was used.

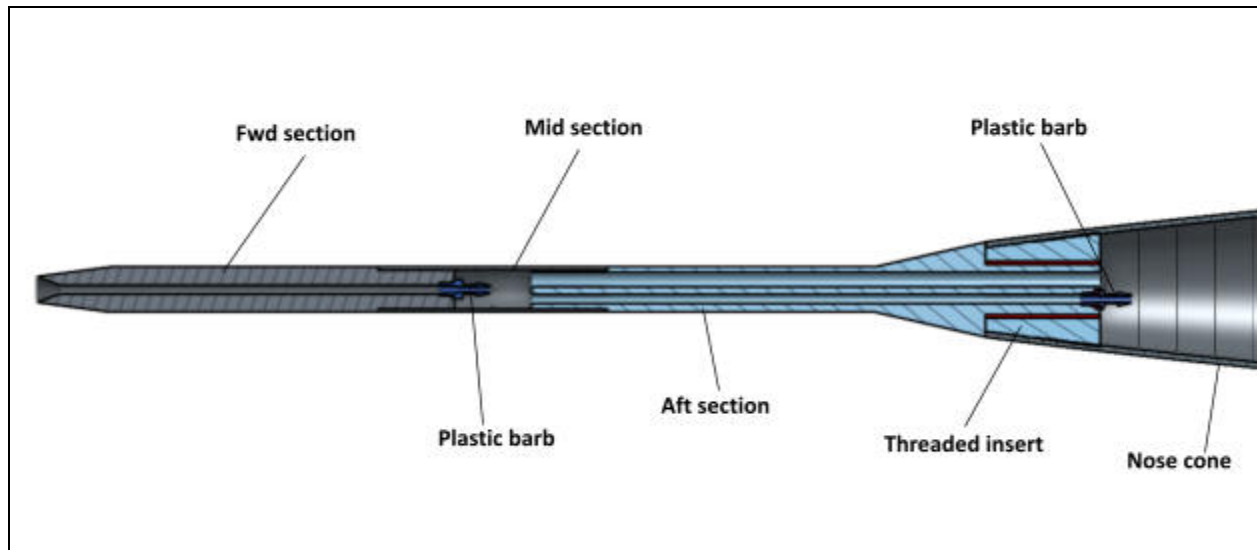


Figure 57: Pitot Tube Assembly Diagram

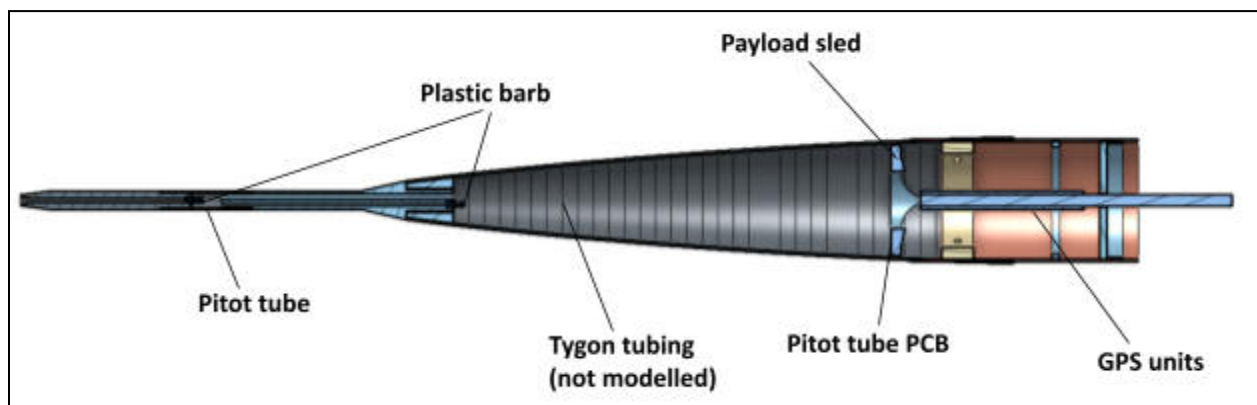


Figure 58: Full Pitot-Static System Assembly Diagram

The pitot tube is retained to the nose cone via a machined aluminum threaded insert. The aft section of the pitot tube has male threads and the insert has female threads creating a modular assembly. This insert has been epoxied to the inside of the nose cone. The assembly of the pitot tube is shown below, followed by a diagram of the complete system.

Each component (aft, mid, and forward sections), are epoxied together using standard, West System Epoxy (105 Resin, 206 Hardener).

16.3 GUEPARD I

The pitot tube board's requirements are to acquire airspeed data from sensors integrated into the pitot tube system. The board includes two pressure transducers, one dedicated for the ram port, and the other for static air, allowing for the retrieval of both pressures and the subsequent calculation of dynamic air pressure. Additionally, a type-K thermocouple and appropriate amplifier are also to be used for the acquisition of temperature data which in conjunction with dynamic air pressure is used to calculate airspeed.

16.3.1 MCU and Sensor Suite

Dubbed GUEPARD I, the pitot board is relatively simple compared to its DAQ counterpart. The board's MCU is a Teensy 4.0, selected for its powerful and fast 600 MHz ARM Cortex-M7 chip and capability to host several SPI peripherals.

The sensor suite incorporates the Honeywell pressure transducers for measuring pressure in ram and static ports. Based on calculated maximum values of 215 kPa and 98 kPa for ram and static pressure respectively, the HSCDANN060PASA5 60 PSI (413 kPa) and HSCDANN1.6BASA5 1.6 Bar (160 kPa) pressure transducers were selected. Apart from operating pressure ranges, both transducers are identical, communicating with the Teensy 4.0 via SPI protocol and operating on 5V logic. Though 3.3V variants exist, they could not be selected due to their unavailability.

Temperature is measured using an Adafruit 270 Type-K thermocouple and MAX31855 thermocouple amplifier. The latter also communicates with the Teensy 4.0 via SPI protocol and through its breakout board, can support 3.3V or 5V logic.

For data storage, the board uses breakout boards for the Winbond W25Q64JVSSIQ flash chip and a microSD connector, both communicating with the Teensy via SPI. Because it only supports 3.3V, the flash chip is powered by the Teensy 4.0's 3.3V output power pin and not directly from the regulator like the microSD reader.

16.3.2 Power and Logic Levels

Power Source

The board's power is derived from a 7.4V 2S LiPo battery and through a TSR 1-2450 switching regulator, supplying the sensors and MCU with 5V (the Teensy 4.0 accepts 5V power though it operates on 3.3V logic). The regulator was chosen as a compact, efficient, drop-in replacement for a linear regulator.

Component	Function	Estimated Max Current (mA)
Teensy 4.0	Main MCU (Running @ 600 MHz)	100.00 [37]
MicroSD Breakout	During Write Bursts	100.00
W25Q64JVSSIQ	64Mb Flash (Page Program/Erase)	25.00 [35]
BOB-12009	Level Shifter Pull-Up Network (All Channels Driven Low)	3.30 [41]
HSCDANN060PASA5	Ram Air Pressure Transducer (Digital SPI)	3.10 [39]
HSCDANN1.6BASA5	Static Air Pressure Transducer (Digital SPI)	3.10 [39]
MAX31855	Thermocouple Amplifier (Active Conversion)	1.50 [40]
Total 5V Load	Maximum Concurrent Draw	~236.00 mA

Table 24: Estimated Total Current Draw from GUEPARD I

With an output rating of 1 A, the TSR 1-2450 would see a maximum concurrent load of roughly 24% of its capacity based on these estimates. This margin allows the regulator to supply the full system load without a heatsink or risk of voltage droop, consistent with its rated full-load operation at ambient temperatures of up to +60°C. The estimate for the BOB-12009 represents the worst case in which all four channels are simultaneously driven low, causing both the 5V and 3.3V 10 kΩ pull-up resistors on each channel to conduct (0.83 mA per channel).

The regulator's minimum input voltage of 6.5 V sets the battery's usable discharge floor at approximately 3.25 V per cell, slightly above the standard 3.0 V per cell safe cutoff. Because the large majority of a LiPo cell's capacity is delivered above this voltage, the effective reduction in runtime is minimal. Additionally, because the W25Q64 flash chip is powered from the Teensy 4.0's onboard 3.3 V regulator rather than directly from the 5V rail, its draw was checked against that regulator's 250 mA external load rating, of which it consumes only 10%.

The battery selected is a 7.4 V 1350 mAh 2S LiPo, identical to that of KRAKENBANE I. While the TSR 1-2450 reaches efficiencies of up to 96%, a worst case of 85% is once again assumed for consistency with the KRAKENBANE I analysis. The total power and current drawn from the battery and its lifetime are calculated as follows.

Power Required by GUEPARD I:

$$P_{3.3V} = 5.0 V \times 0.236 A \approx 1.180 W$$

Total Power Drawn from Battery:

$$P_{Battery} = \frac{P_{5V}}{\eta} \approx \frac{1.180W}{0.85} \approx 1.388 W$$

Total Current Drawn from Battery:

$$I_{Battery} = \frac{P_{Battery}}{7.4V} = \frac{1.388 W}{7.4V} = 0.1876 A$$

Battery Lifetime:

$$Max Time = \frac{1350 mAh}{187.6 mA} = 7.20 Hours$$

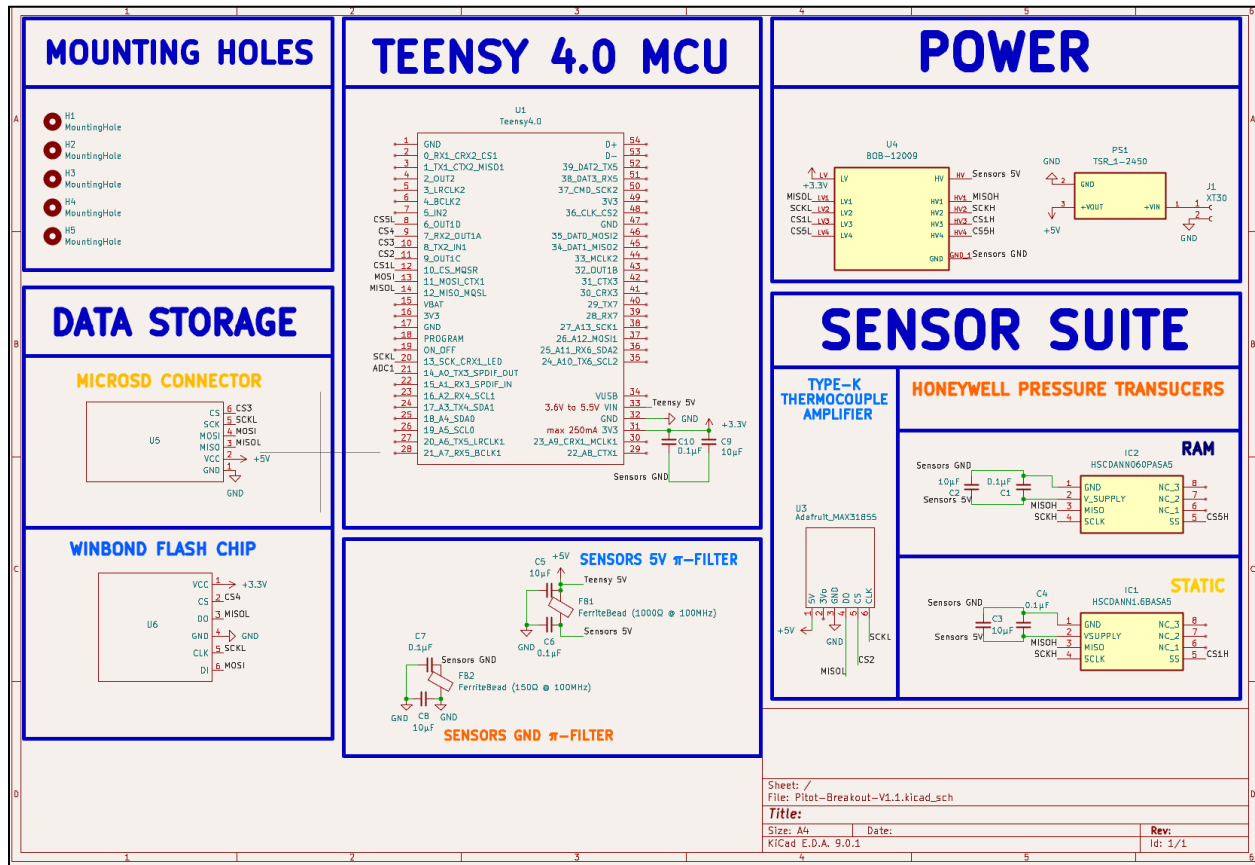
Thus, the battery's lifetime far exceeds the combined duration of pad standby and flight, even accounting for reduced capacity in cold conditions at altitude.

Logic-Level Shifter

Because the Honeywell pressure transducers both operate on 5V logic, a BOB-12009 bidirectional logic-level shifter was also incorporated to convert the high speed SPI lines from 5V to 3.3V and vice versa, and protect the Teensy 4.0's I/O pins.

For the Honeywell sensors, π -filters were also included to serve as additional protection against noise which could easily skew the transducers' data. The ferrite bead for the 5V π -filter is of a higher impedance of 1000 Ω at 100 MHz while that for GND only reaches 150 Ω at the same frequency. Both ferrite beads offer incredibly low DCR values of 75 and 45 m Ω respectively, and because the Honeywell sensors draw only a few mA of current, the voltage drop across the beads is well under 1 mV which is negligible. While the former bead has a higher impedance to more aggressively drop AC noise, the latter's is much lower because voltage developing across the bead risks bouncing the sensors' ground relative to the board's main GND.

Firmware and Validation



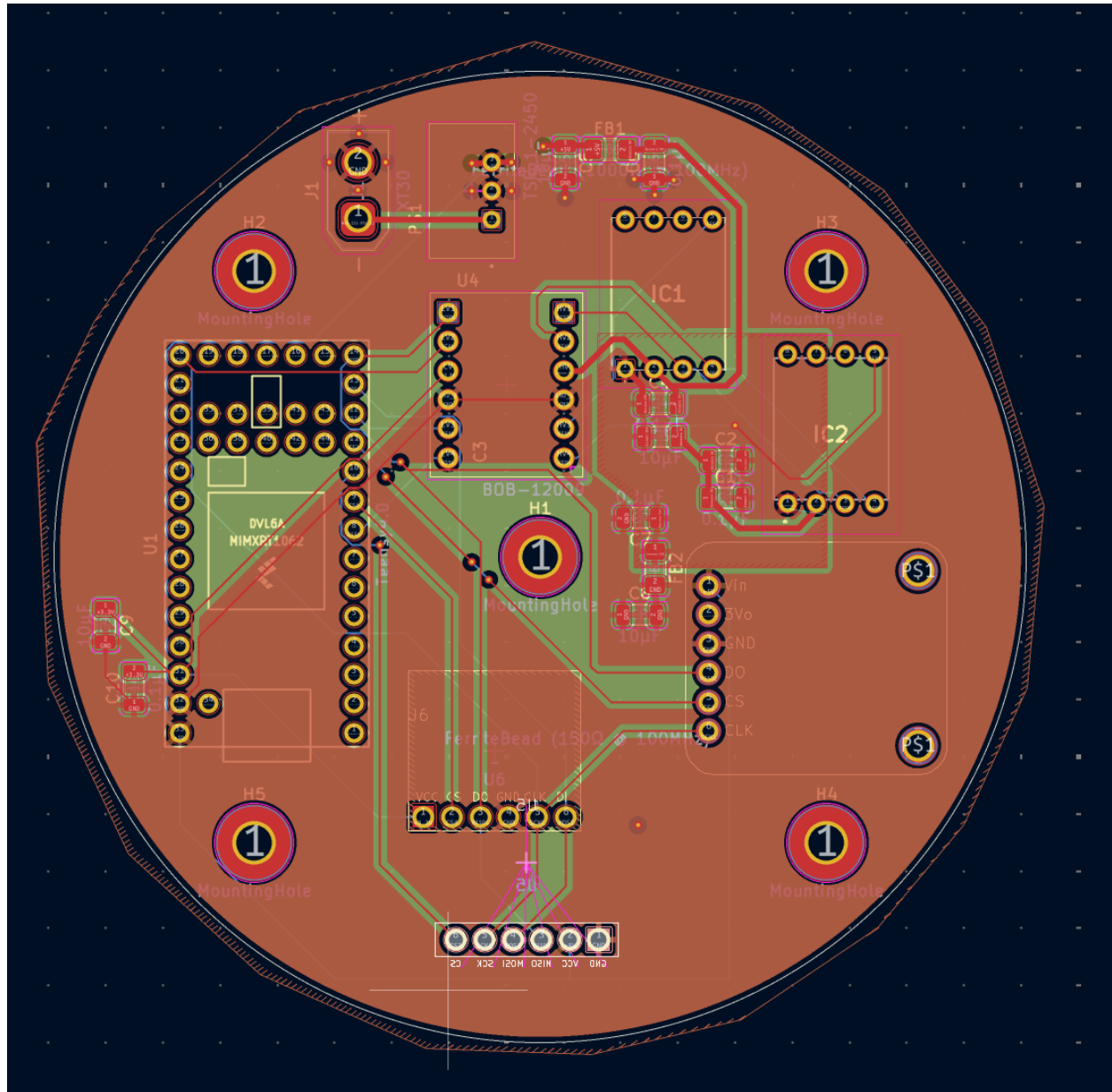


Figure 60: GUEPARD I PCB Layout

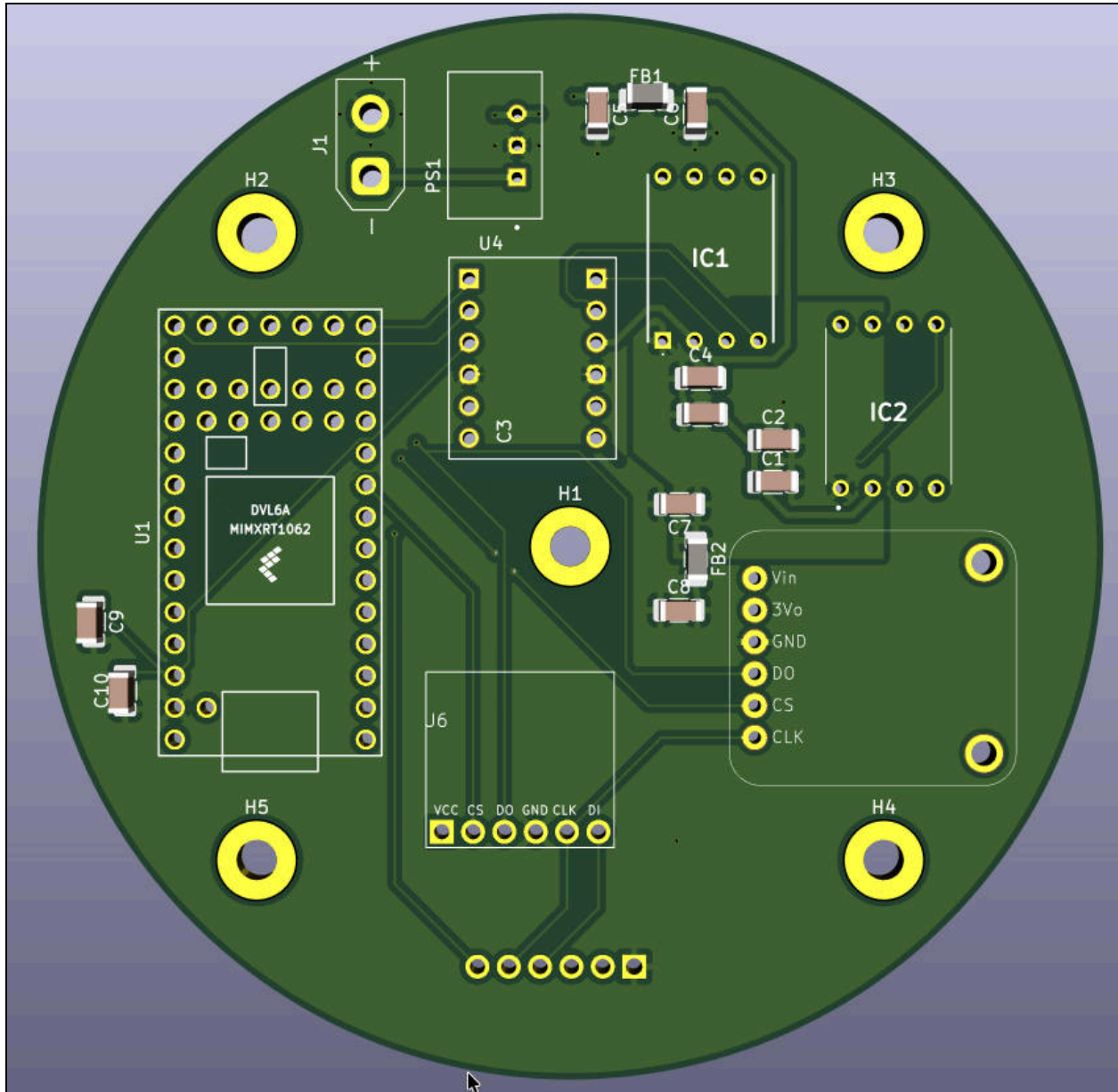


Figure 61: 3D View of GUEPARD I PCB

16.3.4 Component Testing, Challenges, and Future Considerations

Because the board mainly uses THT connections, the components were all bench-tested on a breadboard with the exception of the microSD connector. One issue that was discovered was the behaviour of the level-shifter on the Teensy's SPI line whereby it would never truly let go of the MISO line and allow peripherals other than the two pressure transducers to communicate with the MCU. To avoid using the SMT pads for simplicity, the immediate solution during testing was to assign the pressure transducers and in turn the level shifter, to random GPIO pins instead of the main SPI line and manually code them to behave as an SPI bus using a "bit bang" approach (See Appendix D). The bit bang SPI has a severely limited bus speed of up to 5 MHz compared to its real counterpart's 30+ MHz. While this would prove

detrimental for the storage devices, it is sufficient for the Honeywell sensors and is a change that may be applied to the PCB. With the exception of this, the performance of the components was nominal and proved the board's capability to collect data from the pressure transducers and thermocouple amplifier, and store it in the flash chip for post-flight analysis.

17 ELECTRONICS & SOFTWARE

17.1 COTS Avionics

The COTS Avionics, which refers to both COTS flight computers, as well as the COTS telemetry systems, are the electronics which will manage the successful flight and recovery of Goldeneye. In other words, SRAD Avionics will not be relied upon in order to recover the flight vehicle as TFRA simply does not have sufficient flight heritage with custom electronics.

17.1.1 Primary Flight Computer

The primary flight computer for Goldeneye is the Blue Raven by Featherweight Electronics. This flight computer was chosen for a variety of reasons. To begin, the Featherweight UI and Bluetooth connectivity greatly enhances the versatility and convenience of this flight computer. Testing parachute charges, checking battery health, and overall vehicle performance is made simple through the Bluetooth interface. Further, the advanced data collection offered by this product, chiefly accurate altitude, velocity, and acceleration, make this a highly-suitable option. This IMU-based (dead-reckoning) data is critical as a benchmark for performance of the pitot-static tube.

The battery sizing for the Blue Raven was also a crucial design choice. In 2025, TFRA experienced approximately 45 minutes of pad idle time prior to launch. Fortunately, the batteries were of sufficient size to keep the electronics powered-on during pad idle, and still function adequately during flight. Thus, a 1S (3.7V) 400 mAh battery was chosen to power the Blue Raven. Critically, the specific battery features no "over-current protection", meaning it lacks a Protection Circuit Module (PCM). This is crucial to flight success, as the current draw during firing of igniters is on the order of a few amps [12].

Measured Blue Raven Current Draw (Idle): 20-30 mA

Battery Capacity: 400 mAh

$$\text{Approximate Runtime} = \frac{400 \text{ mAh}}{25 \text{ mA}} = 16 \text{ hours}$$

16 hours is almost certainly an over-estimation of the idle time as the battery voltage will certainly decrease over time and eventually drop below the threshold required to safely trigger deployment. To guarantee the cell still has enough instantaneous power to cleanly fire the pyrotechnic e-matches at apogee and main deployment without causing a processor brownout, a standard 50% engineering safety margin should be applied. This reduces our dependable, safe pad idle time to a much more realistic 6 to 8 hours. This is sufficient.

17.1.2 Secondary Flight Computer

As a secondary flight computer, simplicity and reliability was prioritized over functionality. As such, a less capable, but simpler flight computer, the RRC3+ was utilized. The relative programming and wiring simplicity of this unit enabled it to safely recover Spectre, at Launch Canada 2025.

Although the RRC3+ offers a voltage input from 3.5 to 10 VDC [13], making it more suited towards 9V battery operation, the team decided to use a 3.7V 1S, 400 mAh Lipo instead. This decision was justified by the weight reduction from a standard 9V Battery (~45 grams) to approximately 15 grams with the Li-Po battery. Furthermore, using the same style of battery as the Primary Flight computer introduces a degree of redundancy, where there is only one type of spare battery that the team needs to carry. This battery also provides sufficient pad-idle time, and does not have current surge protection (a PCM).

Listed RRC3+ Current Draw (Stand-By): 6 mA

Listed RRC3+ Current Draw (Active & Beeping): 35 mA

Battery Capacity: 400 mAh

$$\text{Approximate Runtime} = \frac{400 \text{ mAh}}{21 \text{ mA}} \approx 19 \text{ hours}$$

The runtime estimate for this device is longer than the Blue Raven, due to the fact that it comparatively performs less functions than the Blue Raven. The LED and Buzzer are relatively-low current draw devices.

17.1.3 Primary & Secondary GPS Units

The primary and secondary GPS Units are identical, but are housed in different areas of the flight vehicle. Goldeneye will utilize two Featherweight GPS units in order to facilitate tracking of the vehicle. Akin to the Blue Raven primary flight computer, the Featherweight GPS was preferred due to its exceptional ease-of-use, as well as the sheer amount of data collection that is possible. GPS-based velocity estimates, although often inaccurate, may be able to supplement and act as a benchmark for the pitot-tube payload.

In terms of the battery sizing, a standard 3.7V 1S 400 mAh Li-po cell will be used, with PCM included. These batteries are ideal for this application, as the PCM reduces risk of overcurrent and PCB damage, as well as thermal runaway (especially in a hot, stationary environment, like a rocket interior on the pad). In terms of runtime, the 400 mAh battery enables the Featherweight GPS to operate for approximately 4.5 hours, which gives sufficient operation time for pad-idle time, flight, and recovery operations.

Listed Featherweight GPS Current Draw (Not transmitting): 60 mA

Listed Featherweight GPS Current Draw (While Transmitting): 112 mA

Listed Featherweight GPS Current Draw (Operating Average): 86 mA [14]

Battery Capacity: 400 mAh

$$\text{Approximate Runtime} = \frac{400 \text{ mAh}}{86 \text{ mA}} \approx 4.5 \text{ hours}$$

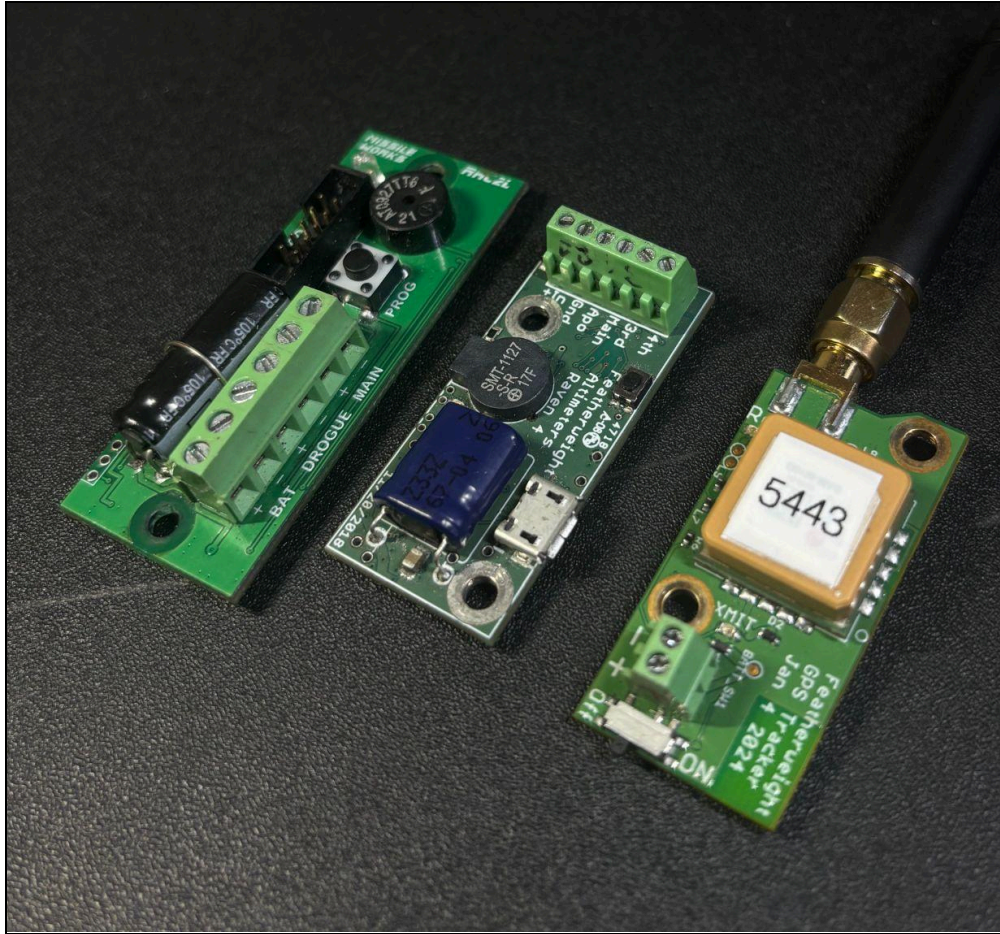


Figure 62: COTS Avionics Suite

17.2 SRAD Avionics

Since the Preliminary Design Review, the SRAD avionics architecture has been deliberately brought down from a complex multi-board, CAN-networked system to a narrower scope encapsulating two main systems: the Data Acquisition (DAQ) Flight Computer and the Pitot Tube Board dubbed KRAKENBANE I and GUÉPARD I, respectively. This drastic change was largely schedule-based, but also reflects the abandonment of higher-risk systems such as the air-brake board, and an overall CAN bus architecture which the team lacks experience designing. Since the remaining two SRAD boards being developed are isolated from each other because of their placement in the rocket, a power board was also deemed unnecessary, as each would have its own dedicated power source.

Neither of the two SRAD boards is flight-critical and is independent of the COTS avionics used for recovery. The DAQ Flight Computer is the primary data-acquisition platform, storing barometric, inertial, orientation, and position data and providing live telemetry. The pitot tube board is a much simpler board critical for the team's pitot tube experiment, incorporating ram and static pressure transducers and a thermocouple for the purpose of measuring airspeed. The rest of this section solely covers the DAQ board's design, as that of the pitot board is documented in the pitot tube section (see pitot tube section).

17.2.1 Power Management

Component	Function	Estimated Max Current (mA)
STM32F407VGT6	Main MCU (Running @ 168MHz)	100.00 [26]
XBP9B-DMSTB002	XBee 900HP (Tx @ 250mW Peak)	290.00 [31]
DM3BT-DSF-PEJS	MicroSD Slot (During Write Bursts)	100.00 [36]
SAM-M10Q-00B	GNSS Module (Continuous Tracking)	100.00
W25Q64JVSSIQ	64Mb Flash (Page Program/Erase)	25.00 [35]
MS560702BA03-50	Barometric Altimeter (Conversion)	1.50 [34]
LSM6DSV32XTR	6-Axis IMU (High-Performance Mode)	1.20 [33]
ADXL375BCCZ	High-G Accelerometer (Active Mode)	0.15 [32]
Total 3.3V Load	Maximum Concurrent Draw	~617.85 mA

Table 25: Estimated Total Current Draw from KRAKENBANE I

Power Source and Regulator Selection

Based on estimated values for maximum current draw, the AP63203WU-7 was selected as an appropriate buck converter for the system due to its simple PCB layout, small form factor, and the substantial margin it provides for the DAQ board in terms of current draw. Being rated for 2A, the system could only ever draw roughly 31% of the regulator's maximum capacity based on estimates. This is an extremely safe margin, ensuring the buck converter will not overheat or experience voltage droop during flight.

Additionally, a 7.4V 1350 mAh 2S LiPo battery is to be selected. For a current draw of 0.543 A, the AP63203WU-7 reaches a maximum efficiency of between 91% and 93%. Assuming a worst case scenario of 85%, the total power and current drawn from the battery and its lifetime are calculated as follows.

Power Required by KRAKENBANE I:

$$P_{3.3V} = 3.3V \cdot 0.61785A \approx 2.0389W$$

Total Power Drawn from Battery:

$$P_{Battery} = \frac{P_{3.3V}}{\eta} \approx \frac{2.0389W}{0.85} \approx 2.3987W$$

Total Current Drawn from Battery:

$$I_{Battery} = \frac{P_{Battery}}{7.4V} = \frac{2.3987W}{7.4V} = 0.3241A$$

Battery Lifetime:

$$Max\ Time = \frac{1350mAh}{324.1mA} = 4.165\ Hours$$

Thus, even with changing environmental factors such as temperature at high altitude, the battery's life time should be far more than sufficient to power the DAQ computer for the duration of the flight.

Buck Converter Circuit Components

Regarding selection for the other components in the power system, namely the input, output, and BST capacitors, and inductor as per the datasheet-based schematic, the following operating conditions were considered:

Parameter	Value	Source
Input voltage (V_{in})	6.0 V – 8.4 V (nominal 7.4 V)	2S LiPo, safe cutoff to full charge
Output voltage (V_{out})	3.3 V (fixed)	AP63203WU fixed-output variant
Switching frequency (F_{sw})	1.1 MHz	AP63203 datasheet
Max load current (I_{out})	0.7 A design ceiling	Maximum of 617.85 mA with headroom

Table 26: Buck Converter Operating Conditions

Regarding the capacitor selection, because the AV bay's temperature likely will not come close to exceeding 85°C, X5R ceramic capacitors were considered acceptable because they have higher capacitance density and better availability for high-voltage ratings at small packages like 0603 and 0805, in comparison to their X7R counterparts, which have greater temperature tolerance.

Inductor (L)

$$L = \frac{V_{out} \cdot (V_{in} - V_{out})}{V_{in} \cdot F_{SW} \cdot \Delta I_L}$$

$$V_{in(max)} = 8.4V$$

$$V_{out} = 3.3V$$

$$F_{SW} = 1.1 MHz$$

$$\text{Target } \Delta I_L = 0.3 \cdot 2.0 A \quad (30\% \text{ of buck converter's max current rating})$$

Calculations with the above equation yield $L \approx 3.0 \mu H$. A $3.3 \mu H$ inductor was selected for the DAQ's power system, which yields a ripple current of 0.55 or 27.5% of the maximum rating of 2 A.

Input Capacitor (CIN)

RMS input current:

$$I_{rms} = I_{out} \times \sqrt{D \times (1 - D)}, \text{ where } D = \frac{V_{out}}{V_{in}}. \text{ This is maximized at } D = 0.5, \text{ which occurs at } V_{in} = 6.6V, \text{ which is inside the 2S range:}$$

$$\text{At } V_{in} = 6.6 V: D = 0.500 \rightarrow I_{rms} = 700 \text{ mA} \times 0.5 = 350 \text{ mA}$$

$$\text{At } V_{in} = 7.4 V: I_{rms} = 348 \text{ mA}$$

$$\text{At } V_{in} = 8.4 V: I_{rms} = 342 \text{ mA}$$

Thus calculated maximum RMS input current is 350 mA. Multi-Layer Ceramic Capacitors (MLCC) were selected for CIN due to their inherently low Equivalent Series Resistance (ESR), which prevents problematic $I^2 R$ heating under high ripple currents. A standard 10 μF MLCC comfortably exceeds this 300 mA thermal dissipation requirement by design [24]. To account for DC bias derating at the maximum 8.4V LiPo input, a capacitor with a 35V rating was selected.

Output Capacitor (COUT)

The output capacitor minimizes voltage ripple, reducing overshoot or undershoot voltage during load transients. This is especially important for the DAQ due to the behaviour of the Xbee module, which draws 229 mA of current when transmitting data and 44 mA while receiving it. Output voltage ripple as calculated by $V_{out} = \Delta I_L \cdot ESR$, is naturally very small and ignored as a deciding factor because the *ESR* (Equivalent Series Resistance) of MLCCs capacitors is inherently a small value. As such, meeting load transient requirements was prioritized.

$$COUT > \frac{L \times (I_{out} + \frac{\Delta I_L}{2})^2}{(\Delta V + V_{out})^2 - V_{out}^2} \text{ (Equation X)}$$

The system experiences significant load steps due to the XBee transceiver's burst transmissions. To restrict the maximum output overshoot ΔV to 50 mV during a sudden load drop, a minimum effective capacitance of 3.28 μF is required, as calculated based on Equation X.

To compensate for the significant DC Bias capacitance derating inherent to MLCCs operating at 3.3V, the nominal capacitance must be scaled up. The datasheet's standard application of two 22 μF X5R ceramic capacitors was selected to ensure the effective capacitance remains well above the 3.28 μF threshold under all flight conditions, ensuring a highly stable 3.3V rail for the DAQ's logic and sensor suite.

Other Components

For the BST capacitor, a standard 100 nF ceramic capacitor was selected as per datasheet recommendations. Additionally, to protect the flight computer from reverse polarity, a Schottky diode was also implemented. The SS34 is rated for 3A of current and 40V of reverse voltage. Moreover, its stated 550 mV drop in forward voltage is negligible considering the DAQ's power supply is a 7.4V 2S LiPo battery.

To connect the power supply to the DAQ board, an XT30 connector was selected for its robustness to minimize the possibility of disconnection during flight.

Part (of Power System)	Part Name	Spec
CIN	GRM21BR6YA106KE43L	10 μF 35V X5R 0805
COUT	KGM21AR51E226MU	22 μF 25V X5R 0805
BST	C0603C104K5RACTU	100nF 50V X7R 0603
L	SPM5030T-3R3M-HZ	26.8m Ω 3.2A 3.3uH

Part (of Power System)	Part Name	Spec
Schottky	SS34	3A 40V 500uA@40V 550mV@3A

Table 27: Switching Regulator Circuit Component Selection

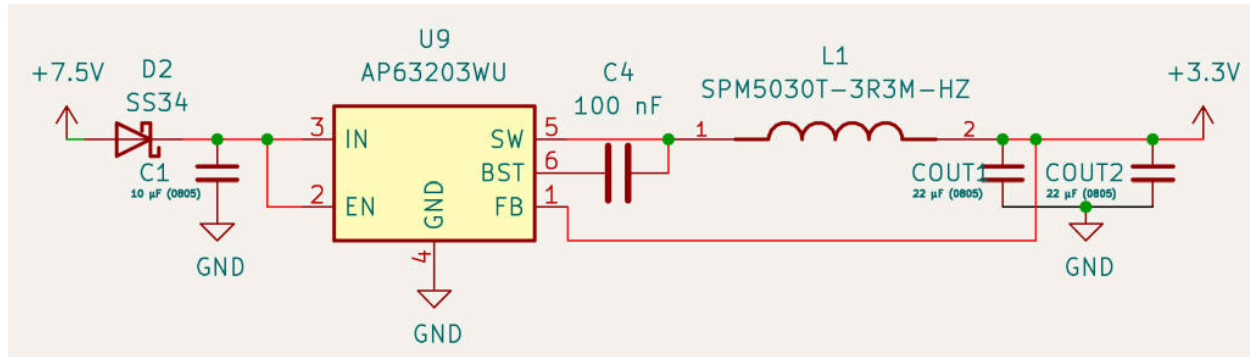


Figure 63: Buck Converter Minimum System

17.2.2 KRAKENBANE I Core System and Sensor Suite

The requirements for our DAQ flight computer were a system that would provide live telemetry regarding orientation, inertial, pressure, temperature, and position data, and store it securely with a dual-redundant system for post-flight analysis. The microcontroller unit (MCU) selected for the board is the STM32F407, which has a 168 MHz maximum clock speed, 1 MB of Flash memory, and 192 KB of RAM. Originally, the board was to include a more powerful STM32H753 with a 480 MHz clock speed and room for more peripherals; however, for simplicity and time's sake, the STM32F407 was deemed a better choice because of its better and more abundant online support and because the scope of our mission would not take advantage of the former's full capabilities.

Regarding sensors, the BNO055 inertial measurement unit (IMU), due to its unavailability, was originally replaced with a newer Bosch sensor, the BNO086, but that too was ultimately sidelined for a 6-axis (accel + gyro) IMU, the LSM6DSV32XTR. The replacement sensor lacks magnetometer data, which is not critical for the project's scope. In turn, it offers advantages such as an accelerometer that can measure up to 32g, well above the simulated peak force on the AV bay, and a much simpler PCB layout than the BNO086. The BMP390 barometric pressure sensor, also unavailable, was similarly replaced with a BMP581, and then an MS560702BA03-50, mainly based on JLCPCB availability, but also due to its widespread use on COTS flight computer applications. The high-g accelerometer selected for the board is the ADXL375 due to its 200g maximum. The IMU, barometric pressure sensor, and high-g accelerometer each communicate with the MCU via the I2C protocol, and each connects to the STM32 through a different bus. This was done to isolate each sensor as a failure point, preventing one from disabling the rest.

For data storage, a DM3BT-DSF-PEJS microSD card connector and a W25Q64JVSSIQ flash chip were both implemented for dual redundancy. Initially, using the STM32F407's flash memory was also considered for storage; however, this may be disregarded if it adversely affects the board's performance or brings unneeded complexity in terms of code. Both storage peripherals communicate with the MCU via the SPI protocol, and each through its own bus.

Finally, the GNSS unit selected for the board is the SAM-M10Q-00B, which is to connect with it via its own dedicated breakout board, through JST-GH 6 pin connectors. A similar connector is also to be used to connect with the XBP9B-DMSTB002, selected for live telemetry. Both peripherals use the UART protocol.

The wiring of all peripherals in this section is relatively simple and based on datasheet specifications.

17.2.3 Microcontroller, Support Circuitry, and Debugging Peripherals

The STM32F407VGT6 provides the DAQ project with a platform capable of handling real-time sensor fusion, filtering, and logging with ample margin. Because the MCU hosts multiple independent I2C, SPI, and UART/USART buses, the aforementioned fault isolation strategy of providing each peripheral its own dedicated bus is possible. The MCU's support circuitry closely follows ST's application note recommendations so as to avoid the common causes of unreliable design that power integrity and clocking shortcuts.

Decoupling and Bypass Capacitors

Each of the microcontroller's VDD (power) pins is locally decoupled via a 100 nF X5R ceramic capacitor and a single 10 μ F capacitor for bulk decoupling. Furthermore, the VDDA (analog power) pin also includes a ferrite bead for additional filtering of noise on the supplied 3.3V. [27] The ferrite bead selected has a low DCR (DC resistance) of 500 m Ω and a high impedance of 600 Ω at 100 MHz frequency. Because the VREF pin takes in very little current, this allows the voltage drop caused by the bead impedance's resistive component to be negligible, while allowing any significant noise to be filtered and dropped through its reactive component.

For both VCAP capacitors, the requirement for each to have an ESR of less than 2 Ω [27] is easily met by any 2.2 μ F ceramic capacitor, which, as previously stated, naturally has ESRs much lower than this. X5R capacitors rated for 50V were selected for small decoupling to minimize change in capacitance value with voltage. Due to limitations in availability, capacitors rated for 25V and 10V were selected for 1 μ F and 10 μ F bulk decoupling, respectively (for clarification, the latter is used with the ADXL375). The application note is followed for NRST, BOOT0, and VREF pins.

Crystal Oscillator and Load Capacitor Selection

The MCU support circuitry does not include a low speed external crystal as recording real clock time was deemed unnecessary for the project. However, it does include a high speed external (HSE) crystal as the STM32's internal RC oscillator can prove unreliable due to inaccuracy and clock drift caused by temperature changes at higher altitudes. To select the crystal, transconductance and gain margin were critical deciding factors as dictated by ST's crystal oscillator application note. According to the note, gain margin is represented by $gain_{margin} = \frac{g_m}{g_{m_{crit}}}$ and should be greater than 5. The STM32's Transconductance, g_m , is a datasheet value for the MCU, and in the DAQ's case, is equivalent to 5 mA/V. The crystal's minimum transconductance, $g_{m_{crit}}$ is calculated by the following equation.

$$g_{m_{crit}} = 4 \cdot ESR \cdot (2\pi F)^2 \cdot (C_0 + C_L)^2$$

Here for the crystal in question, C_0 = shunt capacitance, F = nominal frequency, C_L = load capacitance, and ESR = equivalent series resistance. [24]

For the crystal selected, the NX3225GA-16.000M-STD-CRG-2m the following values are provided.

C_L	8 pF
F	16 MHz
ESR	80 Ω

Table 28: Selected Crystal Oscillator Values [28]

Because a value for C_0 was unavailable from the datasheet, that of a crystal with a similar package, being 1.2 pF was used as a baseline estimate and the point of failure (in meeting ST's requirements) was calculated from there. [29]

C_0 assumed	$g_{m_{crit}}$	$gain_{margin}$
0 pF (impossible best)	0.207 mA/V	24.2
1.2 pF (NDK nominal [1])	0.274 mA/V	18.3
1.5 pF (NDK worst [1])	0.292 mA/V	17.1
3 pF	0.391 mA/V	12.8
5 pF	0.547 mA/V	9.1
7 pF	0.728 mA/V	6.9
9.6 pF — failure point	1.0 mA/V	5.0

Table 29: Calculated Values for $g_{m_{crit}}$ and $gain_{margin}$

To have an unsatisfactory value for $gain_{margin}$, C_0 would have to be impossibly high for the package's limitation, proving the NX3225GA-16.000M-STD-CRG-2 to be a viable HSE crystal option.

Regarding the two load capacitors, denoted C_{L1} and C_{L2} , the following formula relates load capacitance to their individual Capacitances assuming $C_{L1} = C_{L2}$.

$$C_L = \frac{C_{L1}}{2} + C_S \Rightarrow C_{L1} = 2 \cdot (C_L - C_S) \quad [24]$$

C_S represents stray capacitance and is estimated based on the MCU's datasheet I/O pin capacitance value of 5 pF [26] and a rough estimate of 1 pF for trace capacitance. Because the oscillator is connected to two pins, OSC_IN, and OSC_OUT, the total 6 pF estimated for each pin, now with both pins being connected in series, yields a total stray capacitance of 3 pF at most.

Using $C_S = 3 \text{ pF}$ and $C_L = 8 \text{ pF}$ from the crystal's datasheet with Equation Y, we have $C_{L1} = 10 \text{ pF}$.

Again, to be aware of the effect variation in value would have on our frequency, a pulling sensitivity of 19.1 ppm/pF yields the following table where $error (ppm) = 19.1 \cdot (\frac{C_{L1}}{2} + C_S - 8 pF)$.

C_S (pF)	$C_{L1} = 5.6 pF$	$C_{L1} = 6.8 pF$	$C_{L1} = 8.2 pF$	$C_{L1} = 10 pF$	$C_{L1} = 12 pF$
2.0	+61	+50	+36	+19	0
2.5	+52	+40	+27	+10	-10
3.0	+42	+31	+17	0	-19
3.5	+33	+21	+8	-10	-29
4.0	+23	+12	-2	-19	-38
5.0	+4	-8	-21	-38	-57
7.0	-34	-46	-59	-76	-96

Table 30: Estimated Error with Variation in Values for Load and Stray Capacitance

The table reveals that even an impossible worst case scenario would result in a drift of 96 ppm or 0.0096%, virtually insignificant to the operation of the computer or the accuracy of data and physics calculations.

Ultimately, a 10 pF C0G ceramic capacitor was selected. With an extremely low temperature coefficient and 50V voltage rating, it would provide a consistent load capacitance for the crystal oscillator circuitry.

Programming and Debugging Peripherals

For programming and debugging, a SWD/JTAG 10 pin layout was selected with a 2x5 male pin header to connect an ST-Link V3 to the STM32. The board also includes LEDs and an electromagnetic magnetic buzzer, with an N-Channel MOSFET and Schottky diode as supporting circuitry.

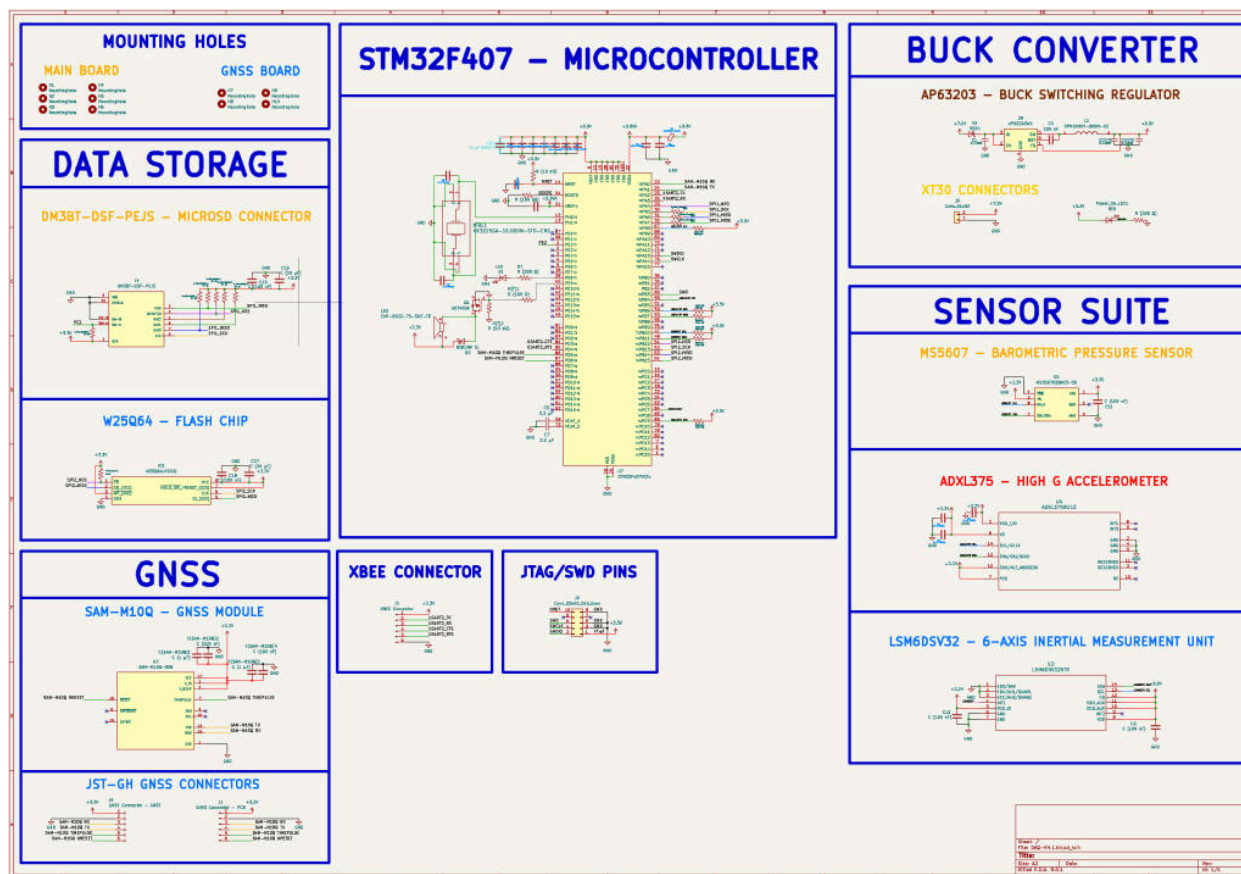


Figure 64: KRAKENBANE I KiCAD Schematic

17.2.4 PCB (Printed Circuit Board) Design Layout

The PCB has a form factor of roughly 1.5 by 4 inches. It utilizes a four-layer stackup, including top and bottom signal planes and dedicated ground and 3.3V power planes, to which power pins are connected. Because JLCPCB is relied on for soldering, components are allowed to be placed quite close to each other, following the company's guidelines regarding minimum margin between them. Component placement follows ST's guidelines with decoupling capacitors placed as closely as possible to power pins. The crystal oscillator is located close to its two pins and includes a guard ring for protection against noise.

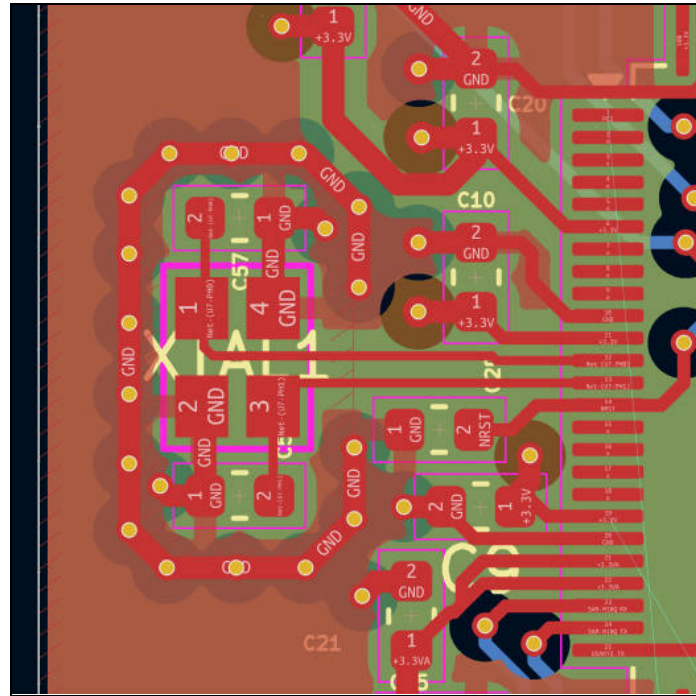


Figure 65: Crystal Oscillator and Guard Ring PCB Layout

Since the board is long, the buck converter and its circuitry are placed on the opposite side from the MCU and most of its peripherals to isolate the latter and peripherals from electrical switching noise that could corrupt sensor data. The top layer also takes advantage of pours for the regulator system and is filled entirely with a large GND pour, providing traces with quick return paths and acting as a further shield against electromagnetic interference.

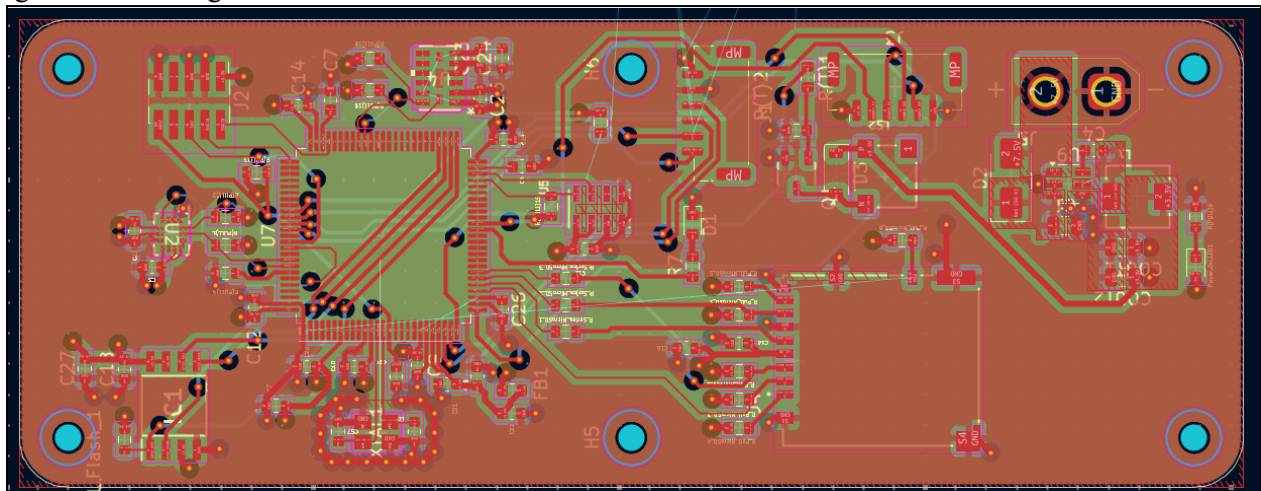


Figure 66: Sideways view of KRAKENBANE I PCB Layout

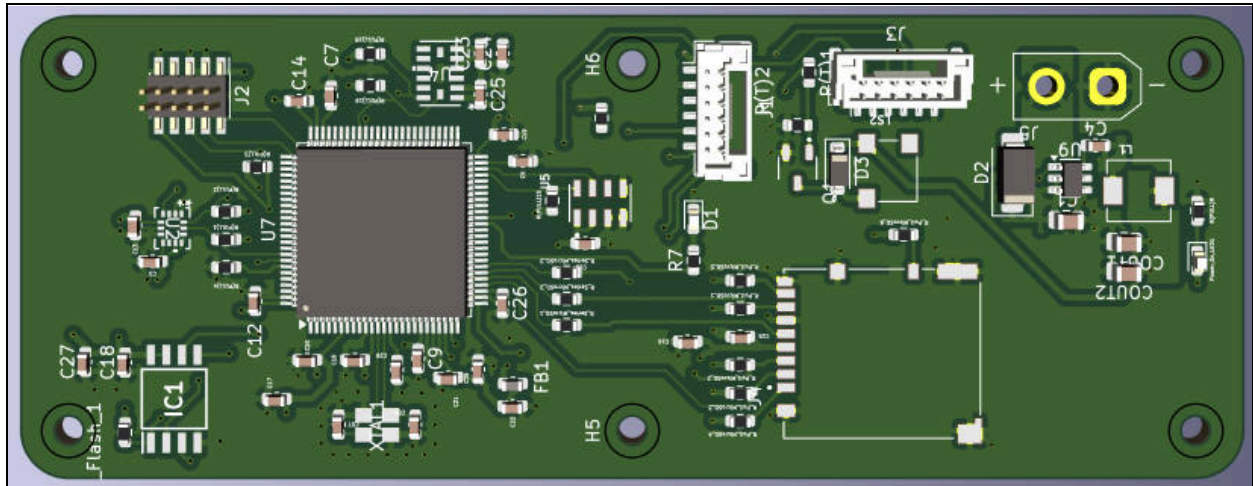


Figure 67: 3D Sideways view of KRAKENBANE I PCB

The GNSS unit is to be included in its own breakout board and will connect to the main board via 6 pin JST-GH connectors. The XBEE module is connected similarly to the main DAQ PCB.

17.3 Telemetry

The SRAD telemetry system provides a radio link between the rocket and the ground station. It displays data tracked from the DAQ system, such as altitude, orientation, acceleration, velocity, GPS location, and pressure.

The ground station requires the XBee-PRO 900HP, which is mounted on an XBee shield and connected to a computer via USB. Additionally, the HG909Y-NF Yagi antenna is connected to the XBee via an N-type RF cable. This Yagi antenna was selected due to its high directional gain of 9 dBi, resulting in strong reception of the omni antenna and a lower probability of signal loss over long distances. Furthermore, its targeted frequency range of 890-960 MHz allows the radio waves to pass through the rocket's airframe effortlessly.

On the rocket side, an STM32 will be attached to a second XBee-PRO 900HP, which is screwed into an A09-HSM-7 omnidirectional antenna. The XBee will also be on a shield consisting of 5 header pins soldered on, allowing the male to female jumper wires to connect the two electronics. The wiring between the XBee and STM32 is as follows: 5V to 5V, GND to GND, TX to TX, and RX to RX. Normally, the TX is wired to the RX; however, the XBee shield causes a change in wiring. An omnidirectional antenna was chosen since a directional antenna would raise a concern regarding signal loss. This is due to the fact that the rocket will change orientation throughout the flight, and a directional antenna would eventually point away from the Yagi antenna, resulting in a loss between radio links.

The XBees were configured using the XCTU software to set them to the same baud rate and Network ID, which ensures communication. It is critical that the XBee's are configured to these same parameters so that the Yagi antenna can receive the data from the omni antenna. The yagi antenna will then send this information to the XBee on the ground station, which will then display the data on the computer using a webpage that the team has previously coded.

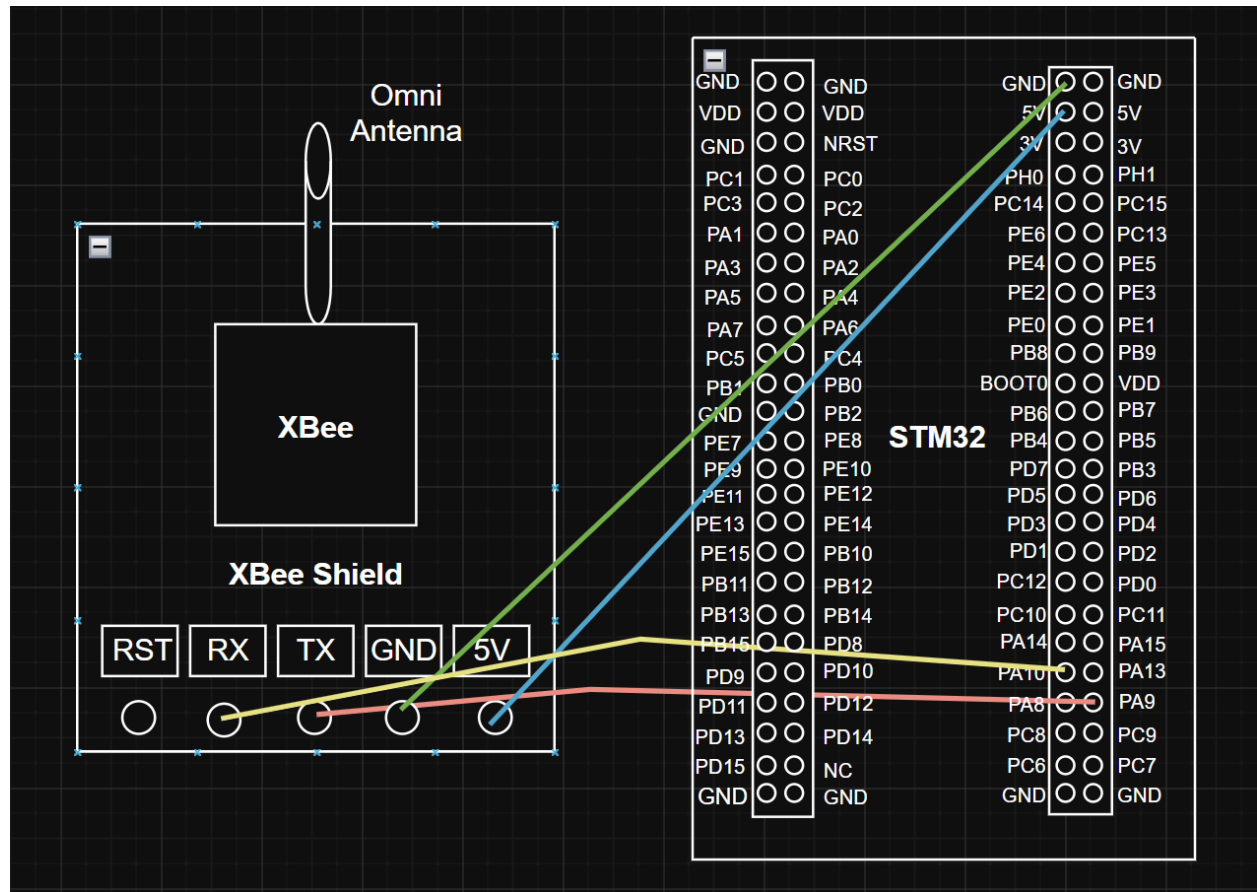


Figure 68: Telemetry System Wiring Diagram

18 DATA ANALYSIS PLAN

18.1 Pitot Tube Data Analysis

The pitot tube will collect three sets of data: static pressure, total pressure (stagnation pressure), and static temperature (ambient air temperature). This data will be important in calculating the airspeed of the vehicle with respect to time. There are three equations which must be applied. Each equation is responsible for making the conversion from pressure data to velocity for different speed regimes.

18.1.1 Subsonic incompressible ($M < 0.3$)

$$V = \sqrt{\frac{2(P_{Total} - P_{Static})}{\rho}}$$

The first equation is the incompressible pitot tube velocity equation, derived from Bernoulli's Principle [15]. This equation will be used to find the airspeed for Mach numbers below 0.3, where air can be approximated as incompressible. The numerator term takes the difference of the total and static pressure, which produces the value for dynamic pressure. The denominator term takes the static air density. The team has approximated this value to stay constant at 1.10 kg/m³ according to local conditions in Timmins, ON. [16]

18.1.2 Subsonic compressible ($0.3 < M < 0.8$)

$$M = \sqrt{\frac{2}{k-1} \left[\left(\frac{P_{Total}}{P_{Static}} \right)^{\frac{k-1}{k}} - 1 \right]}$$

For speeds between Mach 0.3 and Mach 0.8, the assumption of constant density fails, and compressibility effects must be accounted for using isentropic flow relations. In the equation above, k represents the ratio of specific heats for air (approximately 1.4). Notice that this equation gives the Mach number, therefore the local speed of sound must be used to convert to the true airspeed value.

$$a = \sqrt{kRT_{Static}}$$

Where a is the local speed of sound, and R is the specific gas constant (287 J/kg·K).

18.1.3 Supersonic compressible ($M > 0.8$)

As the vehicle transitions through the transonic regime, a detached bow shock or will form ahead of the pitot tube. The total pressure measured by the sensor is now the stagnation pressure behind the bow shock (p_{02} on the diagram), meaning the isentropic assumption is no longer valid due to entropy increase across the shockwave. Furthermore, across the shockwave the Mach number drops from supersonic to subsonic. We are interesting in finding the free-stream velocity, V_1 .

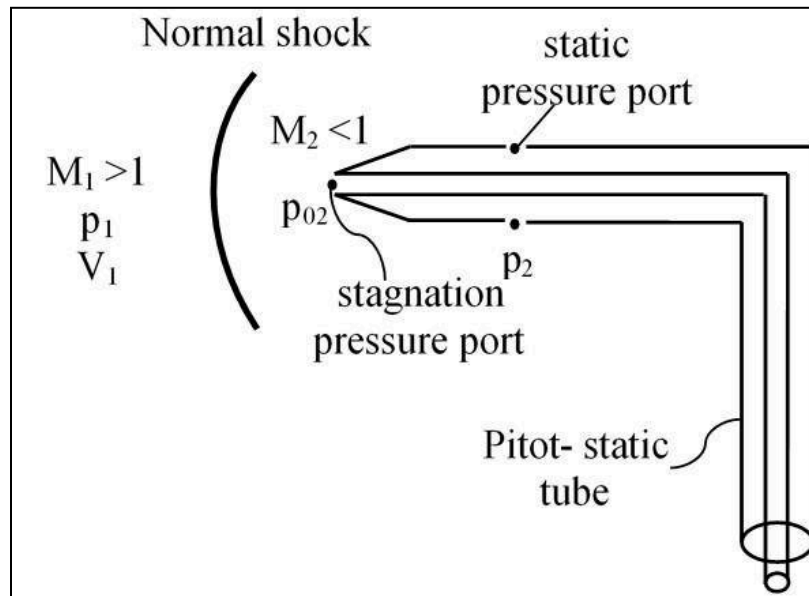


Figure 69: Pitot Tube Bow Shock Illustration

To find the Mach number in this regime, the Rayleigh Pitot Tube Formula must be utilized. Note this equation is expressed in many different ways based on where it is sourced from. All formats of the equation are mathematically equivalent.

$$\frac{P_{Total}}{P_{Static}} = \left[\frac{(k+1)^2 M^2}{4kM^2 - 2(k-1)} \right]^{\frac{k}{k-1}} \cdot \frac{1-k+2kM^2}{k+1}$$

Because this equation cannot be isolated for M algebraically, the software post-processing will utilize a numerical root-finding method (such as the Newton-Raphson method) to iteratively solve for the supersonic Mach number. Once M is approximated, airspeed is again found using the local speed of sound. It should be noted that while standard supersonic flow begins at Mach 1.0, treating data above Mach 0.8 with trans/supersonic logic ensures that shockwaves forming at high subsonic speeds do not skew the pressure readings.

18.1.4 Software Development

The conversion of the raw flight data collected by the pitot tube into airspeed will rely on a custom software developed by the team. This post-processing script is currently in development. The software is being authored in Python with prototypes being developed in the Thonny IDE.

The final software will process the data through a multi-stage pipeline:

1. First, the script will parse the raw .csv or .txt logs exported from the Pitot Board, filtering out noise and organizing the static pressure, total pressure, and temperature data.
2. Because the mathematical relationship between pressure and airspeed changes as the rocket approaches Mach 1.2, the software will run an initial check to estimate the Mach number. Based on this estimate, the data point will be routed to the appropriate function (incompressible, subsonic compressible, or supersonic).
3. For supersonic data points requiring the Rayleigh Pitot Tube formula, the software will utilize numerical root-finding algorithms to solve for the Mach number.
4. Finally, the processed arrays will be exported for graphical visualization.

18.1.5 Benchmarking & Performance Validation

Although the raw data from the pitot tube would be intriguing, without a known baseline to compare against, it is not useful. However, Goldeneye is equipped with several sources of data that can be utilized in different ways in order to characterize the performance of the pitot tube.

Velocity-based Benchmarking

The main sources of velocity data from Goldeneye is the Blue Raven IMU and the DAQ IMU (LSM6DSV32XTR). The integration and subsequent filtering of IMU data yields highly accurate velocity data. Other, less favorable sources of velocity data include differentiating GPS position with time, and differentiating barometric altitude with time. In any case, the presence of several distinct, dissimilar sources of velocity will be critical in forming an accurate estimation of velocity over Goldeneye's flight. This can be done with the use of a standard complementary filter, with the more accurate data source (filtered IMU data), being weighted more heavily.

$$V_{@Given\ time} = (0.90)(V_{IMU}) + (0.05)(V_{GPS}) + (0.05)(V_{Barometer})$$

This process can be repeated until an accurate, complete velocity profile is formed. However, limitations and challenges do exist to this approach. For instance, syncing timestamps from various sensors sampling at widely different rates will be challenging.

In any case, the filtered pitot tube data can be benchmarked against this composite dataset from all the aforementioned sensors. This is critical in evaluating the effectiveness of the pitot tube.

Position-based Benchmarking

The alternative approach to the need for benchmarking is to use position estimates. Although the pitot tube is intended to provide airspeed measurements, these data points can be integrated with time to form a position-time estimate. Although this will certainly be inaccurate, as tiny errors in airspeed will be compounded through integration, it complements the prior velocity-based benchmarking approach. Accurate position estimates include the GPS and barometer (both SRAD and COTS sources). As such, another complementary filter can be created, in order to find position at a given time, and compare it against the position from the integrated pitot tube estimate.

$$x @ Given\ time = (0.45)(x_{Barometer}) + (0.45)(x_{GPS}) + (0.10)(x_{IMU})$$

Thus, by using a dual-pronged approach of utilizing position-based and velocity-based benchmarking, the performance, accuracy, and effectiveness of the pitot tube system can be evaluated.

19 RANGE SAFETY

19.1 Hazard Analysis

Safety has been addressed in the design process, manufacture, testing, transport, and operations of Goldeneye. As with any rocket, there are a number of hazards identified that could potentially cause harm to people and equipment. A formal hazard analysis was performed in order to determine possible failure modes and mitigate the risks.

Even though the motor is commercially manufactured and certified, there are three main risks that might occur related to the propulsion system: accidental ignition, improper motor installation, and exposure to hot exhaust gases or motor components. To minimize these risks, the process of installing the motor will be done solely by trained team members with an established procedure for motor assembly. The motor will not come into contact with ignition sources until launch. The igniter will be installed when the rocket is transferred to the launch site and the launch area is clear. Additionally, the control system used for ignition will be disconnected until the actual launch. After ignition, enough cool down time will be allowed before the motor is recovered. Personnel will be required to stand at an appropriate distance while the motor fires.

The main risks involved in the recovery procedure involve accidental activation and lack of parachute deployment. In order to minimize the risk of accidental release, all flight computers will remain unpowered throughout the assembly and transport process. Black powder charges will only be placed shortly before launch by using the approved checklist. Shear pins will be checked for secure and proper separation of the airframe sections. Ground tests will be performed (using proper PPE) on the recovery system prior to launching in order to make sure that there is enough pressure created by the ejection charges to separate the airframe and activate the recovery mechanism.

Static discharge, electrical sources, or incorrect handling procedures could result in accidental activation of the black powder ejection charges. For this, black powder handling will only be done by trained members of the team, and will only be installed while flight computer systems are disconnected. Additionally, the avionics bay has redundant flight computers, allowing for flight operation to continue even in the case of a full failure of one flight computer.

High speed flight hazards have been accounted for in the design process, as Goldeneye includes three through-the-wall mounted fins that maintain proper stability. Flight simulations were conducted to ensure the quality of these measures.

The pitot tube presents risks in transport and handling, as it protrudes from the front of the nose cone. The installation method of threading the tube into the nose cone creates a strong support system. An inspection will also be done before flight to ensure no damage has been done.

19.2 Risk Assessment

Mission Phase	Failure Mode	Probability	Severity	Risk Mitigation
Assembly	Incorrect motor installation	Low	Critical	Motor installation performed by trained members using checklist procedures
Assembly	Accidental black powder charge activation	Very Low	Critical	Charges installed immediately before launch, avionics remain unarmed during installation
Pad Setup	Unintended motor ignition	Very Low	Catastrophic	Igniter installed only after vehicle is secured
Pad Setup	Battery short circuit	Low	High	Battery inspection, insulated connectors, power-off during assembly
Launch	Structural failure during boost	Very Low	Catastrophic	Carbon fibre airframe, through-the-wall fin mounting, structural analysis
Powered Flight	Loss of aerodynamic stability	Very Low	High	Stability margin maintained at 2.40 cal; flight simulations performed
Powered Flight	Avionics failure	Low	High	Redundant flight computers and GPS trackers
Coast	Failure to detect apogee	Low	High	Dual deployment electronics with independent flight computers
Recovery	Drogue parachute failure	Very Low	Critical	Ground deployment testing and parachute inspection
Recovery	Main parachute failure	Very Low	Critical	Redundant retention checks and pre-flight packing inspection

Recovery	Vehicle lands outside expected area	Low	High	GPS tracking and recovery procedures
Recovery	Damaged pitot tube during landing	Medium	Low	Protective cover during transport and post-flight inspection

Table 31: Risk Assessment Matrix

Any risks that were classified as “unacceptable” were neutralized through design modifications of Goldeneye. The remaining risks were determined to be acceptable under the LC regulations.

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APPENDIX A: PAYLOAD

A.1 Data Analysis Software Prototype

The following prototype software was created to give the team a better understanding of the order of operations for the data analysis software that will allow the pitot tube data to be converted to readable airspeed.

```
import math
# import scipy.optimize # <-- for Newton-Raphson solver at Mach > 0.8

# Constants
RHO = 1.10 # kg/m^3 (approx static air density for Timmins)
K = 1.4    # Ratio of specific heats for air
R = 287.05 # Specific gas constant

def get_incompressible_v(ptot, pstat):
    # Calculates airspeed for M < 0.3 using Bernoulli's
    dyn_p = ptot - pstat

    # Quick fix: sometimes the sensor spits out negative pressure diffs on the pad due to
    noise
    if dyn_p <= 0:
        return 0.0

    return math.sqrt((2 * dyn_p) / RHO)

def get_subsonic_compressible_v(ptot, pstat, tstat):
    # Calculates airspeed for 0.3 < M < 0.8
    # TODO: Write out the isentropic math here.
    # The formula is derived in the FDR, just need to code the pressure ratio logic.
    pass

def get_supersonic_v(ptot, pstat, tstat):
    # Calculates airspeed for M > 0.8 (Rayleigh Pitot formula)
    # TODO: This one needs the root finder since we can't isolate M algebraically.
    pass

def process_flight_data(filepath):
    print(f"Loading flight data from: {filepath}")
    print("Warning: Only incompressible speed calculations are currently active.")

    # Blueprint for the main loop:
    # 1. Open the CSV log file from the Pitot Board
    # 2. Read row by row: extract time, ptot, pstat, tstat
    # 3. Run a quick estimate to see what Mach regime we are in
    # 4. Route variables to the correct math function above
    # 5. Save results to a new array and export for graphing
    pass
```

B.1 Fin Flutter Calculations

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B.2 Fin Surface Prep Testing

B.2.1: Testing sample



B.2.2: Testing Apparatus



B.2.3: Failure Modes



B.3 Bending Loads Calculations

Instructions

1. Clear all the cells in the figure. This is easily done by high-lighting all the cells within the dashed lines, then hit **delete** key.
2. Enter all pertinent dimensions. If no transition or boattail, leave these cells **blank**.

LN = length of nosecone
d = diameter at base of nosecone
dF = diameter at front of transition
dR = diameter at rear of transition
LT = length of transition
XP = distance from tip of nose to front of transition
CR = fin root chord
CT = fin tip chord
N = number of fins (3 or 4)
S = fin semispan
R = radius of body at aft end
XR = distance between fin root leading edge and fin tip leading edge parallel to body
XB = distance from nose tip to fin root chord leading edge
dFB = diameter at front of boattail
dRB = diameter at rear of boattail
LTB = length of boattail
XPB = distance from tip of nose to front of boattail

Note: conical transition can be either shape shown below:

It is recommended to clear all cells 1st (see instructions)

Units must be millimetres

If there is some random error, make sure that the labels on each of the green boxes match the blank version. These labels were edited afterwards, as they had original Google comments from Richard.

NOSECONE	
Shape =	parabolic < use drop-down menu
LN =	584.0 mm
d =	101.6 mm
(CN) _N =	2 per radian Normal force coefficient, nosecone
X _N /LN =	0.5
X _N =	292.0 mm Distance along body to C.P. of nosecone
CONICAL TRANSITION	
Has a conical transition?	no < use drop-down menu
dF =	mm
dR =	mm
LT =	mm
XP =	
(CN) _T =	0.000 per radian Normal force coefficient, transition
X _T =	0.0 mm Distance along body to C.P. of transition
CONICAL BOATTAIL	
Has a conical boattail?	yes < use drop-down menu
dFB =	101.6 mm
dRB =	76.5 mm
LTB =	106.7 mm
XPB =	2159.0
(CN) _{TB} =	-0.866 per radian Normal force coefficient, boattail
X _{TB} =	2209.8 mm Distance along body to C.P. of boattail
FINS	
N =	3 < use drop-down menu
R =	51.1 mm
S =	127.0 mm
CR =	191.7 mm
CT =	71.1 mm
X _R =	193.0 mm
X _B =	1967.0 mm
	0.8073 rad. (mid-chord sweep angle)
	46.26 deg.
LF =	183.7 mm (length of fin mid-chord line)
(CN) _F =	8.875 per radian Normal force coefficient, fins
X _F =	2083.9 mm Distance along body to C.P. of fins
(CN) _R =	10.01 Normal force coefficient, rocket
X _{CP} =	1714.9 mm Distance along body to C.P. of rocket

REFERENCE FIGURES			From "Barrowman" spreadsheet		No Density		REFERENCE METRIC	
NOSE CONE			CUSTOMER FIGURE CALCS		rho (kg/m³)		SA	
Shape = parabolic < use drop-down menu			MAX Q (pa)		rho = Pressure / (ideal gas * temp)		METAR AT SPECTRE LAUNCH	
LN =	584.0 mm	FALSE	111480.4371		Pressure @ Max Q		https://www.aginet.com/metas-portfolio	
LN =	101.6 mm	FALSE	MAXIMUM DYNAMIC PRESSURE (max q)		Source: ORK, Motor Burnout		METAR CYR 1014002 150000T 110V100 11000 H	
LN/N =	0.5	Normal force coefficient, nosecone	$q = \frac{1}{2} \rho v^2$		Temp @Max Q			
LN =	250.0 mm	Distance along body to C.P. of nosecone	Velocity @Max Q		Source: ORK, Motor Burnout			
CONICAL BOATTAIL			Air Density @Max Q		Altitude at Max Q			
Has a conical booyes < use drop-down menu			1.195 kg/m³		Not used in calc, but useful			
FTB =	101.6 mm		Source: ORK, Motor Burnout		Source: ORK, Motor Burnout			
FTB =	76.5 mm		This is tested velocity					
FTB =	106.7 mm	FALSE	Not only upward velocity					
FTB =	2159.0		Air Density @Max Q					
FTB =	0.8603 per radian	Normal force coefficient, boatail	1.195 kg/m³					
FTB =	2209.8 mm	Distance along body to C.P. of boatail	Source: Cell to the TOP RIGHT					
FIN			Maximum Angle		AREA			
FIN =	3 < use drop-down menu		6 degrees		Calculate the frontal area			
FIN =	51.5 mm		This number is ENTIRELY arbitrary - must be fixed at a later date.		0.00817 m²			
FIN =	127.0 mm		Maximum Area (radians)		Simple ass circle area			
FIN =	191.7 mm		Barrowman uses rad.		0.102 meters			
FIN =	71.5 mm							
FIN =	199.0 mm	FALSE						
FIN =	1967.0 mm							
FIN =	0.8073 rad. (mid-chord sweep angle)							
FIN =	46.26 deg.							
FIN =	183.7 mm (length of fin mid-chord line)							
FIN =	8.875 per radian	Normal force coefficient, fin						
FIN =	2061.9 mm	Distance along body to C.P. of fin						
FIN =	10.91	Normal force coefficient, rocket						
FIN =	1714.3 mm	Distance along body to C.P. of rocket						

Latest Revision	Mar 7, 2026	7:20 PM
FIN SECTION NORMAL FORCE		
$N_{FINS} = q A \alpha (C_N \alpha)_F$ [Equation 3]		
N_FINS	847.567055	
Max Pressure(q)	111480.4371	pa
Area (A)	0.00817	m²
AoA (a)	0.1048	RADIANS
(CnA)F	8.875	N per Radian
NOSE SECTION NORMAL FORCE		
$N_{NOSE} = q A \alpha (C_N \alpha)_N$ [Equation 2]		
N_NOSE	190.9933639	
Max Pressure(q)	111480.4371	pa
Area (A)	0.00817	m²
AoA (a)	0.1048	RADIANS
(CnA)F	2.0	N per Radian
BENDING MOMENT CALCS		
FIN BENDING MOMENT 416.91 Newton-meter		
Moment Arm	0.492	
FIN LEVER ARM		
Fin CP (from Nose)	2.084	meters
Source: Barrowman Spreadsheet		
Rocket CG (from Nose)	1.592	meters
Source: ORK		
NOSE CONE BENDING MOMENT 304.06 Newton-meter		
MOMENT ARM	1.300	meters
NOSE LEVER ARM		
Nose CP (From nose tip)	0.292	meters
Source: Barrowman Spreadsheet		
Rocket CG (from Nose)	1.592	meters
Source: ORK		
TOTAL BENDING MOMENT 720.98 Newton-meter		

B.4 Monte Carlo RocketPy Code

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Ellipse
from datetime import datetime
from rocketpy import Environment, SolidMotor, Rocket, Flight

# =====
# =====
print("Initializing baseline GoldenEye model...")
env = Environment(latitude=48.5612, longitude=-81.3758, elevation=295)
env.set_date(datetime(2026, 8, 15, 12))
env.set_atmospheric_model(type='standard_atmosphere')

# Base High-Fidelity Motor Profile
motor = SolidMotor(
    thrust_source="L2375.eng",
    burn_time=1.9,
    dry_mass=1.841,
    dry_inertia=(0.115, 0.115, 0.002),
    center_of_dry_mass_position=0.31,
    grains_center_of_mass_position=0.32,
    grain_number=4,
    grain_separation=0.005,
    grain_density=1815,
    grain_outer_radius=0.033,
    grain_initial_inner_radius=0.015,
    grain_initial_height=0.12,
    nozzle_radius=0.014,
    throat_radius=0.0075,
)

# Compile the GoldenEye geometry and aerodynamics ONCE
goldeneye = Rocket(
    radius=0.051,
    mass=12.450,
    inertia=(14.5, 14.5, 0.05),
    power_off_drag=0.48,
    power_on_drag=0.42,
    center_of_mass_without_motor=1.65,
)
goldeneye.add_nose(length=0.61, kind="ogive", position=0)
goldeneye.add_trapezoidal_fins(n=4, root_chord=0.22, tip_chord=0.11, span=0.12,
position=2.2)
goldeneye.add_parachute(name="Main", cd_s=8.5, trigger="apogee")
goldeneye.add_motor(motor, position=2.6)

# =====
# =====
num_simulations = 600
landing_x = []
landing_y = []
apogees = []

print(f"Starting optimized simulation loop ({num_simulations} flights)...")

# Define structural and launch tolerances
mass_std = 0.250      # +/- 250g structural variance
cm_std = 0.02         # +/- 2cm CG shift margin
```

```

inc_std = 1.0          # +/- 1 deg rail inclination
heading_std = 2.0      # +/- 2 deg rail heading

# Store baseline values to alter dynamically
base_mass = goldeneye.mass
base_cm = goldeneye.center_of_mass_without_motor

for i in range(num_simulations):
    # Print progress status every 20 flights so you can see it moving fast
    if (i + 1) % 20 == 0:
        print(f" -> Simulated {i + 1}/{num_simulations} flights...")

    # Dynamically tweak structural attributes without re-compiling aero fins
    goldeneye.mass = np.random.normal(base_mass, mass_std)
    goldeneye.center_of_mass_without_motor = np.random.normal(base_cm, cm_std)

    # Randomize launch rail parameters
    rand_inc = np.random.normal(84, inc_std)
    rand_heading = np.random.normal(0, heading_std)

    try:
        # Run 6-DOF trajectory integration
        sim_flight = Flight(
            rocket=goldeneye,
            environment=env,
            rail_length=3.6,
            inclination=rand_inc,
            heading=rand_heading
        )

        # Save structural results
        landing_x.append(sim_flight.x[-1])
        landing_y.append(sim_flight.y[-1])
        apogees.append(sim_flight.apogee)
    except Exception:
        continue

# Reset goldeneye rocket to default baseline configuration
goldeneye.mass = base_mass
goldeneye.center_of_mass_without_motor = base_cm

landing_x = np.array(landing_x)
landing_y = np.array(landing_y)

# =====
# 3. STATISTICAL COMPILATION & VISUALIZATION
# =====
print("\n=== RANGE SAFETY STATISTICS ===")
print(f"Total Successful Flights Evaluated: {len(landing_x)}")
print(f"Mean Apogee Altitude: {np.mean(apogees):.2f} m")
print(f"Max Apogee Altitude: {np.max(apogees):.2f} m")
print(f"Min Apogee Altitude: {np.min(apogees):.2f} m")
print(f"Landing Dispersion Center (X, Y): ({np.mean(landing_x):.1f}m, {np.mean(landing_y):.1f}m)")

# Plotting the Range Safety Ellipses
fig, ax = plt.subplots(figsize=(10, 8))
ax.scatter(landing_x, landing_y, color='red', alpha=0.6, label='Parachute Landing Points')
ax.scatter(0, 0, color='black', marker='X', s=150, label='Launch Pad (0,0)')

mean_x, mean_y = np.mean(landing_x), np.mean(landing_y)

```

```
std_x, std_y = np.std(landing_x), np.std(landing_y)

ellipse1 = Ellipse((mean_x, mean_y), width=std_x*2, height=std_y*2, angle=0,
                   edgecolor='blue', facecolor='blue', alpha=0.15, lw=2, label='1-Sigma Zone
(68.3%)')
ellipse2 = Ellipse((mean_x, mean_y), width=std_x*4, height=std_y*4, angle=0,
                   edgecolor='darkblue', facecolor='none', linestyle='--', lw=2,
label='2-Sigma Zone (95.5%)')

ax.add_patch(ellipse1)
ax.add_patch(ellipse2)

ax.axhline(0, color='gray', linestyle=':', alpha=0.5)
ax.axvline(0, color='gray', linestyle=':', alpha=0.5)
ax.set_title('GoldenEye Launch Canada Flight Dispersion Footprint (L2375 Motor)',
fontsize=14)
ax.set_xlabel('East-West Displacement (meters)', fontsize=12)
ax.set_ylabel('North-South Displacement (meters)', fontsize=12)
ax.legend(loc='best')
ax.grid(True, linestyle='--', alpha=0.5)
plt.axis('equal')

plt.show()
```

B.5 Recovery Shock Loading Python Code

```
import numpy as np
from scipy.integrate import odeint
import csv
import os

# === SOLVER PARAMETERS (UVic style) ===
mr = 16.01      # Rocket mass [kg]
mp = 0.39      # Parachute mass [kg]
ms = 0.48      # Shock cord mass [kg]
k = 7297       # Shock cord spring constant [N/m]
rho = 1.15     # Air density [kg/m^3]
wind = 0.0     # Wind speed [m/s]
cd = 2.20      # Drag coefficient []
aeff = 2.55    # Parachute effective area [m^2]
xr_dot = 30.00 # Rocket velocity at inflation [m/s]
xp_dot = 0.0   # Parachute velocity at inflation [m/s]

# === DERIVED PARAMETERS ===
g = 9.81       # Gravity [m/s^2]
c = 0.5 * rho * cd * aeff # Parachute damping coefficient [N/m/s]
mp_eff = mp + ms/3 # Effective parachute mass including 1/3 shock cord

# === SIMULATION TIME ===
t_start = 0.0
t_end = 10.0
nt = 1000
t = np.linspace(t_start, t_end, nt)

# === INITIAL CONDITION VECTOR [xp, xr, yr, yp] ===
y0 = [0.0, 0.0, xr_dot, xp_dot] # parachute pos, rocket pos, rocket vel, parachute vel
```

```
# === ODE SYSTEM (UVic form) ===
def dy_dt(y, t, mp_eff, mr, c, k, g, wind):
    xp, xr, yr, yp = y
    dydt = [
        yp,                # parachute velocity
        yr,                # rocket velocity
        (mr*g - k*(xr - xp))/mr,  # rocket acceleration
        (mp_eff*g - c*(yp + wind)**2 - k*(xp - xr))/mp_eff # parachute acceleration
    ]
    return dydt

# === SOLVE ODEs ===
sol = odeint(dy_dt, y0, t, args=(mp_eff, mr, c, k, g, wind))
xp = sol[:,0]
xr = sol[:,1]
yr = sol[:,2]
yp = sol[:,3]

# === CALCULATE FORCES ===
shock_cord_tension = k * (xr - xp)                # Shock cord tension [N]
ap = np.gradient(yp, t)                        # Parachute acceleration [m/s^2]
shroud_line_tension = (mp - ms) * ap + c * (yp + wind)**2 # Shroud line tension [N]

# === PRINT RESULTS ===
print("Using the following solver parameters:\n")
print(f"Rocket Mass:\t\t\t {mr:.2f} \t[kg]")
print(f"Parachute Mass:\t\t\t {mp:.2f} \t[kg]")
print(f"Shock Cord Mass:\t\t\t {ms:.2f} \t[kg]")
print(f"Shock Cord k:\t\t\t {k:.0f} \t[N/m]")
print(f"Air Density:\t\t\t {rho:.2f} \t[kg/m^3]")
print(f"Wind Speed:\t\t\t {wind:.1f} \t[m/s]")
print(f"Drag Coefficient:\t\t\t {cd:.2f} \t[]")
print(f"Effective Area:\t\t\t {aeff:.2f} \t[m^2]")
print(f"Rocket velocity at inflation:\t {xr_dot:.2f} \t[m/s]")
print(f"Parachute velocity at inflation: {xp_dot:.2f} \t[m/s]\n")

print("Produced the following results:\n")
print(f"Parachute Damping Coeff:\t {c:.2f} \t[N/m/s]")
print(f"Maximum Shock Cord Tension:\t {np.max(shock_cord_tension):.0f} \t[N]")
print(f"Maximum Shroud Line Tension:\t {np.max(shroud_line_tension):.0f} \t[N]")
print(f"Maximum Rocket G load:\t\t {np.max(shock_cord_tension)/mr/g:.2f} \t[G]\n")

# === SAVE CSV ===
script_dir = os.path.dirname(os.path.abspath(__file__)) # Save in same folder as script
csv_file = os.path.join(script_dir, "load_data.csv")
with open(csv_file, 'w', newline='') as f:
    writer = csv.writer(f)
    writer.writerow(["time [s]", "shock cord tension [N]", "shroud line tension [N]"])
    writer.writerows(zip(t, shock_cord_tension, shroud_line_tension))

print(f"Results exported to {csv_file}")
```

APPENDIX C: CHECKLISTS

C.1 Subsystem Checklists

C.1.1 Recovery System Checklist

Inventory and Component Prep

- ☐ Verify all components are present: 72" Main Parachute, 24" Drogue Parachute, 1" Tubular Nylon Shock Cords, Nomex Protectors (9"x9" and 18"x18"), Stainless Steel Swivels, 1/4" Quicklinks, Black Powder, J-TEK igniters, and Nitrile gloves.
- ☐ Inspect all parachutes, chute protectors, and shock cord for any frays or damage.

Shock Cord and Quicklink Assembly

- ☐ Route and securely tie the shock cords to their respective hardpoints (the 3/8-16 eyebolts and U-bolts).
- ☐ Wrap all shock cord knots tightly with aluminum tape.
- ☐ Attach the 3000 lbs stainless steel swivels to the shock cords using the 1/4" quicklinks.
- ☐ Execute Quicklink FR (flight ready) protocol for every quicklink connection:
 - Apply Blue Loctite to the threads.
 - Torque the quicklink down hand-tight.
 - Fully tighten using wrench.
 - Wrap the secured quicklink in orange tape to designate it as FR

Drogue Parachute Assembly

- ☐ Connect the drogue parachute to the drogue shock cord swivel.
- ☐ Fold the drogue parachute and pack it tightly inside the 9"x9" chute protector.
- ☐ Prepare the drogue copper charge wells:
 - Load Primary Charge: 1.7g FFFF black powder.
 - Load Secondary Charge: 2.7g FFFF black powder.
- ☐ Seal the black powder within nitrile glove fingertips, secure firmly with rubber bands, and insert the igniters.

Main Parachute Assembly

- ☐ Connect the main parachute to the main shock cord swivel.
- ☐ Fold the main parachute and pack it tightly inside the 18"x18" chute protector.
- ☐ Prepare the main charge wells:
 - Load Primary Charge: 2.0g FFFF black powder.
 - Load Secondary Charge: 2.7g FFFF black powder.
- ☐ Seal the black powder within nitrile fingertips, secure with rubber bands, and insert the igniters.

Airframe Integration and Final Checks

- ☐ Slide the aft and forward airframe sections over their respective couplers.
- ☐ Install the shear pins through the airframe and into their backing rings to secure separation points.
- ☐ Verify that the pressure equalization holes are completely clear of internal hardware.

C.1.2 Payload Checklist

- ☐ Verify the preassembled pitot tube section is fully cured and structurally sound.
- ☐ Apply Blue Loctite and firmly thread the pitot tube into the nose cone's threaded insert.
- ☐ Route the Tygon tubing internally and connect it securely to the Pitot board PCB.
- ☐ Confirm the Pitot board PCB is tightly secured to the payload sled.
- ☐ Thread the payload sled onto the internal rod and lock it in place using the eyenut on the bulkhead.
- ☐ Slide the nose cone coupler into the airframe and fasten it with the retaining bolts.
- ☐ Conduct a firm tug test on the entire integrated assembly to check for any loose points or rattling.
- ☐ Keep all payload electronics powered OFF; they must only be activated via external switches during pre-launch.

C.2 Launch Checklists

C.2.1 Pre-Launch Checklist (At Assembly Area)

- ☐ Verify all external power switches are in the OFF position.
- ☐ Connect all internal batteries and power sources to the avionics.
- ☐ Ensure that the Featherweight GPS units, Blue Raven flight computer, RRC3+ altimeter, DAQ board, and Pitot Board are switched on locally to the board.
- ☐ Attach the avbay panels.
- ☐ Slide the forward onto the avbay and secure using retaining bolts.
- ☐ Slide on the aft airframe and secure using shear pins.
- ☐ Slide on the nose cone and secure using shear pins.

C.2.2 Pre-Launch Checklist (At Pad)

- ☐ Mount the fully integrated rocket securely onto the launch rail.
- ☐ Power on all electronics using the external switches.
- ☐ Verify connectivity.
- ☐ LC inserts motor igniter.

C.2.3 During Launch Checklist

- ☐ Monitor primary and secondary GPS data continuously at the ground stations.
- ☐ Track the launch using the designated flight app and keep the ground station antenna perpendicular to the rocket's position (vertical while on the pad, angled during launch, and horizontal at apogee).
- ☐ Record the descent rate under the drogue parachute (Target: ~100 ft/s).
- ☐ Record the descent rate under the main parachute (Target: ~20 ft/s).
- ☐ Take note of the last reported coordinates and the direction of travel every 5-10 minutes in case of a lost connection.

C.2.4 Post-Launch Checklist

- ☐ Assemble the designated recovery team and ensure members are equipped with walkie-talkies tuned to team frequency.
- ☐ Verify all recovery personnel have the required safety gear.
- ☐ Enter the most recent coordinates from the flight app into an offline mapping app.
- ☐ Upon visually locating the landed rocket, conduct a visual safety analysis (wear safety glasses) from a distance to ensure both deployment charges have fired, parachutes are deployed, and the rocket is safe to approach.
- ☐ In the event that a black powder charge fails to deploy it will be immediately disposed of and neutralised at the recovery site.
- ☐ Power off electronics using external switches.
- ☐ Collect all rocket components, pack them securely, and return to the launch site.

C.2.5 Launch Abort Checklist

- ☐ Remove the motor igniter.
- ☐ Power off the electronics via external power switches.
- ☐ Remove the rocket from the launch rail.
- ☐ Return to the assembly area.
- ☐ Disassemble the avbay panels and disconnect battery power to avionics.
- ☐ Remove shear pins and disassemble the rocket from all separation points.
- ☐ Dispose of and neutralise all four black powder charges located in the ejection wells.
- ☐ Continue disassembly for inspection as needed.

APPENDIX D: GEOPARD I SOFTWARE PROTOTYPE

```
#include <SPI.h>

// ===== SENSORS: BIT-BANGED SPI =====
const int SOFT_SCK = 2;
const int SOFT_MISO = 3;
const int HALF_US = 3;

// How often to grab a sample and write a record. 50 ms = 20 Hz.
// Lower it for faster capture; ~60 bytes/record, 8 MB flash ~= 140k records
// (~1.9 h at 20 Hz, ~23 min at 100 Hz).
const uint32_t SAMPLE_INTERVAL_MS = 50;

struct HscConfig {
    uint8_t csPin;
    float pMin, pMax;
    float toKpa;
    const char *role, *name, *unit;
};

HscConfig sensorRam = { 10, 0.0f, 60.0f, 6.89476f, "RAM", "HSC 060PA", "psiA" };
HscConfig sensorStatic = { 6, 0.0f, 1.6f, 100.0f, "STATIC", "HSC 1.6BA", "barA" };

const uint16_t OUT_MIN = 1638;
const uint16_t OUT_MAX = 14746;

struct HscReading {
    uint8_t status;
    uint16_t rawP, rawT;
    float pressure, kPa, tempC;
};

struct TcReading {
    bool ok;
    uint8_t fault;
    float tempC;
    float internalC;
};

uint8_t softSpiReadByte() {
    uint8_t v = 0;
    for (int i = 0; i < 8; i++) {
        digitalWriteFast(SOFT_SCK, HIGH);
        delayMicroseconds(HALF_US);
        v = (v << 1) | (digitalReadFast(SOFT_MISO) ? 1 : 0);
        digitalWriteFast(SOFT_SCK, LOW);
        delayMicroseconds(HALF_US);
    }
    return v;
}

HscReading hscRead(const HscConfig &cfg) {
    uint8_t b[4];
    digitalWriteFast(SOFT_SCK, LOW);
    digitalWriteFast(cfg.csPin, LOW);
    delayMicroseconds(5);
    for (int i = 0; i < 4; i++) b[i] = softSpiReadByte();
    digitalWriteFast(cfg.csPin, HIGH);
    delayMicroseconds(5);

    HscReading r;
    r.status = b[0] >> 6;
    r.rawP = ((uint16_t)(b[0] & 0x3F) << 8) | b[1];
    r.rawT = ((uint16_t)b[2] << 3) | (b[3] >> 5);
    r.pressure = ((float)r.rawP - OUT_MIN) * (cfg.pMax - cfg.pMin)
```

```

        / (float)(OUT_MAX - OUT_MIN) + cfg.pMin;
    r.kPa    = r.pressure * cfg.toKpa;
    r.tempC  = (float)r.rawT * 200.0f / 2047.0f - 50.0f;
    return r;
}

bool sensorUsable(const HscReading &r) {
    if (r.status != 0) return false;
    if (r.rawP == 0 && r.rawT == 0) return false;
    if (r.rawP == 0x3FFF && r.rawT == 0x7FFF) return false;
    return true;
}

// ===== FLASH: HARDWARE SPI =====
const int FLASH_CS = 7;
SPISettings flashSpi(2000000, MSBFIRST, SPI_MODE0);

enum {
    CMD_WRITE_ENABLE = 0x06, CMD_READ_STATUS1 = 0x05,
    CMD_READ_DATA     = 0x03, CMD_PAGE_PROGRAM = 0x02,
    CMD_SECTOR_ERASE  = 0x20, CMD_JEDEC_ID    = 0x9F,
};

const uint32_t CHIP_SIZE = 0x800000;    // W25Q64 = 8 MByte
const uint32_t LOG_BASE  = 0x000000;

inline void flashSelect() { digitalWriteFast(FLASH_CS, LOW); }
inline void flashDeselect() { digitalWriteFast(FLASH_CS, HIGH); }

uint32_t flashReadJedecId() {
    SPI.beginTransaction(flashSpi);
    flashSelect();
    SPI.transfer(CMD_JEDEC_ID);
    uint8_t mfr = SPI.transfer(0), type = SPI.transfer(0), cap = SPI.transfer(0);
    flashDeselect();
    SPI.endTransaction();
    return ((uint32_t)mfr << 16) | ((uint32_t)type << 8) | cap;
}

uint8_t flashStatus() {
    SPI.beginTransaction(flashSpi);
    flashSelect();
    SPI.transfer(CMD_READ_STATUS1);
    uint8_t s = SPI.transfer(0);
    flashDeselect();
    SPI.endTransaction();
    return s;
}

bool flashWaitBusy(uint32_t timeout_ms = 800) {
    const int NEED = 3;
    int clean = 0;
    uint32_t t0 = millis();
    while (millis() - t0 < timeout_ms) {
        if ((flashStatus() & 0x01) == 0) {
            if (++clean >= NEED) return true;
        } else {
            clean = 0;
        }
        delayMicroseconds(50);
    }
    return false;
}

void flashWriteEnable() {
    SPI.beginTransaction(flashSpi);

```

```

    flashSelect(); SPI.transfer(CMD_WRITE_ENABLE); flashDeselect();
    SPI.endTransaction();
}

bool flashSectorErase(uint32_t addr) {
    flashWriteEnable();
    SPI.beginTransaction(flashSpi);
    flashSelect();
    SPI.transfer(CMD_SECTOR_ERASE);
    SPI.transfer((addr >> 16) & 0xFF); SPI.transfer((addr >> 8) & 0xFF); SPI.transfer(addr &
0xFF);
    flashDeselect();
    SPI.endTransaction();
    return flashWaitBusy();
}

bool flashPageProgram(uint32_t addr, const uint8_t *data, size_t len) {
    flashWriteEnable();
    SPI.beginTransaction(flashSpi);
    flashSelect();
    SPI.transfer(CMD_PAGE_PROGRAM);
    SPI.transfer((addr >> 16) & 0xFF); SPI.transfer((addr >> 8) & 0xFF); SPI.transfer(addr &
0xFF);
    for (size_t i = 0; i < len; i++) SPI.transfer(data[i]);
    flashDeselect();
    SPI.endTransaction();
    return flashWaitBusy();
}

uint8_t flashReadByte(uint32_t addr) {
    SPI.beginTransaction(flashSpi);
    flashSelect();
    SPI.transfer(CMD_READ_DATA);
    SPI.transfer((addr >> 16) & 0xFF); SPI.transfer((addr >> 8) & 0xFF); SPI.transfer(addr &
0xFF);
    uint8_t b = SPI.transfer(0x00);
    flashDeselect();
    SPI.endTransaction();
    return b;
}

// Find where to resume logging = first blank page after existing data.
// Records always start on a page boundary (see the padding logic in
// logRecord), and a record's first byte is a digit, never 0xFF. So the
// first byte of a page is 0xFF iff that page is erased. Data is written
// contiguously from address 0, so a binary search finds the boundary fast
// (~15 reads) instead of scanning all 32768 pages.
uint32_t findLogStart() {
    const uint32_t numPages = CHIP_SIZE / 256;
    // If the very first page is blank, the whole log is empty.
    if (flashReadByte(LOG_BASE) == 0xFF) return 0;

    uint32_t lo = 0, hi = numPages;          // lo = written, hi = assumed blank/end
    while (hi - lo > 1) {
        uint32_t mid = (lo + hi) / 2;
        if (flashReadByte(LOG_BASE + mid * 256) == 0xFF) hi = mid;    // blank
        else lo = mid;        // has data
    }
    return hi * 256;    // first blank page (== CHIP_SIZE if flash is full)
}

// ===== THERMOCOUPLE: MAX31855 =====
const int TC_CS = 9;
SPISettings tcSpi(4000000, MSBFIRST, SPI_MODE0);

TcReading tcRead() {

```

```

TcReading r;
uint32_t raw = 0;

SPI.beginTransaction(tcSpi);
digitalWriteFast(TC_CS, LOW);
delayMicroseconds(1);
for (int i = 0; i < 4; i++) raw = (raw << 8) | SPI.transfer(0x00);
digitalWriteFast(TC_CS, HIGH);
SPI.endTransaction();

r.fault = raw & 0x07;

int16_t internalRaw = (raw >> 4) & 0x0FFF;
if (internalRaw & 0x0800) internalRaw |= 0xF000;
r.internalC = internalRaw * 0.0625f;

int16_t tcRaw = (raw >> 18) & 0x3FFF;
if (tcRaw & 0x2000) tcRaw |= 0xC000;
r.tempC = tcRaw * 0.25f;

r.ok = (r.fault == 0) && (((raw >> 16) & 0x1) == 0) && (raw != 0xFFFFFFFF) && (raw != 0);
return r;
}

// ===== LOGGING STATE =====
bool    flashOk = false;
bool    logFull = false;
uint32_t logAddr = 0;

// Append one CSV record. Returns true on success. Same on-flash format as
// pres_temp_flash1 so check_flash / the CSV columns line up exactly:
//   t_ms, ram_kPa, ram_T, static_kPa, static_T, q_kPa, tc_C, tc_int_C, tc_fault
bool logRecord(uint32_t t_ms, const HscReading &ram, const HscReading &sta,
               float q_kPa, const TcReading &tc) {
    if (logFull) return false;

    char rec[128];
    int n = snprintf(rec, sizeof(rec), "%lu,%.3f,%.2f,%.3f,%.2f,%.4f,%.2f,%.2f,%u\n",
                    (unsigned long)t_ms, ram.kPa, ram.tempC,
                    sta.kPa, sta.tempC, q_kPa,
                    tc.tempC, tc.internalC, (unsigned)tc.fault);
    if (n <= 0) return false;
    if (n > (int)sizeof(rec)) n = sizeof(rec);

    // A page program can't cross a 256-byte page boundary; if this record
    // won't fit in what's left of the page, skip to the next page.
    uint32_t pageRemain = 256 - (logAddr % 256);
    if ((uint32_t)n > pageRemain) logAddr += pageRemain;

    if (logAddr + n > CHIP_SIZE) { logFull = true; return false; }

    // Erase a sector the first time we step into it (fresh flash is already
    // erased, so this is a no-op there; it matters when reusing space).
    if ((logAddr % 4096) == 0) {
        if (!flashSectorErase(LOG_BASE + logAddr)) return false;
    }

    if (!flashPageProgram(LOG_BASE + logAddr, (const uint8_t*)rec, n)) return false;

    logAddr += n;
    return true;
}

void setup() {
    pinMode(SOFT_SCK, OUTPUT);    digitalWriteFast(SOFT_SCK, LOW);
    pinMode(SOFT_MISO, INPUT);

```

```
pinMode(sensorRam.csPin, OUTPUT);    digitalWriteFast(sensorRam.csPin, HIGH);
pinMode(sensorStatic.csPin, OUTPUT); digitalWriteFast(sensorStatic.csPin, HIGH);
pinMode(FLASH_CS, OUTPUT);          digitalWriteFast(FLASH_CS, HIGH);
pinMode(TC_CS, OUTPUT);              digitalWriteFast(TC_CS, HIGH);

SPI.begin();
delay(10);

flashOk = (flashReadJedecId() == 0xEF4017);
if (flashOk) {
    logAddr = findLogStart();          // resume after any existing data
    if (logAddr >= CHIP_SIZE) logFull = true;
}
}

void loop() {
    static uint32_t nextSample = 0;

    // Nothing we can do without flash; idle. (No serial by design.)
    if (!flashOk || logFull) { delay(500); return; }

    uint32_t now = millis();
    if ((int32_t)(now - nextSample) < 0) return;
    nextSample = now + SAMPLE_INTERVAL_MS;

    HscReading ram = hscRead(sensorRam);
    HscReading sta = hscRead(sensorStatic);
    TcReading tc = tcRead();

    if (!(sensorUsable(ram) && sensorUsable(sta))) return; // skip bad cycle

    float q_live = ram.kPa - sta.kPa;
    logRecord(now, ram, sta, q_live, tc);
}
```